The CSP Dichotomy Theorem

A formalization blueprint

Abstract

The Constraint Satisfaction Problem over a finite constraint language is solvable in polynomial time if the language admits a weak near-unanimity polymorphism, and is NP-complete otherwise. This document rewrites Zhuk's simplified proof [13] — together with the prerequisites it imports — as a blueprint for a formalization in Lean 4 and Mathlib. It states precisely what a formalization can and cannot claim (§1); fixes the vocabulary at a granularity a proof assistant can consume (§§2–5); records the theory's interface (§6) and its summit (§8); states the algorithm and the hard half (§§9–10); and reports what Mathlib is missing, what has been built, and what the remaining work is (§§12–13).

Throughout, *formalization notes* mark the places where the informal proof has latitude that a formal one does not: a suppressed side condition, a quantifier whose scope is ambiguous, an object asserted to exist without an argument, or a step whose well-foundedness is left to the reader.

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Introduction

The dichotomy conjecture of Feder and Vardi [6] was settled independently by Bulatov [5] and Zhuk [12] in 2017. Both proofs are long. Zhuk's, in the form published in 2020, is seventy-eight journal pages, and its central induction alternates between global and local information in a way its author later described as forcing "a very complicated induction to prove most of the claims".

In 2024 Zhuk published a simplified proof [13]. The simplification is a redesign of the theory of strong subalgebras so that *every reduction is either strong or global*: for each domain D_x one can build a chain $D_x \supsetneq D_x^{(1)} \supsetneq \cdots \supsetneq \{a\}$ in which every step is either a strong subset or the intersection with a block of an equivalence relation with very strong properties. The types of the intermediate sets need not be tracked — only that such a chain exists, written $C \lhd\lhd\lhd B$. This is what makes the argument tractable, and it is what makes it a plausible formalization target.

Why this route. Among the available proofs, [13] is the only complete, recent, rigorously written one whose author has already removed the obstruction that made its predecessor unformalizable. Bulatov's proof requires tame congruence theory and centralizer theory, neither of which is formalized anywhere. Restricting to minimal Taylor algebras — which [13] itself suggests would simplify its §2 — turns out to be a bad trade: it would simplify perhaps a quarter of one section and none of the CSP argument, at the cost of importing the cyclic term theorem, which the present route does not otherwise need. Brady's notes [2] are the clearest prose in the subject and are the best source for several of the imported prerequisites, but they stop short of the dichotomy. The route taken here is [13] as the spine, with [2] for the imports, on top of an existing formalization of Zhuk's centre theorem.

What is claimed. Not that the dichotomy has been formalized. §13.5 estimates the transitive closure of what must be proved at 100–130 printed pages, which extrapolates to tens of thousands of lines of Lean, none of it attempted in any proof assistant. That is a large number but not, at current throughput, a long time; what governs the schedule is the critical path and the defects of §11, not the line count. What is offered is its design: a route, precise statements, a vocabulary pinned down where the sources leave it loose, an inventory of the missing Mathlib layers, and a Lean development whose foundational modules are complete and whose target theorems are stated.

What the formalization notes are for. A blueprint's value is mostly in its failures of nerve — the places where writing the statement down carefully shows that it does not quite say what it appeared to say. The predecessor project found four such places in twenty-four pages, and all four were in the foundations rather than in the mathematics. The notes here record the same kind of thing: that $\mathbf{Z}_p$ lies in $\mathcal{V}_n$ only when $p \mid n-1$ (Hazard 3.3); that σ^* need not be a congruence (Hazard 3.22); that images do not commute with intersection, and that one proof needs them to (Hazard 3.14); that "we cannot weaken forever" needs a measure that is not the obvious one (Hazard 5.11); that the main induction quantifies over the instance *inside* (Hazard 8.2). §11 collects the harder cases separately: ten places where the source cannot be transcribed at all, of which one — [13]'s restatement of a lemma from [14], with two hypotheses dropped — is false as printed, with a two-line counterexample.

1 What is being proved

1.1 The theorem

Fix a finite set A and a finite set Γ of relations on A, the *constraint language*. An instance of CSP(Γ) is a conjunction $R_1(\bar{x}_1) \wedge \cdots \wedge R_s(\bar{x}_s)$ with each $R_i \in \Gamma$; the question is whether it is satisfiable.

Theorem 1.1 (Dichotomy; Bulatov, Zhuk)**.** *If some weak near-unanimity operation preserves* Γ*, then* CSP(Γ) *is solvable in polynomial time. Otherwise* CSP(Γ) *is* NP*-complete.*

Both halves are conditional on a model of computation, and this is not a formality that a formalization may wave through. "Solvable in polynomial time" quantifies over an encoding of instances as strings, a machine, and a polynomial bound on its step count; "NP-complete" additionally quantifies over all problems in NP and over polynomial-time many-one reductions. Mathlib supplies Turing machines (`Mathlib/Computability/TuringMachine/`) and a notion of computability, but it has *no* complexity class, no polynomial-time predicate, and no notion of polynomial-time reduction.

1.2 Where the work actually is

An earlier draft of this section concluded from that gap that the complexity-theoretic half should be deferred because building it would dominate the project. That inference was wrong, and it is worth recording why, because the corrected version is what organises the rest of the document.

The complexity vocabulary itself — languages, a cost model, P, NP by verifiers, $\leq_p$, NP-hardness, NP-completeness, and a generic NP-hard problem — has since been built, in about 850 lines. That is consistent with the two existing Cook–Levin formalizations, whose class layers are 702 lines (Isabelle) and 992 lines (Coq) against totals of 56 514 and $\approx$ 16 500. Defining the classes is cheap in every development that has done it.

What is expensive is not complexity theory. It is any statement that requires one to *exhibit a program*. In a formalization, "f is computable in polynomial time" is not a property one observes of a mathematical function; it is an existential over programs, discharged only by writing one and proving a bound on its step count. The dividing line runs along that axis, and it cuts across the algebra/complexity split rather than agreeing with it:

Target	Exhibit a program?	Cost
(T0) the algebraic core	no	the mathematics
(T1) SOLVE is correct	no (but see Hazard 1.2)	moderate
(T2) SOLVE is polynomial	**yes — the largest one here**	**large**
(H0) Γ pp-interprets NAE	no	the mathematics
(H1) pp-interpretation $\Rightarrow \leq_p$	yes, a syntactic substitution	small

So the document still separates the machine-free statements from the rest, but for a sharper reason than before, and with a different verdict on (T2).

1.3 The layered targets

Write $\mathbf{A} = (A; w)$ for a finite idempotent algebra whose basic operation w is a special weak near-unanimity operation, and $\mathcal{V}_n$ for the class of such algebras with w of arity n (Convention 2.1).

Tractable half. Three layers.

(T0) ***The algebraic core.*** The three theorems of §8 — the existence of a strong subuniverse or a linear congruence, the safety of a reduction to a strong subuniverse, and the codimension-one theorem — together with everything they rest on. These are statements about finite algebras, congruences, and CSP instances regarded as finite combinatorial objects. No computation appears in them.

(T1) ***Correctness of the algorithm.*** Zhuk's algorithm written as a total function SOLVE from instances to Booleans, and the proposition

$$\textsc{Solve}(\mathcal{I}) = \texttt{true} \iff \mathcal{I} \text{ has a solution.}$$

This is a statement about a recursive function, and it is provable from (T0) with no cost model. It is the honest formal content of "the algorithm works" — *provided* SOLVE actually computes; see Hazard 1.2.

(T2) ***The complexity wrapper.*** That SOLVE runs in polynomial time in the size of the instance, for fixed Γ. This is the one place a cost model is needed on this side, and it is a large undertaking rather than a wrapper: see Hazard 1.3.

Hard half. Two layers.

(H0) ***The algebraic core.*** If no weak near-unanimity operation preserves Γ then Γ primitively positively interprets every finite constraint language — in particular one whose CSP is NP-hard. This is again pure algebra: the Galois connection between polymorphisms and primitive positive definability, the Maróti–McKenzie theorem that a Taylor term yields a weak near-unanimity term, and the Bulatov–Jeavons–Krokhin reduction.

(H1) ***The complexity wrapper.*** That a primitive positive interpretation yields a polynomial-time many-one reduction, and hence NP-hardness. This does need a program, but the program performs a syntactic substitution of formulas whose size is linear in the instance, so it is genuinely small.

The claim this document makes is that (T0), (T1) and (H0) are the mathematics and are what the literature actually proves. (H1) is a small addition on top. (T2) is neither small nor mathematics, but it is not *unrelated* to the dichotomy either — it is the assertion that the algorithm one has just proved correct is also fast, and it is the second-largest component of the project. The blueprint is written for (T0), (T1) and (H0); §12 states (T2) and (H1) precisely and says what building them costs.

Formalization note 1.2 ((T1) is vacuous unless SOLVE computes)**.** "SOLVE written as a total function" hides a requirement that no cost model would catch. If SOLVE is defined using classical choice — for instance by deciding "does D_x have a nonempty proper binary absorbing subuniverse?" with `Classical.dec` — then $\textsc{Solve}(\mathcal{I}) = \texttt{true} \iff \mathcal{I}$ has a solution is a true theorem that says nothing algorithmic, because SOLVE does not reduce. To mean what it appears to mean, (T1) needs every predicate the algorithm branches on to carry a genuine decision procedure.

That is a real obligation with a locatable cost. Absorption is defined by an existential over *terms* — an infinite type — so "$C <_{\mathrm{BA}} B$" is not decidable on its face. It becomes decidable through the bound that a witness of arity at most $|A|^{|A|}$ suffices, which is a theorem and not a definition. Hazard 9.1(b) records that the bound exists; what it does not record, and what matters here, is that (T1) is empty without it. The same applies to the searches over congruences and over subsets.

Formalization note 1.3 ((T2) is not a wrapper)**.** Two things make (T2) much larger than the phrase "complexity wrapper" suggests.

The program is enormous. (T2) requires SOLVE — FORCECONSISTENCY, the search for a strong subuniverse, SOLVELINEAR, and the mutual recursion between them — to be expressed in a cost model, with a proved polynomial bound. Every existing measurement says this kind of work dominates: it is the 50 000 lines of Turing-machine engineering in the Isabelle Cook–Levin development, against 702 for the classes themselves.

The statement is non-uniform. Because Γ is fixed, the polynomial may depend on Γ, and the searches over subsets of A, over congruences on A, and over term operations of bounded arity are all constant-time. But "constant" here means the search is compiled away into a program whose *size* depends on $|A|$. So SOLVE is not one program: it is a family SOLVE_Γ with a polynomial p_Γ for each Γ, and (T2) must be stated in that form. Attempting a bound uniform in Γ would be proving something else, and something false.

Remark 1.4. The separation is not a dodge. Layer (T1) is strictly stronger than " CSP(Γ) is decidable", which is trivial for a finite template: it says that one particular, highly non-obvious algorithm — the one whose existence is the content of the dichotomy — is correct. Everything that makes the dichotomy hard lives in (T0). Conversely, no part of (T0)–(T1) becomes easier if a cost model is available.

2 Standing conventions

Zhuk's papers carry their hypotheses in prose: "in this paper we assume that every algebra is a finite idempotent algebra having a WNU term operation". A formalization cannot carry a hypothesis in prose, and an informal reader cannot audit which results actually consume which hypothesis. We therefore fix the hypotheses as a labelled convention and record, for each one, where it is consumed.

Convention 2.1 (Standing hypotheses)**.** Throughout, $n \geq 3$ is a fixed integer and:

(a) every algebra is *finite* and has a *nonempty* universe;
(b) every algebra has exactly one basic operation, of arity n, written w;
(c) w is *idempotent*: $w(x, \dots, x) = x$;
(d) w is a *weak near-unanimity* operation: $w(y, x, \dots, x) = w(x, y, x, \dots, x) = \cdots = w(x, \dots, x, y)$ for all x, y;
(e) w is *special*: $w(x, \dots, x, y) = w(x, \dots, x, w(x, \dots, x, y))$ for all x, y.

The class of algebras satisfying a–e is written $\mathcal{V}_n$. Since only finite algebras are considered, $\mathcal{V}_n$ is not a variety.

Remark 2.2 (Why one basic operation)**.** Restricting to a single basic operation is not a loss. The dichotomy hypothesis is that *some* weak near-unanimity operation preserves Γ; one may then discard the original signature and work in the algebra $(A; w)$ whose only basic operation is that one, because every relation of Γ is a subuniverse of a power of $(A; w)$ and that is all the argument uses. Passing from a WNU w to a *special* one is Lemma 6.20, and costs an increase of arity from n to $n^{n!}$.

The gain is large for a formalization: a term is a finite tree with n-ary branching over a single symbol, so "term operation" needs no signature bookkeeping, and $\mathrm{Clo}(\mathbf{A})$ is the clone generated by one operation.

Formalization note 2.3 (Where each hypothesis is consumed)**.** a is used in every induction on $|A|$ and in every minimality argument (a minimal element of a nonempty family of subsets exists); it cannot be dropped anywhere. c is what makes singletons subuniverses and makes Sg of a nonempty set nonempty. d is consumed only through the results it implies — the existence of absorption-theoretic structure — and never directly in §§5–8. e is used where a unary polynomial must be idempotent under iteration, chiefly in the theory of Sg-closures of two-element sets. A formalization should carry a–e as typeclass instances on a structure and not as ambient hypotheses, so that the audit is mechanical.

Convention 2.4 (The empty set)**.** A *subuniverse* of $\mathbf{A}$ is a subset closed under w; the empty set is a subuniverse under this definition, and we allow it. A *subalgebra* is a nonempty subuniverse; $\mathbf{B} \leq \mathbf{A}$ always carries $B \neq \varnothing$.

The relations $<_T$, $\leq_T$ and $\lhd\lhd\lhd$ of §4 are defined only between *nonempty* sets; in particular $\varnothing \lhd\lhd\lhd A$ never holds. Where a construction can produce the empty set — a projection of an intersection, most often — the dotted variants $\dot{<}_T$, $\dot{\leq}_T$, $\dot{\lhd}\dot{\lhd}\dot{\lhd}$ are used, and they mean "the stated relation holds, or the left-hand side is empty".

Formalization note 2.5 (The dot is not decoration)**.** Conflating $\lhd\lhd\lhd$ with $\dot{\lhd}\dot{\lhd}\dot{\lhd}$ is the single most common way to misread §4. Every propagation statement of §6 that projects or intersects produces a dotted conclusion, and every statement that consumes a $\lhd\lhd\lhd$ hypothesis needs the undotted one. In a formalization the two should be distinct constants, with the dotted one *defined* as `C = ∅ ∨ C ◁◁◁ B`, so that the case split is forced at every use site rather than being resolved by the reader's judgement.

Convention 2.6 (Relations, and indices)**.** A relation is a subset of a product $A_1 \times \cdots \times A_m$ of universes, indexed by $[m] = \{1, \ldots, m\}$; $\mathbf{R} \leq \mathbf{A}_1 \times \cdots \times \mathbf{A}_m$ means additionally that R is a subuniverse of the product algebra. We write $\mathrm{pr}_{i_1,\ldots,i_s}(R)$ for the projection onto the listed coordinates and $\mathbf{R} \leq_{\mathrm{sd}} \mathbf{A}_1 \times \cdots \times \mathbf{A}_m$ when every $\mathrm{pr}_i(R) = A_i$.

Products are indexed by *arbitrary finite index sets*, not only by initial segments of $\mathbb{N}$. Several constructions in §8 form a product over the variable set of an instance, or over a set of subuniverses, and reindexing those to $[m]$ by hand is exactly the kind of bookkeeping that a formalization pays for repeatedly and that buys nothing.

2.1 Legislated readings

Ten places in [13] admit more than one reading, and in each the choice is invisible in prose but forced in a formalization. This subsection fixes all ten. Two of them turn out not to be conventions at all: item i is a theorem, proved here, and item x is a missing lemma, proved in §6. Of the remaining eight, six are decided by internal evidence — one reading makes some statement of the source true or some proof valid and the other does not, so the "choice" is really a reconstruction — and only items vi and vii are free.

(i) **The algebra $\mathbf{Z}_p$.** [13] declares that "every algebra $\mathbf{Z}_p$ belongs to $\mathcal{V}_n$ for a fixed n", with basic operation $x_1 + \cdots + x_n \bmod p$. That operation is idempotent only when $p \mid n-1$, so the declaration is a constraint on p, and the source never says where the constraint comes from. It comes from Convention 2.1e; see Proposition 2.7 below. We therefore keep the two roles of $\mathbf{Z}_p$ apart: the abelian *group* $(\mathbb{Z}_p; +, -)$, which supplies the relation $x_1 - x_2 = x_3 - x_4$ in Definition 4.10 and carries no arity constraint, and the *algebra* $\mathbf{Z}_p = (\mathbb{Z}_p; x_1 + \cdots + x_n) \in \mathcal{V}_n$, which does. Only the second occurs in Definition 3.35, where $\zeta \leq \mathbf{A} \times \mathbf{A} \times \mathbf{Z}_p$ is a subuniverse of a product and therefore needs a common signature.

(ii) σ^* **is a tolerance.** It is a reflexive symmetric subuniverse of $\mathbf{A}^2$ and *not* in general a congruence (Proposition 3.23). That σ^* is a congruence is item (b) of the definition of a linear congruence and conclusion (a) of Lemma 7.39; typing it as a congruence would make the first vacuous and collapse the linear/PC distinction.

(iii) **The empty set.** As in Convention 2.4: subuniverses may be empty, subalgebras may not, and the typed relations hold only between nonempty sets. Two statements of the source must be restated because they are true only under the opposite reading. Read syntactically, $\varnothing$ is vacuously absorbing and vacuously central, so *(a)* the clause defining $<_{\mathrm{s}}$ needs its witness D required nonempty (Definition 4.6), and *(b)* Lemma 7.18 needs an explicit alternative "$C = \varnothing$" — without it the statement is false, as $\mathbf{A} = \mathbf{B} = \mathbf{Z}_p$ with $R = \{(x, x+1)\}$ shows. Likewise Lemma 7.1 concludes "there exists a BA or central subuniverse", which is vacuous because every algebra is a binary absorbing subuniverse of itself; read *proper nonempty*, which is how it is used.

(iv) **The empty family is allowed.** In Definition 3.21 the family $S_1, \ldots, S_k$ may be empty, the empty intersection being A^2. This is not a free choice: the two formulations given there agree *only* under it. Forbid $k = 0$ and the second no longer excludes $\sigma = A^2$ while the first still does. With $k = 0$ allowed, an irreducible congruence is automatically proper — which recovers the word "proper" that [12] carries and [13] drops — and σ^* exists.

(v) $\lhd\lhd\lhd$ **carries its chain.** $C \lhd\lhd\lhd^A B$ is *data*: a list of intermediate sets with their typed steps and witnessing congruences, not merely the proposition that such a list exists. Several statements speak of "the dividing congruences coming from $C \lhd\lhd\lhd^A B$", which is a function of the chain and not of the pair. See Hazard 4.25 for the two places where this is load-bearing.

(vi) **"Dimension".** $\prod_i \mathbb{Z}_{q_i}$ with distinct primes is not a vector space over anything and has no dimension. Use the composition length of the finite abelian group, which agrees with the source wherever the source is meaningful, and prove its additivity along short exact sequences. A free choice; nothing in §§4–6 uses the notion.

(vii) **"Linked instance".** Two inequivalent definitions coexist in the source: global connectivity of the value graph, and the per-variable condition. The second does not imply the first, because a fragmented instance is per-variable linked in each component. We use the per-variable definition and carry non-fragmentation as a separate hypothesis wherever the informal reading was doing that work (Hazard 3.12).

(viii) **Quotients of subsets.** B/δ always means the *image* $\{b/\delta : b \in B\}$ in $\mathbf{A}/\delta$, for B any subset of A; never the quotient of B by a congruence of $\mathbf{B}$. Consequently $(C_1 \cap \cdots \cap C_t)/\delta \subseteq \bigcap_i C_i/\delta$ always, with equality false in general (Hazard 3.14). The reverse inclusion is established on the fly inside the proof of Lemma 6.4(fm) and must be extracted as a standalone lemma with its hypothesis visible.

(ix) **Sentences beginning "Notice that".** Three of them are used as lemmas and never proved. They are promoted here: Lemma 4.11 (the relation S of Definition 4.10(c) is a bridge with $\widetilde{S} = \mathrm{pr}_{1,2}(S) = \mathrm{pr}_{3,4}(S) = \sigma^*$), Lemma 3.25 ($\sigma$ is irreducible, linear or PC exactly when $0_{\mathbf{A}/\sigma}$ is), and Lemma 4.8 (the two formulations of S-freeness agree). Lemma 3.16 belongs to the same list.

(x) **A citation that drops a case.** The source's Lemma 7.18 cites [14, Thm. 6.15] but omits that theorem's third alternative, a nontrivial *projective* subuniverse. The elimination is legitimate under Convention 2.1, but it is a lemma and not a convention, and the source does not state it: see Lemma 7.17 and Lemma 7.18.

Proposition 2.7 (Where $p \mid n-1$ comes from)**.** *Let $\mathbf{U}$ be a finite algebra with a single basic operation w of arity n that is idempotent, weak near-unanimity and special, and suppose $\mathbf{U}$ is affine over $\mathbb{Z}_p$ with $|U| > 1$. Then $p \mid n-1$, and w acts on $\mathbf{U}$ as $x_1 + \cdots + x_n$.*

Proof. Being affine, $\mathbf{U}$ is polynomially equivalent to a $\mathbb{Z}_p$-module, so w acts as $w(\bar{x}) = \sum_i c_i x_i + d$ for scalars $c_i \in \mathbb{Z}_p$ and a constant d. Idempotence gives $\left(\sum_i c_i\right)x + d = x$ for all x, hence $\sum_i c_i = 1$ and $d = 0$.

The weak near-unanimity identities $w(y, x, \ldots, x) = w(x, y, x, \ldots, x) = \cdots = w(x, \ldots, x, y)$ read $c_1 y + (1 - c_1)x = c_2 y + (1 - c_2)x = \ldots$, and comparing coefficients of y gives $c_1 = \cdots = c_n =: c$, so $nc = 1$.

Specialness, $w(x, \ldots, x, y) = w(x, \ldots, x, w(x, \ldots, x, y))$, reads

$$(n-1)c\,x + cy \;=\; (n-1)c\,x + c((n-1)c\,x + cy),$$

and comparing coefficients of y gives $c = c^2$, so $c \in \{0, 1\}$. Now $c = 0$ contradicts $nc = 1$, so $c = 1$; then $n \cdot 1 = 1$ in $\mathbb{Z}_p$, i.e. $p \mid n-1$, and w acts as $x_1 + \cdots + x_n$. $\square$

Corollary 2.8. $\mathbf{Z}_p = (\mathbb{Z}_p; x_1 + \cdots + x_n)$ *lies in $\mathcal{V}_n$ if and only if $p \mid n-1$; and whenever a prime p arises from a linear congruence on some $\mathbf{A} \in \mathcal{V}_n$ — so that some subquotient of $\mathbf{A}$ is affine over $\mathbb{Z}_p$ and nontrivial — one has $p \mid n-1$. Hence the source's standing identification of $\mathbf{Z}_p$ is legitimate.*

Formalization note 2.9 (The proposition is not decoration)**.** Proposition 2.7 is needed twice, and in both places its absence would leave a statement that does not typecheck. It is what makes $\mathbf{Z}_p$ an object of $\mathcal{V}_n$ at all, hence what makes Definition 3.35 well formed; and it is what upgrades the group isomorphism of Lemma 4.17 to an isomorphism of algebras. Verified by exhaustion for $p \in \{2, 3, 5, 7\}$ and $3 \le n < 12$: a special idempotent affine WNU of arity n on $\mathbb{Z}_p$ exists exactly when $p \mid n - 1$, and is then the sum.

3 Vocabulary

This section fixes the universal-algebraic vocabulary. It is Zhuk's §2.1, rewritten so that every definition is unambiguous when read by a machine. The substantive content begins in §4; a reader who knows the subject should skim to Definition 3.21 (irreducible congruences) and Definition 3.27 (bridges), which are the two places where the conventions genuinely matter.

3.1 Algebras, subuniverses, terms

Definition 3.1 (Algebra)**.** An algebra is a pair $\mathbf{A} = (A; w^{\mathbf{A}})$ with A a finite nonempty set and $w^{\mathbf{A}} \colon A^n \to A$, subject to Convention 2.1. We write A for the universe of $\mathbf{A}$ and drop the superscript on w when no confusion arises.

Definition 3.2 (The algebras $\mathbf{Z}_p$)**.** For a prime p with $p \mid n - 1$, $\mathbf{Z}_p$ is the algebra on $\{0, 1, \ldots, p - 1\}$ with $w^{\mathbf{Z}_p}(x_1, \ldots, x_n) = x_1 + \cdots + x_n \bmod p$.

Formalization note 3.3 ($\mathbf{Z}_p$ is only sometimes in $\mathcal{V}_n$)**.** Zhuk defines $\mathbf{Z}_p$ for every prime p and then asserts that "every algebra $\mathbf{Z}_p$ belongs to $\mathcal{V}_n$". That assertion carries a silent side condition. The operation $x_1 + \cdots + x_n \bmod p$ is idempotent exactly when $n \equiv 1 \pmod{p}$: idempotence says $nx \equiv x$ for all x, i.e. $p \mid n-1$. Given that, it is also a WNU (all the identities read $(n - 1)x + y = y$) and special (both sides reduce to y).

So $\mathbf{Z}_p \in \mathcal{V}_n$ if and only if $p \mid n - 1$, and every statement of the form "$\mathbf{A}/\sigma \cong \mathbf{Z}_p$ for some prime p" (Lemma 4.17, Definition 3.35) silently produces a p dividing $n - 1$. A formalization must decide whether $\mathbf{Z}_p$ is defined for all p and merely fails to lie in $\mathcal{V}_n$ for bad p, or is defined only for $p \mid n - 1$. The first is cleaner; the second makes the membership free. Either way the side condition must be visible, because "for some prime p" otherwise reads as an unconstrained existential.

Definition 3.4 (Subuniverse, subalgebra)**.** $B \subseteq A$ is a *subuniverse* of $\mathbf{A}$ if $w(b_1, \ldots, b_n) \in B$ for all $b_1, \ldots, b_n \in B$. If moreover $B \neq \varnothing$ then $\mathbf{B} = (B; w\restriction_{B^n})$ is a *subalgebra*, written $\mathbf{B} \le \mathbf{A}$.

Definition 3.5 (Term operations)**.** $\mathrm{Clo}(\mathbf{A})$ is the least set of finitary operations on A containing all projections and closed under composition with w. Its m-ary part is written $\mathrm{Clo}_m(\mathbf{A})$. Elements of $\mathrm{Clo}(\mathbf{A})$ are the *term operations* of $\mathbf{A}$.

Formalization note 3.6 (Terms versus term operations)**.** Zhuk writes $t \in \mathrm{Clo}(\mathbf{A})$ and treats t as both a syntactic term and the operation it induces. These must be distinguished in a formalization: a term is a finite tree, and distinct terms may induce the same operation. Every statement below that quantifies over $\mathrm{Clo}(\mathbf{A})$ is a statement about *operations*, so the syntactic layer may be quotiented away — but it cannot be skipped, because Sg is characterised through terms (Lemma 3.8) and because the arity bookkeeping in the absorption arguments is syntactic.

Definition 3.7 (Generation)**.** For $X \subseteq A^m$, $\mathrm{Sg}_{\mathbf{A}}(X)$ is the least subuniverse of $\mathbf{A}^m$ containing X.

Lemma 3.8 (Generation by terms)**.** *Let $X = \{x_1, \ldots, x_k\} \subseteq A^m$ be nonempty. Then*

$$\mathrm{Sg}_{\mathbf{A}}(X) = \{\, t^{\mathbf{A}^m}(x_1, \ldots, x_k) \;:\; t \in \mathrm{Clo}_k(\mathbf{A}) \,\}.$$

The generator list is fixed*: the same list $x_1, \ldots, x_k$ in the same order serves every element of the closure, with t varying.*

Formalization note 3.9. The “fixed list” phrasing is what makes Lemma 3.8 usable. The weaker reading — for each y in the closure there are *some* generators and *some* term — is also true but useless, because every later application applies one term simultaneously to several closure elements and needs their generator lists to coincide. This was a defect found in review of the predecessor blueprint and is called out here for the same reason.

3.2 Relations

Definition 3.10 (Relational vocabulary)**.** Let $R \subseteq A_1 \times \cdots \times A_m$.

(a) R is *subdirect* if $\mathrm{pr}_i(R) = A_i$ for every i; $\mathbf{R} \leq_{\mathrm{sd}} \mathbf{A}_1 \times \cdots \times \mathbf{A}_m$ abbreviates “R is a subdirect subuniverse”.
(b) For $R \subseteq A^m$: R is *reflexive* if $(a, \ldots, a) \in R$ for every $a \in A$.
(c) For binary $\delta_1 \subseteq A_1 \times A_2$, $\delta_2 \subseteq A_2 \times A_3$: $\delta_1 \circ \delta_2 = \{(a, c) : \exists b\ (a, b) \in \delta_1 \wedge (b, c) \in \delta_2\}$, and $\delta^{-1} = \{(b, a) : (a, b) \in \delta\}$.
(d) For $B \subseteq A_1$ and $\delta \subseteq A_1 \times A_2$: $B \circ \delta = \{c : \exists b \in B\ (b, c) \in \delta\}$.
(e) A subdirect $\delta \subseteq A \times B$ is *linked* if the bipartite graph on $A \sqcup B$ with edge set δ is connected. It is *bijective* if $|\delta| = |A| = |B|$.
(f) For binary σ and $m \geq 2$, $\sigma^{[m]} = \{(a_1, \ldots, a_m) : (a_i, a_j) \in \sigma\ \forall i, j\}$.

Formalization note 3.11 (Composition is diagrammatic)**.** $\delta_1 \circ \delta_2$ applies δ_1 *first.* This is the order of `Rel.comp` and the opposite of `Function.comp`. Every composite of bridges and of congruences below is written left to right, and silently reversing it produces statements that are true-looking and wrong — in Corollary 6.14(l) the composite $\sigma_k \circ \mathrm{pr}_{k,\ell}(R) \circ \sigma_\ell$ is asymmetric in k and ℓ and the order is the content.

Formalization note 3.12 (“Linked” is used for two different things)**.** Zhuk uses *linked* both for a binary relation (Definition 3.10(e)) and for a CSP instance (Definition 5.8). They are related but not the same predicate, and the coincidence of names causes real confusion in §8, where both occur in one proof. Use two names.

Definition 3.13 (Quotients of relations)**.** For an equivalence σ on A and $a \in A$, a/σ is the class of a; for $B \subseteq A$, $B/\sigma = \{b/\sigma : b \in B\}$; for $R \subseteq A^m$, $R/\sigma = \{(b_1/\sigma, \ldots, b_m/\sigma) : \bar{b} \in R\}$.

Formalization note 3.14 (Images do not commute with intersection)**.** B/σ and R/σ are *images.* Three consequences that the source navigates silently:

(a) B/σ is a subset of A/σ, not the quotient of B by $\sigma \cap B^2$. For B a subuniverse and σ a congruence the two are canonically isomorphic as algebras, and the source moves between them freely. Fix B/σ to be the image and prove the isomorphism once.
(b) In general $(C_1 \cap C_2)/\sigma \subsetneq C_1/\sigma \cap C_2/\sigma$. Lemma 6.4(fm) needs equality for a family $C_1, \ldots, C_t$, and gets it from the extra hypothesis $(C_i \circ \sigma) \cap B = C_i$, i.e. that each C_i is σ-saturated inside B. That is a content step, not notation.
(c) R/σ need not be stable under anything, and need not be a subuniverse of $(\mathbf{A}/\sigma)^m$ unless σ is a congruence.

3.3 Rectangularity

Definition 3.15 (Parallelogram property, rectangularity)**.** Let $R \subseteq A_1 \times \cdots \times A_m$.

(a) The i-th variable of R is *rectangular* if for all $\bar{a}, \bar{b}$,

$$\bar{a}[i \mapsto b_i] \in R, \quad \bar{b}[i \mapsto a_i] \in R, \quad \bar{b} \in R \implies \bar{a} \in R,$$

where $\bar{a}[i \mapsto c]$ is $\bar{a}$ with its i-th entry replaced by c. R is *rectangular* if every variable is.

(b) *R has the parallelogram property* if for every permutation of the coordinates giving R' and every $\ell \in [m-1]$,

$$(a_1, \dots, a_\ell, b_{\ell+1}, \dots, b_m) \in R', \quad (b_1, \dots, b_\ell, a_{\ell+1}, \dots, a_m) \in R', \quad \bar{b} \in R' \implies \bar{a} \in R'.$$

(c) The *rectangular closure* of R is the least rectangular relation containing R.

Lemma 3.16 (Two assertions the source makes in passing)**.** (a) *A relation with the parallelogram property is rectangular.*
(b) *The rectangular closure exists.*

Proof. (a) Apply the parallelogram property to the permutation carrying coordinate i to position 1, with $\ell = 1$. (b) Rectangularity at coordinate i is a Horn implication, so the rectangular relations containing R are closed under arbitrary intersection and the family is nonempty (it contains $A_1 \times \cdots \times A_m$); the intersection of all of them is the least one. □

Formalization note 3.17. Both parts are asserted without proof in [13] — (a) as "as it follows from the definitions", (b) by the definite article in "*the* minimal rectangular relation". Both are easy and both must be present. Note that "R is rectangular" unqualified — the form used in "rectangular closure" — is never defined in the source; it means every variable is rectangular.

Formalization note 3.18 (Quantifying over permutations)**.** The parallelogram property is stated "any permutation of its variables gives a relation R' satisfying …". Formally this is a quantification over $\mathfrak{S}_m$ combined with a quantification over the split point ℓ; equivalently, and more usably, over all ways of splitting $[m]$ into two nonempty blocks. The second formulation is the one to formalize: it avoids transporting R along a permutation, which in a dependently typed setting means an equivalence of index types and a coercion at every use.

3.4 Congruences

Definition 3.19 (Stability)**.** Let $R \subseteq A_1 \times \cdots \times A_m$ and let σ be a binary relation on A_i. The i-th variable of R is *stable under* σ if $\bar{a} \in R$ and $(a_i, b) \in \sigma$ imply $\bar{a}[i \mapsto b] \in R$. R is *stable under* σ if every variable is (each with respect to σ on the corresponding factor, when the factors agree).

Definition 3.20 (Congruence)**.** A *congruence* on $\mathbf{A}$ is an equivalence relation on A that is a subuniverse of $\mathbf{A} \times \mathbf{A}$. It is *nontrivial* if it is neither the equality relation $0_{\mathbf{A}}$ nor A^2. The quotient $\mathbf{A}/\sigma$ is the algebra on A/σ with $w(\bar{a}/\sigma) = w(\bar{a})/\sigma$.

Definition 3.21 (Irreducible congruence and its cover)**.** A congruence σ on $\mathbf{A}$ is *irreducible* if it cannot be written as an intersection of binary subuniverses of $\mathbf{A} \times \mathbf{A}$ each stable under σ and each properly containing σ. Equivalently: there are no $\mathbf{S}_1, \dots, \mathbf{S}_k \leq \mathbf{A}/\sigma \times \mathbf{A}/\sigma$ with $0_{\mathbf{A}/\sigma} = S_1 \cap \cdots \cap S_k$ and $S_i \neq 0_{\mathbf{A}/\sigma}$ for every i.

For irreducible σ, σ^* denotes the least $\delta \leq \mathbf{A} \times \mathbf{A}$ with $\delta \supsetneq \sigma$ and δ stable under σ.

Formalization note 3.22 (Three traps in Definition 3.21)**.** (a) Irreducibility is *not* meet-irreducibility in the congruence lattice. The $\mathbf{S}_i$ range over binary *subuniverses* stable under σ, which need not be equivalence relations. The two notions differ, and every use below is the stated one.

(b) σ^* need not be a congruence. It is a subuniverse of $\mathbf{A}^2$ stable under σ, nothing more; that σ^* *is* a congruence is an extra hypothesis in the definition of a linear congruence (Definition 4.10), which would be redundant otherwise.

(c) Existence and uniqueness of σ^* need an argument, given in Proposition 3.23. Uniqueness is exactly what irreducibility buys; without it there is no "the" cover. A formalization must produce σ^* as data attached to a proof of irreducibility, not as a standalone function.

(d) The family $S_1, \ldots, S_k$ may be *empty* (item iv of §2.1). This is not a free choice: the two formulations above agree only under it.

Proposition 3.23 (The cover exists, and is a tolerance)**.** *Let σ be an irreducible congruence on $\mathbf{A}$. Then*

(a) *$\sigma \neq A^2$;*
(b) *the family $\mathcal{F}$ of $\delta \leq \mathbf{A} \times \mathbf{A}$ with $\delta \supsetneq \sigma$ and δ stable under σ is nonempty and closed under intersection, so it has a least element σ^*;*
(c) *σ^* is reflexive and symmetric.*

Proof. (a) The empty family is allowed, and its intersection is A^2; so if $\sigma = A^2$ then σ is an intersection of a family of relations each properly containing it, vacuously, and σ is reducible.

(b) $\mathcal{F} \ni A^2$ by (a). If $\delta_1, \delta_2 \in \mathcal{F}$ then $\delta_1 \cap \delta_2$ is again a subuniverse of $\mathbf{A}^2$ stable under σ and contains σ; it contains σ *properly*, since otherwise $\sigma = \delta_1 \cap \delta_2$ exhibits σ as an intersection of two members of $\mathcal{F}$, contradicting irreducibility. So $\mathcal{F}$ is closed under binary, hence finite, intersection, and $\sigma^* = \bigcap \mathcal{F} \in \mathcal{F}$ is its least element.

(c) $\sigma^* \supseteq \sigma \supseteq 0_{\mathbf{A}}$ is reflexive. For symmetry, $(\sigma^*)^{-1}$ is a subuniverse of $\mathbf{A}^2$, is stable under σ because σ is symmetric, and properly contains $\sigma^{-1} = \sigma$; so $(\sigma^*)^{-1} \in \mathcal{F}$ and therefore $\sigma^* \subseteq (\sigma^*)^{-1}$, whence equality. ☐

Remark 3.24 (Why not a congruence)**.** Nothing in Proposition 3.23 gives transitivity, and the source supplies the internal evidence that none is available: that σ^* is a congruence is item (b) of Definition 4.10 and conclusion (a) of Lemma 7.39, both of which would be redundant otherwise. Symmetry of σ^*, by contrast, is used silently in [13] — in the proof of Lemma 4.15, where the argument is run again "switching x_1 and x_2".

Lemma 3.25 (Passing to the quotient)**.** *Let σ be a congruence on $\mathbf{A}$. Then $\delta \mapsto \delta/\sigma$ is an inclusion-preserving bijection, commuting with intersections, from the subuniverses of $\mathbf{A}^2$ that contain σ and are stable under σ onto the subuniverses of $(\mathbf{A}/\sigma)^2$, sending σ to $0_{\mathbf{A}/\sigma}$. Consequently σ is irreducible, linear or PC exactly when $0_{\mathbf{A}/\sigma}$ is, and then $\sigma^*/\sigma = (0_{\mathbf{A}/\sigma})^*$.*

Proof. A subuniverse δ of $\mathbf{A}^2$ containing σ and stable under σ is a union of products of σ-blocks, so it is the preimage of δ/σ; conversely the preimage of a subuniverse of $(\mathbf{A}/\sigma)^2$ is a subuniverse containing σ and stable under σ. The two constructions are mutually inverse and preserve inclusion and intersection, and σ is the preimage of $0_{\mathbf{A}/\sigma}$. The statement of Definition 3.21 is a statement about that family alone, so it transports; and both the least element in Proposition 3.23(b) and the data of Definition 4.10 transport with it. ☐

Formalization note 3.26 (This is the licence for "assume $\sigma = 0$")**.** Three proofs in §6 open with "we may factor by σ and assume σ is the equality relation". Lemma 3.25 is what licenses that, and it is one of the sentences [13] states without proof (§2.1, item ix).

3.5 Bridges

Definition 3.27 (Bridge)**.** Let σ_1, σ_2 be congruences on $\mathbf{D}_1, \mathbf{D}_2$. A relation $\delta \leq \mathbf{D}_1^2 \times \mathbf{D}_2^2$ is a *bridge from* σ_1 *to* σ_2 if

(a) the first two variables of δ are stable under σ_1;
(b) the last two variables of δ are stable under σ_2;
(c) $\mathrm{pr}_{1,2}(\delta) \supsetneq \sigma_1$ and $\mathrm{pr}_{3,4}(\delta) \supsetneq \sigma_2$;
(d) $(a_1, a_2, a_3, a_4) \in \delta$ implies $((a_1, a_2) \in \sigma_1 \iff (a_3, a_4) \in \sigma_2)$.

For a bridge δ put $\widetilde{\delta} = \{(x, y) : (x, x, y, y) \in \delta\}$. A bridge $\delta \subseteq D^4$ is *reflexive* if $(a, a, a, a) \in \delta$ for all $a \in D$.

Remark 3.28 (The motivating example)**.** On $\mathbb{Z}_4$, the relation $\delta = \{(a_1, a_2, a_3, a_4) : a_1 - a_2 = 2a_3 - 2a_4\}$ is a bridge from $0_{\mathbb{Z}_4}$ to congruence mod 2. Condition (d) is the content: $a_1 = a_2$ exactly when $a_3 \equiv a_4 \pmod 2$. Bridges are the formal shadow of a centralizer relation; see [11].

Definition 3.29 (Composition of bridges)**.** For bridges δ_1 from σ_0 to σ_1 and δ_2 from σ_1 to σ_2, set

$$(\delta_1 * \delta_2)(x_1, x_2, z_1, z_2) = \exists y_1 \exists y_2\ \delta_1(x_1, x_2, y_1, y_2) \wedge \delta_2(y_1, y_2, z_1, z_2).$$

Lemma 3.30 (Bridge composition; [12, Lem. 6.3])**.** *If $\sigma_0, \sigma_1, \sigma_2$ are irreducible then $\delta_1 * \delta_2$ is a bridge from σ_0 to σ_2, and $\widetilde{\delta_1 * \delta_2} = \widetilde{\delta_1} \circ \widetilde{\delta_2}$.*

Formalization note 3.31. Lemma 3.30 is imported, not proved, in [13]; it is [12, Lemma 6.3]. Both conclusions are used: the first in every chain of bridges, the second — the identity $\widetilde{\cdot}$ commutes with composition — in Lemma 6.22, where a bridge with $\widetilde{\delta}$ linked is powered up until $\widetilde{\delta} = A^2$. The irreducibility hypothesis on *all three* congruences is needed and is easy to drop by accident.

Definition 3.32 (Symmetric and pair-reflexive bridges)**.** Let δ be a bridge from σ to σ on $\mathbf{D}$. Call δ *symmetric* if $\delta(x_1, x_2, x_3, x_4) \iff \delta(x_3, x_4, x_1, x_2)$, and *pair-reflexive* if

$$(x_1, x_2) \in \underset{1,2}{\mathrm{pr}}(\delta) \implies (x_1, x_2, x_1, x_2) \in \delta.$$

Write $\delta^{-1}(x_1, x_2, x_3, x_4) = \delta(x_3, x_4, x_1, x_2)$.

Lemma 3.33 (Symmetrisation)**.** *Let δ be a reflexive bridge from σ to σ on $\mathbf{D}$ and put $\delta^\circ = \delta * \delta^{-1}$, that is*

$$\delta^\circ(x_1, x_2, x_3, x_4) = \exists x_5 \exists x_6\ \delta(x_1, x_2, x_5, x_6) \wedge \delta(x_3, x_4, x_5, x_6).$$

Then δ° is a reflexive, symmetric, pair-reflexive bridge from σ to σ with $\mathrm{pr}_{1,2}(\delta^\circ) = \mathrm{pr}_{3,4}(\delta^\circ) = \mathrm{pr}_{1,2}(\delta)$ and $\widetilde{\delta^\circ} = \widetilde{\delta} \circ \widetilde{\delta}^{-1} \supseteq \widetilde{\delta}$. Moreover any pair-reflexive bridge ε satisfies $\varepsilon \subseteq \varepsilon * \varepsilon$.

Proof. Symmetry is visible in the formula. For pair-reflexivity, if $(x_1, x_2) \in \mathrm{pr}_{1,2}(\delta^\circ)$ then $\delta(x_1, x_2, x_5, x_6)$ for some x_5, x_6, and taking the same witnesses twice gives $(x_1, x_2, x_1, x_2) \in \delta^\circ$; the same computation shows $\mathrm{pr}_{1,2}(\delta^\circ) = \mathrm{pr}_{1,2}(\delta)$, the inclusion $\subseteq$ being immediate, and symmetry then gives $\mathrm{pr}_{3,4}(\delta^\circ) = \mathrm{pr}_{1,2}(\delta^\circ)$. Clause (c) of Definition 3.27 follows, and clause (d) because $(x_1, x_2) \in \sigma \iff (x_5, x_6) \in \sigma \iff (x_3, x_4) \in \sigma$. Stability is inherited from δ, and reflexivity from $(a, a, a, a) \in \delta$. The identity $\widetilde{\delta^\circ} = \widetilde{\delta} \circ \widetilde{\delta^{-1}} = \widetilde{\delta} \circ \widetilde{\delta}^{-1}$ is Lemma 3.30; it contains $\widetilde{\delta}$ because $\widetilde{\delta}$ is reflexive, so $(x, y) \in \widetilde{\delta}$ gives $(x, y) \in \widetilde{\delta} \circ \widetilde{\delta}^{-1}$ by taking y as the intermediate point. Finally, if ε is pair-reflexive and $\varepsilon(x_1, x_2, x_3, x_4)$, then taking (x_1, x_2) itself as the intermediate pair — legitimate because $\varepsilon(x_1, x_2, x_1, x_2)$ — gives $(\varepsilon * \varepsilon)(x_1, x_2, x_3, x_4)$. $\square$

Formalization note 3.34 (Pair-reflexivity is the hypothesis three lemmas need)**.** [13] states pair-reflexivity only once, as a hypothesis of Lemma 7.41, but three further statements require it and do not say so; without it one of them is false. See Hazard 7.35. Lemma 3.33 is what makes this costless: every bridge these statements are ever applied to is a δ°.

Definition 3.35 (Perfect linear congruence)**.** A congruence σ on $\mathbf{A}$ is a *perfect linear congruence* if it is irreducible and there are a prime p and $\zeta \leq \mathbf{A} \times \mathbf{A} \times \mathbf{Z}_p$ with $\mathrm{pr}_{1,2}(\zeta) = \sigma^*$ and

$$(a_1, a_2, b) \in \zeta \implies \big((a_1, a_2) \in \sigma \iff b = 0\big).$$

Perfect linear congruences are the payoff of the whole bridge machinery: they let the passage from σ to σ^* be tracked by a single element of $\mathbf{Z}_p$, which is what turns a CSP instance into a system of linear equations in §8.3.

4 The six types of strong subuniverse

This section is Zhuk's §2.2. It defines six binary relations $C <_T^A B$ between subsets of a fixed algebra $\mathbf{A}$, one for each type T, and the transitive-closure relation $C \lhd\lhd\lhd^A B$. Everything in §§6–8 is expressed in this language, so the definitions repay care.

The informal reading is: $C <_T^A B$ says that C is a *strong* subset of B — strong enough that the CSP algorithm may reduce a variable's domain from B to C without losing all solutions — and T records *why* it is strong. Three of the reasons are absorption-theoretic (BA, C, S) and three are congruence-theoretic (D, L, PC).

4.1 Absorption

Definition 4.1 (Absorbing subuniverse)**.** A subuniverse B of $\mathbf{A}$ is *absorbing* if there is $t \in \mathrm{Clo}_k(\mathbf{A})$, for some k, such that for every $i \in [k]$

$$t(B, \dots, B, \underbrace{A}_{i}, B, \dots, B) \subseteq B.$$

We then say B *absorbs* $\mathbf{A}$ *with the term* t. If t may be chosen binary, B is a *binary absorbing* subuniverse (BA); if ternary, a *ternary absorbing* subuniverse.

Formalization note 4.2 (The tuple form)**.** Definition 4.1 must be formalized over *tuples constrained coordinatewise*, not as "for all $b_1, \dots, b_k \in B$ and $a \in A$, $t(b_1, \dots, b_{i-1}, a, b_{i+1}, \dots, b_k) \in B$". The two agree, but the second phrasing invites the variant in which b_i is quantified and then discarded, which is vacuously true when $B = \varnothing$ and at $k = 1$ where the intended content is not vacuous. This was an actual defect in the predecessor blueprint. State it as: for every $\bar{a} \in A^k$ such that $a_j \in B$ for all $j \neq i$, one has $t(\bar{a}) \in B$.

Definition 4.3 (Central subuniverse)**.** A subuniverse C of $\mathbf{A}$ is *central* (C) if it is absorbing and for every $a \in A \setminus C$,

$$(a, a) \notin \mathrm{Sg}_{\mathbf{A}}((\{a\} \times C) \cup (C \times \{a\})).$$

Lemma 4.4 ([14, Cor. 6.11.1])**.** *A central subuniverse is a ternary absorbing subuniverse.*

Remark 4.5 (Already formalized)**.** Lemma 4.4 is the one deep import of this theory that is already available in Lean: it is the main theorem of the predecessor project (`zhuk_main` in `ZhukLean`), proved there from the definitions via the Barto–Kazda relational description of absorption and the Zhuk–Kozik doubling trick. See §13.1.

Definition 4.6 (BA and centre free; S)**.** $\mathbf{A}$ is *BA and centre free* if it has no proper nonempty binary absorbing subuniverse and no proper nonempty central subuniverse.

For $\varnothing \neq C \lneq B \leq A$ write $C <_S^A B$ if there is a *nonempty* subuniverse $D \subseteq C$ of $\mathbf{B}$ that is simultaneously binary absorbing and central in $\mathbf{B}$. $\mathbf{A}$ is S-*free* if there is no $C <_S \mathbf{A}$.

Formalization note 4.7 (The witness must be required nonempty)**.** $\varnothing$ is vacuously binary absorbing and vacuously central (§2.1, item iii), so without the word "nonempty" the clause defining $<_S$ is universally true and the type S collapses. The global convention that subuniverses are nonempty repairs it, but the clause does not inherit that convention on its own, because D is introduced by an existential inside the clause rather than as a subalgebra of the ambient discourse.

Lemma 4.8 (The two readings of S-freeness agree)**.** $\mathbf{A}$ *is* S-*free if and only if no proper nonempty* $D \leq A$ *is simultaneously binary absorbing and central in* $\mathbf{A}$*.*

Proof. If such a D exists then $D <_{\mathrm{S}} \mathbf{A}$, taking $C = D$. Conversely if $C <_{\mathrm{S}} \mathbf{A}$ then by definition some nonempty $D \subseteq C \subsetneq A$ is binary absorbing and central in $\mathbf{A}$, and D is proper. □

Formalization note 4.9 (S is not the conjunction of BA and C)**.** $C <_{\mathrm{S}} B$ does not say that C is binary absorbing and central in $\mathbf{B}$; it says that C *contains* some D that is. The type S exists precisely so that it is closed under enlarging C, which BA and C are not. Note also the asymmetry with S-freeness, which *is* stated as a conjunction. Getting this backwards makes Lemma 6.4(ft) unprovable.

4.2 Linear and PC congruences

Definition 4.10 (Linear congruence)**.** A congruence σ on $\mathbf{A}$ is *linear* if

(a) σ is irreducible;
(b) σ^* is a congruence;
(c) there are a prime p and $S \leq (\sigma^*)^{[4]}$ such that for every block B of σ^* there is $m \geq 0$ with
$$(B/\sigma;\ S \cap (B/\sigma)^4) \ \cong\ (\mathbb{Z}_p^m;\ x_1 - x_2 = x_3 - x_4).$$

An irreducible congruence that is not linear is a *PC congruence*.

Here $(\sigma^*)^{[4]}$ denotes the set of 4-tuples whose entries lie in a single block of σ^*, which is meaningful because (b) makes σ^* a congruence. The source records the following in a sentence beginning "Notice that" and then uses it as a lemma (§2.1, item ix).

Lemma 4.11 (The witness of (c) is a bridge)**.** *Let σ be a linear congruence and S as in Definition 4.10(c). Then S is a reflexive pair-reflexive bridge from σ to σ with $\widetilde{S} = \mathrm{pr}_{1,2}(S) = \mathrm{pr}_{3,4}(S) = \sigma^*$.*

Proof. Work inside a block B of σ^* and identify B/σ with $\mathbb{Z}_p^m$ so that $S \cap (B/\sigma)^4$ becomes $x_1 - x_2 = x_3 - x_4$; tuples of S meet no other block. Each variable is stable under σ, since the defining equation is an equation between σ-classes. Clause (d) of Definition 3.27 holds because $x_1 - x_2 = 0$ if and only if $x_3 - x_4 = 0$, which says $(x_1, x_2) \in \sigma \iff (x_3, x_4) \in \sigma$. For the projections, any (x_1, x_2) inside a common block satisfies $(x_1, x_2, x_1, x_2) \in S$, so $\mathrm{pr}_{1,2}(S) = \sigma^*$ — which is at once clause (c), since $\sigma^* \supsetneq \sigma$, and pair-reflexivity; symmetry of the defining equation in the two pairs gives $\mathrm{pr}_{3,4}(S) = \sigma^*$. Finally $(x, x, y, y) \in S$ reads $0 = 0$, so $\widetilde{S}$ is the whole of σ^*, and $(a, a, a, a) \in S$ gives reflexivity. □

This is the useful reformulation:

Lemma 4.12 ([13, Lem. 4.12])**.** *For an irreducible congruence σ the following are equivalent:*

(a) *σ is linear;*
(b) *there is a bridge δ from σ to σ with $\widetilde{\delta} \supsetneq \sigma$.*

Formalization note 4.13 (Definition (c) versus criterion (b))**.** Definition 4.10(c) is a per-block isomorphism statement with an existential m *depending on the block*, and it quantifies over blocks of σ^*, not of σ. Lemma 4.12 replaces all of that by one bridge. A formalization should take (b) as the *definition* and prove (c) as a structure theorem, reversing the paper's order: (b) is a Σ_1 statement over finite data and is what every consumer actually uses, while (c) drags in a quotient, a choice of prime, and a family of isomorphisms.

The dichotomy "linear or PC" is then *by definition* exhaustive on irreducible congruences, which is how the paper uses it. Note that PC-ness is a negative condition; every lemma about PC congruences is proved by contradiction against Lemma 4.12(b).

Lemma 4.14 (No bridge crosses the types)**.** *If σ_1 is linear, σ_2 is irreducible, and there is a bridge from σ_1 to σ_2, then σ_2 is linear.*

Lemma 4.15 (Bridges from PC congruences are trivial)**.** *Let σ be a PC congruence on A and δ a reflexive bridge from σ to σ with $\mathrm{pr}_{1,2}(\delta) = \mathrm{pr}_{3,4}(\delta) = \sigma^*$. Then*

$$\delta(x_1, x_2, x_3, x_4) = \sigma(x_1, x_3) \wedge \sigma(x_2, x_4) \quad \text{or} \quad \delta(x_1, x_2, x_3, x_4) = \sigma(x_1, x_4) \wedge \sigma(x_2, x_3).$$

Lemma 4.16 (Bridges into PC congruences)**.** *Let δ be a bridge from a PC congruence σ_1 on $\mathbf{A}_1$ to an irreducible congruence σ_2 on $\mathbf{A}_2$, with $\mathrm{pr}_{1,2}(\delta) = \sigma_1^*$ and $\mathrm{pr}_{3,4}(\delta) = \sigma_2^*$. Then*

(a) *σ_2 is a PC congruence;*
(b) *$\mathbf{A}_1/\sigma_1 \cong \mathbf{A}_2/\sigma_2$;*
(c) *$\{(a/\sigma_1, b/\sigma_2) : (a, b) \in \widetilde{\delta}\}$ is bijective;*
(d) *$\delta(x_1, x_2, x_3, x_4) = \widetilde{\delta}(x_1, x_3) \wedge \widetilde{\delta}(x_2, x_4)$ or $\delta(x_1, x_2, x_3, x_4) = \widetilde{\delta}(x_1, x_4) \wedge \widetilde{\delta}(x_2, x_3)$.*

Two results connect the new vocabulary to the linear and PC subuniverses of the original proof.

Lemma 4.17. *If σ is a linear congruence on $\mathbf{A} \in \mathcal{V}_n$ with $\sigma^* = A^2$, then $\mathbf{A}/\sigma \cong \mathbf{Z}_p$ for a prime p.*

Lemma 4.18. *If σ is a PC congruence on $\mathbf{A}$ with $\sigma^* = A^2$, then $\mathbf{A}/\sigma$ is polynomially complete.*

Formalization note 4.19 (What the two proofs on top actually need)**.** Neither statement is proved in [13] as it stands.

For Lemma 4.17 the source produces the witness ζ of Definition 3.35, sets $\xi(x, z) = \zeta(x, a, z)$ for a fixed a, and says "then ξ is a bijective relation giving an isomorphism" — which is the entire content of the lemma. We prove it instead from Lemma 7.39(c): with $\sigma^* = A^2$ that lemma has a single block, so $\mathbf{A}/\sigma \cong \mathbb{Z}_p^m$ as a group; $m = 0$ would give $\sigma = A^2$, excluded because irreducible congruences are proper; and $m \geq 2$ is excluded by irreducibility, because the m coordinate congruences $\theta_i = \{(x, y) : x_i = y_i\}$ are subuniverses of $(\mathbf{A}/\sigma)^2$ — linear subspaces are preserved by any coordinatewise affine map — each properly above $0_{\mathbf{A}/\sigma}$, with $\bigcap_i \theta_i = 0_{\mathbf{A}/\sigma}$. So $m = 1$. That the resulting bijection is an isomorphism of *algebras*, rather than of groups, is Proposition 2.7. This route also avoids the import [12, Cor. 8.17.1] that the source's route needs.

For Lemma 4.18 two things are used and not stated: that an idempotent algebra is polynomially complete exactly when every *reflexive* subuniverse of a power is a conjunction of equalities — those being precisely the invariants of the clone generated by the basic operations together with all constants — and that $\mathbf{A}/\sigma$ is BA and centre free, without which the step "$\{a_i\} <_{\mathrm{PC}} \mathbf{A}/\sigma$" is not licensed by Definition 4.21.

Formalization note 4.20 (Polynomial completeness)**.** An algebra is *polynomially complete* if the clone generated by its basic operations together with all constants is the clone of *all* operations on its universe. Lemma 4.18 is where the name "PC" comes from, and it is the only place in the development where polynomial completeness is used as more than a label. Formalizing it needs the Istinger–Kaiser characterization [8]; formalizing everything else needs only the *name*. A staged development should therefore treat "PC congruence" as "irreducible and not linear" throughout, and prove Lemma 4.18 last or not at all — it is used only to connect with [12], not by any argument here.

4.3 The types

Definition 4.21 (The six types)**.** Let $\varnothing \neq C \lneq B \leq A$. We write:

- $C <^A_{\mathrm{BA}} B$ if C is a binary absorbing subuniverse of $\mathbf{B}$;
- $C <^A_{\mathrm{C}} B$ if C is a central subuniverse of $\mathbf{B}$;
- $C <^A_{\mathrm{D}} B$ if there is an irreducible congruence σ on $\mathbf{A}$ with
  (i) $B^2 \subseteq \sigma^*$,
  (ii) $C = B \cap E$ for some block E of σ,
  (iii) $\mathbf{B}/\sigma$ is BA and centre free;
- $C <^A_{\mathrm{L}} B$ if $C <^A_{\mathrm{D}} B$ with the witnessing σ linear;
- $C <^A_{\mathrm{PC}} B$ if $C <^A_{\mathrm{D}} B$ with the witnessing σ a PC congruence;
- $C <^A_{\mathrm{S}} B$ as in Definition 4.6.

When the witnessing congruence matters we write $C <^A_{T(\sigma)} B$; for $T \in \{\mathrm{BA}, \mathrm{C}, \mathrm{S}\}$, where no congruence is involved, σ is taken to be A^2.

Formalization note 4.22 (The superscript A is not decoration)**.** For $T \in \{\mathrm{D}, \mathrm{L}, \mathrm{PC}\}$ the witnessing congruence lives on $\mathbf{A}$, not on $\mathbf{B}$, and condition (i) relates the two: $B^2 \subseteq \sigma^*$ says B sits inside a single block of σ^*. This is why the relation carries the ambient algebra as a superscript. For $T \in \{\mathrm{BA}, \mathrm{C}, \mathrm{S}\}$ the superscript is genuinely irrelevant and Zhuk drops it. A formalization should not overload: define $<^A_T$ with A always present, and prove the independence for the three absorption types as a lemma.

Note also that $\mathbf{B}/\sigma$ in (iii) means B/σ with the induced algebra structure, which requires B to be a subuniverse and σ restricted to B; since $B^2 \subseteq \sigma^*$ and not necessarily $B^2 \subseteq \sigma$, the quotient B/σ is in general *not* a single point, and its being BA and centre free is a real condition.

Definition 4.23 (Multi-types)**.** For $T \in \{\mathrm{L}, \mathrm{PC}, \mathrm{D}\}$ write $C <^A_{\mathcal{M}T} B$ if $C \neq \varnothing$ and $C = C_1 \cap \cdots \cap C_t$ with $C_i <^A_T B$ for every $i \in [t]$.

Definition 4.24 (The strong-reduction relation)**.** Write $C \lhd\lhd\lhd^A B$ if there are $B_0, \ldots, B_m \subseteq B$ and $T_1, \ldots, T_m \in \{\mathrm{BA}, \mathrm{C}, \mathrm{S}, \mathrm{D}\}$ with

$$C = B_m <^A_{T_m} B_{m-1} <^A_{T_{m-1}} \cdots <^A_{T_1} B_0 = B.$$

Here $m = 0$ is allowed, so $\lhd\lhd\lhd^A$ is reflexive. A congruence *comes from* $C \lhd\lhd\lhd^A B$ if it is one of the witnessing congruences in such a chain. We write $B \lhd\lhd\lhd A$ for $B \lhd\lhd\lhd^A A$, and $C \leq^A_{T(\sigma)} B$ for "$C = B$ or $C <^A_{T(\sigma)} B$".

Formalization note 4.25 ($\lhd\lhd\lhd$ is data, not a proposition)**.** It is tempting to read $C \lhd\lhd\lhd^A B$ as the reflexive transitive closure of $\bigcup_T <^A_T$ — in Lean, `Relation.ReflTransGen` — and to forget the chain. That is wrong, and the error is not cosmetic: Definition 4.24 also defines "the congruences coming from $C \lhd\lhd\lhd^A B$", which is a function of the chain and not of the pair. Two places consume it, and in both the *choice* of chain is what makes the argument work.

(a) In the "moreover" clause of Lemma 7.30(d) the conclusion is $\delta \supseteq \delta_1 \cap \cdots \cap \delta_s$ where the δ_i are the congruences of the chains for $B \lhd\lhd\lhd A$ and $C \lhd\lhd\lhd A$ *fixed at the top of that proof*; the proof establishes exactly $\sigma \cap \omega \subseteq \delta$ for those two intersections.
(b) In the type-D case of Lemma 7.45 the clause is applied to C_2 rather than to B_2, and its output is the intersection of ω with σ_2 — which is correct only because the chain witnessing $C_2 \lhd\lhd\lhd A$ is chosen to be the chain witnessing $B_2 \lhd\lhd\lhd A$ extended by the single step $C_2 <_{\mathrm{D}(\sigma_2)} B_2$. A different chain gives a different family and the step fails.

So $\lhd\lhd\lhd^A$ should be an inductive family carrying, at each step, the intermediate set, its type, and its witnessing congruence; extension by one step is an operation on that data.

The `ReflTransGen` version is its propositional truncation and may be used only where no congruence is named, which is most but not all of the development. What *is* true, and is the paper's central simplification over [12], is that the intermediate *types* may be forgotten: L and PC are absent from the list in Definition 4.24 because D subsumes them.

Convention 4.26 (Dotted variants)**.** $C \mathrel{\dot{<}}_T^A B$, $C \mathrel{\dot{\leq}}_T^A B$ and $C \mathrel{\dot{\lhd}\dot{\lhd}\dot{\lhd}}^A B$ mean that the undotted relation holds *or* $C = \varnothing$. See Hazard 2.5.

5 Instances

This section is Zhuk's §3.1. It is entirely combinatorial: no algebra is used beyond the fact that each domain carries a w and each constraint relation is a subuniverse of a product of domains. It is also where a formalization must make the most decisions, because the paper's instances are informal syntactic objects manipulated by operations — weakening, covering, renaming — that are described in words.

5.1 Instances and solutions

Definition 5.1 (Instance)**.** Fix a finite set V of variables and, for each $x \in V$, an algebra $\mathbf{D}_x \in \mathcal{V}_n$. A *constraint* is a pair $C = (\bar{x}, R)$ where $\bar{x} = (x_1, \ldots, x_m)$ is a tuple of variables and $R \leq \mathbf{D}_{x_1} \times \cdots \times \mathbf{D}_{x_m}$. An *instance* $\mathcal{I}$ is a finite list of constraints; we write $C \in \mathcal{I}$, and $\mathrm{Var}(C)$, $\mathrm{Var}(\mathcal{I})$ for the variables occurring. A *subinstance* is a sublist.

A *solution* is a map s with $s(x) \in D_x$ for all x such that $(s(x_1), \ldots, s(x_m)) \in R$ for every constraint. The solution set of $\mathcal{I}$ is *subdirect* if for every x and every $a \in D_x$ there is a solution with $s(x) = a$.

Formalization note 5.2 (Lists, not sets; tuples, not sets of variables)**.** Two representation choices are forced by the way the paper argues, and both differ from the naive one.

Constraints form a list, not a set. Proofs delete "a constraint $C \in \mathcal{I}$" and reason about $\mathcal{I} \setminus \{C\}$, and the same relation may occur twice on different variable tuples — and, in the expanded coverings of §5.5, the same relation on the *same* tuple can legitimately occur twice. Multiset or list semantics is required; `List` with membership by index is the safe choice.

A constraint's scope is a tuple, not a set. Repeated variables in a scope are allowed and occur — $\zeta(x, x', z)$ in §8 arises by splitting one variable into two, and $R(x, x, \ldots, x)$ appears explicitly in Definition 5.14. So $\mathrm{Var}(C)$ is the image of the scope, and $|\mathrm{Var}(C)|$ may be smaller than the arity.

Definition 5.3 (Relations defined by an instance)**.** For $x_1, \ldots, x_m \in \mathrm{Var}(\mathcal{I})$, $\mathcal{I}(x_1, \ldots, x_m)$ is the set of $(a_1, \ldots, a_m)$ such that $\mathcal{I}$ has a solution with $s(x_i) = a_i$ for all i. It is a subuniverse of $\mathbf{D}_{x_1} \times \cdots \times \mathbf{D}_{x_m}$, being defined by a primitive positive formula over the constraint relations.

5.2 Reductions

Definition 5.4 (Reduction)**.** A *reduction* for $\mathcal{I}$ is a map $D^{(\top)}$ assigning to each $x \in \mathrm{Var}(\mathcal{I})$ a subuniverse $D_x^{(\top)} \subseteq D_x$, possibly empty. It is *nonempty* if every $D_x^{(\top)} \neq \varnothing$. The instance $\mathcal{I}^{(\top)}$ is $\mathcal{I}$ with each domain replaced by $D_x^{(\top)}$ and each constraint relation intersected with the corresponding box. For reductions we write $D^{(\perp)} \lhd\lhd\lhd\, D^{(\top)}$ and $D^{(\perp)} \leq_T D^{(\top)}$ when the corresponding relation holds at every variable.

Formalization note 5.5 (Empty reductions)**.** Reductions are allowed to be empty at some or all variables — the maximal 1-consistent reduction below $D^{(B,\top)}$ in the proof of Theorem 8.1 is constructed exactly so that it may come out empty, and the case split on whether it does is the branch point of the whole argument. But $D^{(\perp)} \lhd\lhd\lhd\, D^{(\top)}$ requires nonemptiness at every variable (Convention 2.4). Both notions are needed and they must be different objects.

5.3 Consistency

Definition 5.6 (Paths)**.** A *path* in $\mathcal{I}$ is an alternating sequence $z_1 - C_1 - z_2 - \cdots - C_{l-1} - z_l$ with $z_i, z_{i+1} \in \mathrm{Var}(C_i)$. It *connects b and c* if there are $a_i \in D_{z_i}$ with $a_1 = b$, $a_l = c$, and $(a_i, a_{i+1}) \in \mathrm{pr}_{z_i, z_{i+1}}(C_i)$ for every i. $\mathcal{I}$ is a *tree-instance* if it has no path with $l \geq 3$, $z_1 = z_l$ and $C_1, \ldots, C_{l-1}$ pairwise distinct.

Definition 5.7 (Consistency)**.** $\mathcal{I}$ is 1-*consistent* if $\mathrm{pr}_z(C) = D_z$ for every constraint C and every $z \in \mathrm{Var}(C)$. A reduction $D^{(\top)}$ is 1-consistent for $\mathcal{I}$ if $\mathcal{I}^{(\top)}$ is 1-consistent. $\mathcal{I}$ is *cycle-consistent* if it is 1-consistent and every path from z to z connects a to a, for every variable z and every $a \in D_z$.

Definition 5.8 (Linked, fragmented, irreducible)**.** $\mathcal{I}$ is *linked* if for every $z \in \mathrm{Var}(\mathcal{I})$ and every $a, b \in D_z$ some path from z to z connects a and b. $\mathcal{I}$ is *fragmented* if $\mathrm{Var}(\mathcal{I})$ splits into two disjoint nonempty sets X_1, X_2 with $\mathrm{Var}(C) \subseteq X_1$ or $\mathrm{Var}(C) \subseteq X_2$ for every C. $\mathcal{I}$ is *irreducible* if there is no instance $\mathcal{I}'$ with

(i) $\mathrm{Var}(\mathcal{I}') \subseteq \mathrm{Var}(\mathcal{I})$;
(ii) every constraint of $\mathcal{I}'$ is a projection of a constraint of $\mathcal{I}$ onto some of its variables;
(iii) $\mathcal{I}'$ is not fragmented;
(iv) $\mathcal{I}'$ is not linked;
(v) the solution set of $\mathcal{I}'$ is not subdirect.

Formalization note 5.9 (Irreducibility is a Π_1 condition over a large family)**.** Conditions (i)–(v) quantify over *all* instances built from projections of constraints of $\mathcal{I}$ — a finite but exponentially large family (any multiset of projections of any constraints onto any subsets of their variables). It is finite, so the predicate is decidable, but it is not something to unfold. Every use in §8 goes through Lemma 8.12 and *never* through the definition. Formalize it once, prove the two consequences, and never unfold it again.

The typo in the source at this point ("$\mathrm{Var}(C) \subseteq X_1$ or $\mathrm{Var}(C) \subseteq X_1$") is corrected above to X_2.

5.4 Weakening and cruciality

Definition 5.10 (Weakening)**.** A constraint $R_1(y_1, \ldots, y_t)$ is *weaker or equivalent to* $R_2(z_1, \ldots, z_s)$ if $\{y_i\} \subseteq \{z_j\}$ and $R_2(\bar{z})$ implies $R_1(\bar{y})$; it is *weaker* if in addition $R_1(\bar{y})$ does not imply $R_2(\bar{z})$. *Weakening* a constraint C in $\mathcal{I}$ means replacing C by all constraints weaker than it. An instance $\mathcal{I}'$ is a *weakening* of $\mathcal{I}$ if $\mathrm{Var}(\mathcal{I}') \subseteq \mathrm{Var}(\mathcal{I})$ and every constraint of $\mathcal{I}'$ is weaker or equivalent to a constraint of $\mathcal{I}$.

Formalization note 5.11 ("Replace by all weaker constraints")**.** Read literally this is an infinite operation: there are $2^{|D|^t}$ relations weaker than a given one on a given scope, and one must also range over all sub-scopes. It is finite for a fixed instance, but the resulting instance is astronomically large, and no algorithm performs this operation — it appears only inside proofs, as a device for reaching a crucial instance (Remark 5.13).

Two consequences for a formalization. First, weakening must be defined as a *relation between instances* ("$\mathcal{I}'$ is a weakening of $\mathcal{I}$"), not as a function producing one. Second, the descending chain argument in Remark 5.13 needs a well-founded measure; the natural one is the total number of tuples in all constraint relations, which strictly decreases when a constraint is properly weakened, together with the fact that only finitely many scopes are available. State that measure explicitly — "we cannot weaken forever" occurs three times in §8 and is the kind of step that silently fails to be well-founded if the scope is allowed to grow.

Definition 5.12 (Crucial)**.** Let $D^{(\top)}$ be a reduction for $\mathcal{I}$. A variable y_i of a constraint $R(y_1, \ldots, y_t)$ is *dummy* if R does not depend on its i-th coordinate. A constraint $C \in \mathcal{I}$ is *crucial in* $D^{(\top)}$ if it has no dummy variables, $\mathcal{I}^{(\top)}$ has no solution, and weakening C in $\mathcal{I}$ yields an instance with a solution in $D^{(\top)}$. The instance $\mathcal{I}$ is *crucial in* $D^{(\top)}$ if it has at least one constraint and all of its constraints are crucial in $D^{(\top)}$.

Remark 5.13 (Reaching a crucial instance)**.** If $\mathcal{I}^{(\top)}$ has no solution, then iteratively replacing each constraint by all weaker constraints without dummy variables terminates in an instance crucial in $D^{(\top)}$. Every relation so introduced is still a subuniverse of the corresponding product of domains.

5.5 Coverings

Definition 5.14 (Expanded covering)**.** $\mathcal{I}'$ is an *expanded covering* of $\mathcal{I}$, written $\mathcal{I}' \in \mathrm{Exp}(\mathcal{I})$, if there is $S\colon \mathrm{Var}(\mathcal{I}') \to \mathrm{Var}(\mathcal{I})$ with

(i) $S(x) = x$ for $x \in \mathrm{Var}(\mathcal{I}) \cap \mathrm{Var}(\mathcal{I}')$;
(ii) $D_x = D_{S(x)}$ for every $x \in \mathrm{Var}(\mathcal{I}')$;
(iii) for every constraint $R(x_1, \ldots, x_m)$ of $\mathcal{I}'$, either $S(x_1), \ldots, S(x_m)$ are pairwise distinct and $R(S(x_1), \ldots, S(x_m))$ is weaker or equivalent to a constraint of $\mathcal{I}$, or $S(x_1) = \cdots = S(x_m)$ and $\{(a, \ldots, a) : a \in D_{x_1}\} \subseteq R$.

It is a *covering* if moreover $R(S(x_1), \ldots, S(x_m)) \in \mathcal{I}$ for every constraint; a *tree-covering* if it is a covering and a tree-instance. $S(x)$ is the *parent* of x, and x a *child*.

Proposition 5.15 (Basic properties)**.** *Let $\mathcal{I}'$ be an expanded covering of $\mathcal{I}$.*

(a) *Replacing every x by $S(x)$ and deleting the constraints $R(x, \ldots, x)$ yields a weakening of $\mathcal{I}$.*
(b) *A weakening is an expanded covering with $S = \mathrm{id}$.*
(c) *Every solution of $\mathcal{I}$ lifts to a solution of $\mathcal{I}'$.*
(d) *If $\mathcal{I}$ is 1-consistent and $\mathcal{I}'$ is a tree-covering, the solution set of $\mathcal{I}'$ is subdirect.*
(e) *The union of two expanded coverings is an expanded covering.*
(f) *An expanded covering of an expanded covering is an expanded covering.*
(g) *An expanded covering of a cycle-consistent irreducible instance is cycle-consistent and irreducible.*
(h) *Every reduction of $\mathcal{I}$ extends to $\mathcal{I}'$, and a 1-consistent one extends to a 1-consistent one.*

Formalization note 5.16 (Variable renaming is the hidden cost)**.** Expanded coverings are where a formalization pays. The paper writes “we rename the variables so that the only common variable of Υ_x and $\mathcal{J}$ is x”, and forms conjunctions $\bigwedge_{\mathcal{J}\in\Omega} \bot(\mathcal{J})$ of instances whose variable sets must first be made disjoint. Doing this with a concrete variable type means a fresh-name supply and a renaming operation, with a lemma that renaming preserves each of 1-consistency, cycle-consistency, cruciality, and being a covering — eight lemmas that the paper does not state.

The cheaper design is to make the variable type a parameter and build instances by *disjoint union* of variable types: Υ_x and $\mathcal{J}$ are placed over $V_1 \oplus V_2$ quotiented by identifying the single shared variable. Renaming then disappears into the coproduct, and the eight lemmas become transport along an equivalence. This is the single most important representation decision in §5, and it should be made before anything downstream is written.

5.6 Induced congruences and connectedness

Definition 5.17 (Con_1)**.** For R of arity m and $i \in [m]$, $\mathrm{Con}_1(R, i)$ is the binary relation

$$\{(y, y') \ : \ \exists\, \bar{x} \ \ R(\bar{x}[i \mapsto y]) \wedge R(\bar{x}[i \mapsto y'])\}.$$

For a constraint $C = R(x_1, \ldots, x_m)$ we write $\mathrm{Con}_1(C, x_i)$ for $\mathrm{Con}_1(R, i)$. For an instance, $\mathrm{Con}(\mathcal{I}, x) = \{\mathrm{Con}_1(C, x) : C \in \mathcal{I}\}$ and $\mathrm{Con}(\mathcal{I}) = \bigcup_x \mathrm{Con}(\mathcal{I}, x)$.

Lemma 5.18. *The i-th variable of R is rectangular if and only if R is stable under* $\mathrm{Con}_1(R, i)$*. If R is subdirect and its i-th variable is rectangular, then* $\mathrm{Con}_1(R, i)$ *is a congruence.*

Formalization note 5.19. $\mathrm{Con}_1(R, i)$ is always reflexive and symmetric but is *not* transitive in general, hence not always a congruence; Lemma 5.18 is what makes it one, and its hypotheses (subdirect *and* rectangular at i) are both needed. The notation Con_1 invites the reader to assume congruence-hood; a formalization should give the raw relation a neutral name and reserve Con_1 for the case where the lemma applies. Note also $\mathrm{Con}_1(C, x_i)$ is ill-defined if x occurs twice in the scope: it depends on the *position* i, not the variable.

Definition 5.20 (Types of relations and instances)**.** R is *of PC type* (resp. *linear type*) if R is rectangular and every $\mathrm{Con}_1(R, i)$ is a PC (resp. linear) congruence. An instance is of PC or linear type if all its constraints are.

Definition 5.21 (Adjacency, connectedness)**.** Two congruences σ_1, σ_2 on $\mathbf{D}_x$ are *adjacent* if there is a reflexive bridge from σ_1 to σ_2. Two rectangular constraints C_1, C_2 are *adjacent in a common variable x* if $\mathrm{Con}_1(C_1, x)$ and $\mathrm{Con}_1(C_2, x)$ are adjacent. An instance $\mathcal{I}$ is *connected* if all its constraints are rectangular, all congruences in $\mathrm{Con}(\mathcal{I})$ are irreducible, and the graph on the constraints with edges the adjacent pairs is connected.

Note that every proper congruence is adjacent to itself, via $\delta(x_1, x_2, x_3, x_4) = \sigma(x_1, x_3) \wedge \sigma(x_2, x_4)$.

6 Properties of strong subuniverses

This section is Zhuk's §2.3. It is the *interface* between the algebraic theory and the CSP argument: §8 uses the results below and nothing else from §§4. Proofs are deferred to §5 of [13], which is not reproduced here; §13 budgets it.

Its shape is worth naming. Four of the five groups say that the relation $\lhd\lhd\lhd$ and the types T *propagate*: forward and backward along surjective homomorphisms, down to quotients, across products of a congruence, and through subdirect relations. The fifth — Theorem 7.53 — is the one genuinely hard statement, and it is what converts a failure of consistency into a bridge. Everything the CSP proof does with bridges traces back to it.

Convention 6.1. Where a type is not specified, T ranges over $\{\text{BA}, \text{C}, \text{S}, \text{PC}, \text{L}, \text{D}\}$. Where a multi-type $\mathcal{M}T$ appears, T ranges over $\{\text{PC}, \text{L}, \text{D}\}$.

6.1 Ubiquity

Lemma 6.2 (Ubiquity)**.** *If $B \lhd\lhd\lhd A$ and $|B| > 1$, then $C <_T^A B$ for some C and some $T \in \{\text{BA}, \text{C}, \text{L}, \text{PC}\}$.*

Lemma 6.2 is the formal content of Informal Claim 8.1: every domain of size at least 2 that has been reached by a chain of strong reductions admits another strong reduction. It is what guarantees the outer loop of the algorithm makes progress, and it is the reason the algorithm terminates with singleton domains or a linear instance.

Formalization note 6.3. The hypothesis is $B \lhd\lhd\lhd A$, not $B \leq A$. A subalgebra reached by no chain of strong reductions need admit nothing. In the algorithm this hypothesis is maintained as an invariant of the reduction loop; in a formalization it should be carried in the state, not re-derived.

6.2 Propagation

Lemma 6.4 (Propagation along a homomorphism)**.** Let $f\colon \mathbf{A} \to \mathbf{A}'$ be a surjective homomorphism. Then

(f) $C \lhd\lhd\lhd^A B \Rightarrow f(C) \lhd\lhd\lhd f(B)$;
(b) $C' \lhd\lhd\lhd^{A'} B' \Rightarrow f^{-1}(C') \lhd\lhd\lhd f^{-1}(B')$;
(ft) $C <_{T(\sigma)}^A B \lhd\lhd\lhd A \Rightarrow (f(C) = f(B)$ or $f(C) <_{\text{S}} f(B)$ or $f(C) <_T^{A'} f(B))$;
(bt) $C' <_{T(\sigma)}^{A'} B' \Rightarrow f^{-1}(C') <_{T(f^{-1}(\sigma))}^A f^{-1}(B')$;
(fs) $T \in \{\text{BA}, \text{C}, \text{S}\}$ and $C <_T B \Rightarrow f(C) \leq_T f(B)$;
(fm) $C \leq_{\mathcal{M}T}^A B \lhd\lhd\lhd A$ and $f(B)$ S-free $\Rightarrow f(C) \leq_{\mathcal{M}T}^{A'} f(B)$;
(bm) $C' \leq_{\mathcal{M}T}^{A'} B' \lhd\lhd\lhd A' \Rightarrow f^{-1}(C') \leq_{\mathcal{M}T}^A f^{-1}(B')$.

Formalization note 6.5 (The three-way disjunction in (ft))**.** Clause (ft) is the characteristic shape of this theory and the reason it is usable. Pushing a strong subset forward along a homomorphism can

- collapse it ($f(C) = f(B)$: the reduction becomes vacuous),
- degrade it ($f(C) <_{\text{S}} f(B)$: still strong, but the type is lost),
- or preserve it ($f(C) <_T^{A'} f(B)$: same type).

The type S exists to be the value of the middle case. Every downstream proof that pushes a reduction through a quotient does a three-way case split here, and the middle case is the one that is easy to forget; in [12] it is the source of the "go forward and backward from global to local" induction that this paper eliminates.

Note the asymmetry: the backward clauses (b), (bt), (bm) are clean implications with no disjunction. Preimages behave; images do not.

Corollary 6.6 (Propagation to a quotient)**.** *Let δ be a congruence on* $\mathbf{A}$*. Then* (f) $C \lhd\lhd\lhd^{A} B \Rightarrow C/\delta \lhd\lhd\lhd^{A/\delta} B/\delta$*;* (t) $C <^{A}_{T(\sigma)} B \lhd\lhd\lhd A \Rightarrow (C/\delta = B/\delta$ *or* $C/\delta <_{\mathrm{S}} B/\delta$ *or* $C/\delta <^{A/\delta}_{T} B/\delta)$*;* (s) *for* $T \in \{\mathrm{BA}, \mathrm{C}, \mathrm{S}\}$*,* $C <_T B \Rightarrow C/\delta \leq_T B/\delta$*;* (m) $C \leq^{A}_{\mathcal{M}T} B \lhd\lhd\lhd A$ *and* B/δ S*-free* $\Rightarrow C/\delta \leq^{A/\delta}_{\mathcal{M}T} B/\delta$*.*

Corollary 6.7 (Reflection from a quotient)**.** *Let δ be a congruence on* $\mathbf{A}$ *and* $B, C \leq \mathbf{A}$*. Then* $C/\delta \lhd\lhd\lhd^{A/\delta} B/\delta \iff C \circ \delta \lhd\lhd\lhd^{A} B \circ \delta$*, and* $C/\delta <^{A/\delta}_{T} B/\delta \iff C \circ \delta <^{A}_{T} B \circ \delta$*.*

Corollary 6.8 (Multiplying by a congruence)**.** *Let δ be a congruence on* $\mathbf{A}$*. Then* (f) $C \lhd\lhd\lhd^{A} B \Rightarrow C \circ \delta \lhd\lhd\lhd^{A} B \circ \delta$*;* (t) $C <^{A}_{T(\sigma)} B \lhd\lhd\lhd^{A} A \Rightarrow (C \circ \delta = B \circ \delta$ *or* $C \circ \delta <^{A}_{\mathrm{S}} B \circ \delta$ *or* $C \circ \delta <^{A}_{T} B \circ \delta)$*;* (e) *if* $\delta \subseteq \sigma$ *and* $C <^{A}_{T(\sigma)} B \lhd\lhd\lhd^{A} A$ *then* $C \circ \delta <^{A}_{T} B \circ \delta$*.*

Corollary 6.9 (Propagation to relations)**.** Let $R \leq_{\mathrm{sd}} \mathbf{A}_1 \times \cdots \times \mathbf{A}_m$ and $B_i \lhd\lhd\lhd A_i$. Then

(r) $R \cap (B_1 \times \cdots \times B_m) \dot{\lhd}\dot{\lhd}\dot{\lhd} R$;
(r1) $\mathrm{pr}_1(R \cap (B_1 \times \cdots \times B_m)) \dot{\lhd}\dot{\lhd}\dot{\lhd} A_1$;
(b) if $C_i \lhd\lhd\lhd^{A_i} B_i$ for all i then $R \cap (C_1 \times \cdots \times C_m) \dot{\lhd}\dot{\lhd}\dot{\lhd}^{R} R \cap (B_1 \times \cdots \times B_m)$;
(b1) under the same hypothesis, $\mathrm{pr}_1(R \cap \prod C_i) \dot{\lhd}\dot{\lhd}\dot{\lhd}^{A_1} \mathrm{pr}_1(R \cap \prod B_i)$;
(m) if $C_i \leq^{A_i}_{\mathcal{M}T} B_i$ for all i then $R \cap \prod C_i \dot{\leq}^{R}_{\mathcal{M}T} R \cap \prod B_i$;
(m1) if in addition $\mathrm{pr}_1(R \cap \prod B_i)$ is S-free, then $\mathrm{pr}_1(R \cap \prod C_i) \dot{\leq}^{A_1}_{\mathcal{M}T} \mathrm{pr}_1(R \cap \prod B_i)$.

For the two absorption types the conclusion is stronger — no chain, no dot on the type, and no subdirectness hypothesis:

Lemma 6.10 ([14, Cor. 6.1.2, 6.9.2])**.** *Let* $R \leq \mathbf{A}_1 \times \cdots \times \mathbf{A}_m$ *with* $\mathrm{pr}_1(R) = A_1$*, and let* $C_i \leq_T A_i$ *for every i, where* $T \in \{\mathrm{BA}, \mathrm{C}\}$ *— and, when* $T = \mathrm{BA}$*, with a* single *binary term t witnessing all m absorptions. Then* $\mathrm{pr}_1(R \cap (C_1 \times \cdots \times C_m)) \dot{\leq}_T A_1$*, again with the same t when* $T = \mathrm{BA}$*.*

Formalization note 6.11 (The source's restatement is false; this is the repair)**.** Lemma 6.10 is the workhorse of case (2) of Theorem 8.1, and [13, Lemma 19] restates it from [14] *dropping two hypotheses*: the subdirectness $\mathrm{pr}_1(R) = A_1$, and — in the BA case — that the C_i absorb with a common term. Both are present in the cited source ([14, Cor. 6.1.2] reads "Suppose $R \leq A_1 \times \cdots \times A_n$, $\mathrm{pr}_1(R) = A_1$, $B_i \leq_{\mathrm{BA}(t)} A_i$ for every i"), and without the first the statement is false.

Counterexample to the printed form. Take $R = \{(0, 0)\} \leq \mathbf{Z}_p \times \mathbf{Z}_p$ — a subuniverse, by idempotence — and $C_i = A_i$, which is allowed because $\leq_T$ admits equality. Then $R \cap (C_1 \times C_2) = R$ and $\mathrm{pr}_1(R) = \{0\}$, which is neither empty nor all of $\mathbf{Z}_p$, so the conclusion asserts $\{0\} <_{\mathrm{BA}} \mathbf{Z}_p$ — contradicting Lemma 6.24, the paper's own Lemma 29. The failure is exactly the missing subdirectness: $\mathrm{pr}_1(R) = \{0\} \neq \mathbf{Z}_p$.

The repair is to restore both hypotheses, as above. A formalization must also check that they hold at each of the call sites — in particular in the proof of Theorem 8.1(2), where the lemma is applied to the relation computing E_2 from E_1. The common-term condition is the reason [14] writes $\leq_{\mathrm{BA}(t)}$ with the term in the notation, and carrying t in the type is the cheapest way to keep it.

This is the single highest-value import to formalize first: small, self-contained, used everywhere, and adjacent to what the predecessor project already proves.

6.3 Intersections

Lemma 6.12 (Intersecting with a strong subset). *If $B \lhd\lhd\lhd A$ and $D \lhd\lhd\lhd A$ then* (i) $B \cap D \dot{\lhd}\dot{\lhd}\dot{\lhd} A$, *and* (t) $C <^{A}_{T(\sigma)} B \Rightarrow C \cap D \dot{\leq}^{A}_{T(\sigma)} B \cap D$.

The next theorem is the heart of §6. Read it as follows: if a family of strong reductions is *jointly* inconsistent but *minimally* so — dropping any one of them restores consistency — then the types must agree, and in the linear case one gets bridges between the witnessing congruences. Bridges are produced nowhere else.

Theorem 6.13 (Stable intersection). Suppose

(i) $C_i <^{A}_{T_i(\sigma_i)} B_i \lhd\lhd\lhd A$ with $T_i \in \{\textsc{ba}, \textsc{c}, \textsc{s}, \textsc{l}, \textsc{pc}\}$, for $i \in [m]$, $m \geq 2$;
(ii) $\bigcap_{i\in[m]} C_i = \varnothing$;
(iii) $B_j \cap \bigcap_{i\neq j} C_i \neq \varnothing$ for every $j \in [m]$.

Then one of:

(ba) $T_1 = \cdots = T_m = \textsc{ba}$;
(l) $T_1 = \cdots = T_m = \textsc{l}$, and for all $k, \ell \in [m]$ there is a bridge δ from σ_k to σ_ℓ with $\widetilde{\delta} = \sigma_k \circ \sigma_\ell$;
(c) $m = 2$ and $T_1 = T_2 = \textsc{c}$;
(pc) $m = 2$, $T_1 = T_2 = \textsc{pc}$ and $\sigma_1 = \sigma_2$.

Corollary 6.14 (Stable intersection, relational form). Suppose $R \leq_{\mathrm{sd}} \mathbf{A}_1 \times \cdots \times \mathbf{A}_m$, $C_i <^{A_i}_{T_i(\sigma_i)} B_i \lhd\lhd\lhd A_i$ with $T_i \in \{\textsc{ba}, \textsc{c}, \textsc{s}, \textsc{l}, \textsc{pc}\}$ and $m \geq 2$, $R \cap (C_1 \times \cdots \times C_m) = \varnothing$, and $R \cap (C_1 \times \cdots \times B_j \times \cdots \times C_m) \neq \varnothing$ for every j. Then one of: (ba) all $T_i = \textsc{ba}$; (l) all $T_i = \textsc{l}$ and for all k, ℓ a bridge δ from σ_k to σ_ℓ with $\widetilde{\delta} = \sigma_k \circ \mathrm{pr}_{k,\ell}(R) \circ \sigma_\ell$; (c) $m = 2$ and $T_1 = T_2 = \textsc{c}$; (pc) $m = 2$, $T_1 = T_2 = \textsc{pc}$, $\mathbf{A}_1/\sigma_1 \cong \mathbf{A}_2/\sigma_2$, and $\{(a/\sigma_1, b/\sigma_2) : (a, b) \in R\}$ is bijective.

Remark 6.15. Several applications need more than one restriction on a single coordinate of R. The statement is kept simple; to apply it, duplicate the coordinate (form $R' = \{(\bar{a}, a_k) : \bar{a} \in R\}$) and restrict the copies separately.

Formalization note 6.16 (Minimality is a hypothesis, not a construction). Hypotheses (ii)–(iii) of Theorem 7.53 say the family is *critically* inconsistent. In applications one starts with an inconsistent family and shrinks it to a critical subfamily. That shrinking step is left implicit in the paper ("let $\mathcal{M}$ be a minimal set of children of z we need to restrict"). It is easy — a minimal subset of a finite set with a monotone property — but it must be stated as a lemma, because it is performed at least four times and each time the property being minimised is slightly different. State once: *if P is a monotone predicate on subsets of a finite set X and $P(X)$ holds, there is $M \subseteq X$ minimal with $P(M)$*, and instantiate.

6.4 Multi-types

Lemma 6.17. *If $C \leq^{A}_{\mathcal{M}T} B$ then $C <^{A}_{T} \ldots <^{A}_{T} B$ and $C \lhd\lhd\lhd^{A} B$.*

That is: an intersection of type-T subsets, though not itself obtained by a single reduction of type T, is reachable by a *chain* of type-T reductions. This is what makes $\mathcal{M}T$ compatible with $\lhd\lhd\lhd$, and it is used whenever the algorithm reduces several variables at once.

Lemma 6.18 (Preserving linkedness). *Let $R \leq_{\mathrm{sd}} \mathbf{A}_1 \times \mathbf{A}_2$, $C_i \leq^{A_i}_{\mathcal{M}\mathrm{D}} B_i \lhd\lhd\lhd A_i$ for $i = 1, 2$, let S be the rectangular closure of R, and suppose $R \cap (B_1 \times B_2) \neq \varnothing$ and $S \cap (C_1 \times C_2) \neq \varnothing$. Then $R \cap (C_1 \times C_2) \neq \varnothing$.*

Lemma 6.19 (Maximal multi-type extension). *Let $C_1 <^{A}_{\mathcal{M}T} B_1 \lhd\lhd\lhd A$, $B_2 \lhd\lhd\lhd A$, $C_1 \cap B_2 = \varnothing$, $B_1 \cap B_2 \neq \varnothing$, and let σ be a maximal congruence on $\mathbf{A}$ with $(C_1 \circ \sigma) \cap B_2 = \varnothing$. Then $\sigma = \omega_1 \cap \cdots \cap \omega_s$ where each ω_i is a congruence of type T on $\mathbf{A}$ with $\omega_i^* \supseteq B_1^2$.*

6.5 Auxiliary imports

The following are used but not proved in [13].

Lemma 6.20 ([9, Lem. 4.7]). *If w is an idempotent WNU operation of arity n on a finite set A, then* $\mathrm{Clo}(w)$ *contains a special idempotent WNU operation of arity $n^{n!}$.*

Lemma 6.21 ([12, Cor. 8.17.1]). *If σ is an irreducible congruence on $\mathbf{A} \in \mathcal{V}_n$ and δ is a bridge from σ to σ with $\widetilde{\delta} = A^2$, then σ is a perfect linear congruence.*

Lemma 6.22. *If σ is an irreducible congruence on $\mathbf{A} \in \mathcal{V}_n$ and δ is a bridge from σ to σ with $\widetilde{\delta}$ linked, then σ is a perfect linear congruence.*

Proof. Let $\delta^{-1}(y_1, y_2, x_1, x_2) = \delta(x_1, x_2, y_1, y_2)$, which is again a bridge from σ to σ. Since $\widetilde{\delta}$ is linked, $(\widetilde{\delta} \circ \widetilde{\delta^{-1}})^N = A^2$ for N large. Composing $2N$ bridges $\delta, \delta^{-1}, \ldots$ by Lemma 3.30 gives a bridge δ' from σ to σ with $\widetilde{\delta'} = A^2$; apply Lemma 6.21. □

Formalization note 6.23 ("for sufficiently large N"). The step $(\widetilde{\delta} \circ \widetilde{\delta^{-1}})^N = A^2$ deserves a lemma of its own: if a reflexive symmetric relation ρ on a finite set A has connected graph, then $\rho^N = A^2$ for $N \geq |A|$. Reflexivity is what makes the powers increase monotonically; $\widetilde{\delta}$ is reflexive because $(a, a, a, a) \in \delta$ is not assumed — so the reflexivity must come from δ being a bridge from σ to σ with σ reflexive, via $\mathrm{pr}_{1,2}(\delta) \supsetneq \sigma$. Check this: it is the sort of step where the informal proof is right for a reason it does not give.

Lemma 6.24. *If $B \leq \mathbf{Z}_p \times \cdots \times \mathbf{Z}_p$ then there is no $C <_T B$ with $T \in \{\text{BA}, \text{C}\}$.*

Proof. It suffices to check that for every term τ, if the last variable of $\tau^{\mathbf{Z}_p}(a, \ldots, a, x)$ is not dummy then it takes all values; hence no absorbing subuniverse exists. □

7 Proofs of the strong-subalgebra theory

This section proves the statements of §6. It renders §5 of [13], the part its author calls the most technical. Fourteen places there are wrong, incomplete, or state less than their own proofs give; each repair is marked **[R]** and carries a note saying what the source has. Everything else is expansion: about two dozen steps that the source performs silently and that a formalization cannot.

Throughout, Convention 2.1 is in force — every algebra is finite, idempotent, and has a WNU term operation — and §2.1 fixes the readings.

7.1 What is imported

The rendering is self-contained except for the following, which are proved in the literature and used here as black boxes. Two are substantial; the rest are routine.

Theorem 7.1 (Absorption Theorem; [2, Thm. 3.11.1]). Let $R \lneq_{sd} \mathbf{A} \times \mathbf{B}$ be a proper subdirect relation between finite idempotent Taylor algebras. If R is linked then $\mathbf{A}$ or $\mathbf{B}$ has a proper nonempty binary absorbing or central subuniverse.

Formalization note 7.2 (Two words the source omits). [13] states this as "there exists a BA or central subuniverse on $\mathbf{A}$ or $\mathbf{B}$", with no properness. Read literally it is vacuous, since every algebra is a binary absorbing subuniverse of itself; every use is the proper nonempty one (§2.1, item iii). This is the single heaviest import in the section.

Lemma 7.3 ([4, Lem. 2.9], [14, Lem. 6.1, Thm. 6.9]). *Let $R \leq A^n$ be defined by a pp-formula Φ in which a relation S occurs, and let R' be defined by the formula obtained from Φ by replacing* every *occurrence of S by some $S' <_T S$, where $T \in \{\text{BA}, \text{C}\}$. Then $R' \leq_T R$.*

Lemma 7.4 ([4, Prop. 2.14], [14, Lem. 3.2]). *For $B \leq \mathbf{A}$ and $n \geq 2$: B absorbs $\mathbf{A}$ with an operation of arity n if and only if there is no $S \leq A^n$ with $S \cap B^n = \varnothing$ and $S \cap (B^{i-1} \times A \times B^{n-i}) \neq \varnothing$ for every i.*

Lemma 7.5 ([14, Lem. 6.8]). *If $B \leq_T \mathbf{A}$ with $T \in \{\text{BA}, \text{C}, \text{S}\}$ and σ is a congruence on $\mathbf{A}$, then $B/\sigma \leq_T \mathbf{A}/\sigma$.*

Formalization note 7.6 (The BA and S cases are not imports). [13] writes "for $T = \text{BA}$ it is straightforward, for $T = \text{C}$ see [14, Lem. 6.8], for $T = \text{S}$ it is just a combination". The first is indeed one line — if $t(B, A) \subseteq B$ and $t(A, B) \subseteq B$ then the same t works on images — and the third follows from the other two by Definition 4.6. Only C is imported.

Lemma 7.7 ([14, Lem. 6.24]). *If R is a nontrivial strong subuniverse of $\mathbf{A}_1 \times \cdots \times \mathbf{A}_n$ of a type $T \neq \text{PC}$, then some $\mathbf{A}_i$ has a nontrivial subuniverse of type T. In particular $B <_T \mathbf{A}^n$ with $T \in \{\text{BA}, \text{C}, \text{S}\}$ yields some $C <_T \mathbf{A}$.*

Formalization note 7.8 (The source undersells its own citation). [13] proves the case $T = \text{S}$ by "just repeat the same proof word to word replacing BA by S". No repetition is needed. The cited lemma is already stated for every type $T \neq \text{PC}$, and its proof is *type-uniform*: it takes $\mathrm{pr}_1(R)$ if that is proper and otherwise a fibre $R' = \{(a_2, \ldots, a_n) : (a_1, \ldots, a_n) \in R\}$, and neither construction mentions the type — the type enters only through the facts that projections and fibres preserve BA and preserve C. So a single run of the proof produces one witness carrying *every* strong type that R has, and a subuniverse that is simultaneously binary absorbing and central yields one that is too.

Theorem 7.9 ([14, Thm. 6.15]). Let $R \leq_{\text{sd}} \mathbf{A} \times \mathbf{B}$ and $C = \{c \in A : \{c\} \times B \subseteq R\}$. Then C is a central subuniverse of $\mathbf{A}$, or $\mathbf{B}$ has a nontrivial binary absorbing subuniverse, or $\mathbf{B}$ has a nontrivial *projective* subuniverse.

Formalization note 7.10 (To do). Theorem 7.9 is the one import we are not yet comfortable taking on faith, and it should eventually be proved here. Its proof in [14] is two pages and uses only the Barto–Kazda criterion (Lemma 7.4) and the theory of projective subuniverses; none of that theory is needed anywhere else in this document, which is why absorbing it has been deferred.

Lemma 7.11 ([7]). *An algebra is Abelian if and only if some congruence of its square has the diagonal as a block; and a finite algebra with a WNU term operation is Abelian if and only if it is affine.*

Lemma 7.12. *$\mathbf{A}$ is Abelian if and only if there is a bridge δ from $0_{\mathbf{A}}$ to $0_{\mathbf{A}}$ with $\widetilde{\delta} = \mathrm{pr}_{1,2}(\delta) = \mathrm{pr}_{3,4}(\delta) = A^2$.*

Proof. A congruence of $\mathbf{A}^2$ with the diagonal as a block is such a bridge. Conversely, composing a bridge with itself enough times yields a reflexive, symmetric, transitive relation on $\mathbf{A}^2$, whose diagonal class is the diagonal by clause (d); now apply Lemma 7.11. □

7.2 Absorption preliminaries

Lemma 7.13. *If $B \leq_T \mathbf{A}$ with $T \in \{\text{BA}, \text{C}\}$ and $C \leq \mathbf{A}$, then $B \cap C \dot{\leq}_T C$.*

Proof. C is defined by the pp-formula $A(x) \wedge C(x)$; replacing the occurrence of the full unary relation A by B yields $B \cap C$, so Lemma 7.3 applies. □

Lemma 7.14 (No absorbing diagonal)**.** *Let* $\mathbf{C}$ *be a finite idempotent algebra with* $|C| \geq 2$. *Then* $0_\mathbf{C}$ *is not an absorbing subuniverse of* $\mathbf{C}^2$*, of any arity; in particular* $0_\mathbf{C} \not\leq_{\text{BA}} \mathbf{C}^2$ *and* $0_\mathbf{C} \not\leq_{\text{C}} \mathbf{C}^2$.

Proof. Suppose t is k-ary and absorbs at every position. Fix i, take $a_j \in C$ for $j \neq i$ and $b, c \in C$, and apply t coordinatewise to the tuples $(a_j, a_j) \in 0_\mathbf{C}$ for $j \neq i$ and $(b, c) \in C^2$ at position i. The result $(t(\bar{a}; b),\, t(\bar{a}; c))$ lies in $0_\mathbf{C}$, so $t(\bar{a}; b) = t(\bar{a}; c)$. As b, c were arbitrary, t does not depend on its i-th argument; and this holds for every i, so t is constant. Idempotence then forces $|C| = 1$. A central subuniverse is absorbing by Definition 4.3, so the central case is included. □

Lemma 7.15 (Absorption preserves linkedness, subrelation form)**.** *Let* $W \leq_{\text{sd}} \mathbf{P} \times \mathbf{A}$ *be linked and let* $\varnothing \neq W' \leq_T W$ *with* $T \in \{\text{BA}, \text{C}\}$. *Put* $P' = \mathrm{pr}_1(W')$ *and* $A' = \mathrm{pr}_2(W')$. *Then* $W' \leq_{\text{sd}} \mathbf{P}' \times \mathbf{A}'$ *is linked.*

Proof. Let $\Phi_k(x, y)$ be the pp-formula in one binary relation symbol obtained by iterating $x \sim y \iff \exists z\, (W(x, z) \wedge W(y, z))$ k times. Because W is subdirect, $W \circ W^{-1}$ is reflexive and symmetric on P, so for $k \geq |P|$ the formula Φ_k interpreted in W defines the transitive closure of $W \circ W^{-1}$, which is P^2 since W is linked. Interpreted in W' and for the same $k \geq |P| \geq |P'|$ it defines $\lambda := \mathrm{LeftLinked}(W')$. By Lemma 7.3, replacing every occurrence of W by W' gives $\lambda \leq_T P^2$. Since $\lambda \subseteq (P')^2 \leq P^2$, Lemma 7.13 gives $\lambda \leq_T (P')^2$.

Now λ is a congruence on $\mathbf{P}'$: reflexive because W' is subdirect over P', symmetric by construction, transitive because the powers stabilise. So $\lambda \times \lambda$ is a congruence on the algebra $(P')^2$, and the image of λ under $(P')^2 \twoheadrightarrow (P'/\lambda)^2$ is exactly $0_{\mathbf{P}'/\lambda}$; by Lemma 7.5, $0_{\mathbf{P}'/\lambda} \leq_T (\mathbf{P}'/\lambda)^2$. Lemma 7.14 forces $|P'/\lambda| = 1$, i.e. $\lambda = (P')^2$, which says W' is linked. □

Formalization note 7.16 (The box form is not enough)**.** **[R]** [13] has only the *box* form of this principle, [3, Prop. 2.15(i)]: if $W \leq_{\text{sd}} \mathbf{A}_1 \times \mathbf{A}_2$ is linked, B_i absorbs $\mathbf{A}_i$, and $W \cap (B_1 \times B_2)$ is subdirect in $B_1 \times B_2$, then $W \cap (B_1 \times B_2)$ is linked. It cites that form for a restriction that is not a box, in the proof of Lemma 7.41, and the two really do differ: for the meet-semilattice on $\{0, 1, 2\}$ with $B = \{0, 1\}$, the subalgebra $K = \{(0, 0), (1, 2)\}$ of A^2 meets B^2 and has $\mathrm{pr}_1(K) \cap B = \{0, 1\}$ while $\mathrm{pr}_1(K \cap B^2) = \{0\}$. Lemma 7.15 is the form the paper actually needs; it also discharges the subdirectness hypothesis of the box form for free. Checked by exhaustive search over nine small Taylor algebras and about 10^5 pairs (W, W'): no counterexample.

Lemma 7.17 (A WNU forbids essentially unary quotients)**.** *Let* $\mathbf{U}$ *be a finite algebra with a WNU term operation and* $|U| \geq 2$. *Then some term operation of* $\mathbf{U}$ *depends on more than one variable. Consequently, if* $\mathbf{A}$ *has a WNU term operation then no* $\mathbf{U} \in \mathrm{HS}(\mathbf{A})$ *with* $|U| \geq 2$ *is essentially unary.*

Proof. Suppose every term operation of $\mathbf{U}$ has at most one non-dummy variable, and let w be a WNU term operation of arity $k \geq 2$; it is idempotent. If w has no non-dummy coordinate it is constant, and idempotence gives $|U| = 1$. Otherwise $w(\bar{x}) = g(x_i)$ for a unary g, and $g(x) = w(x, \dots, x) = x$, so w is the i-th projection. The WNU identities equate the k terms in which a single y sits at each position against a background of x's; two of them are w with y at position i, which evaluates to y, and w with y at some position $\neq i$, which evaluates to x. Hence $x = y$ for all x, y and $|U| = 1$. The consequence follows because idempotence and the WNU identities are identities, hence hold in every member of $\mathrm{HS}(\mathbf{A})$. □

Lemma 7.18 (Central relations)**. [R]** Let $\mathbf{A}, \mathbf{B}$ satisfy Convention 2.1, let $R \leq_{\mathrm{sd}} \mathbf{A} \times \mathbf{B}$, and put $C = \{c \in A : \{c\} \times B \subseteq R\}$. Then

(a) $C = \varnothing$, or
(b) C is a central subuniverse of $\mathbf{A}$, or
(c) $\mathbf{B}$ has a proper nonempty binary absorbing subuniverse.

Proof. Apply Theorem 7.9. Its first alternative gives (a) or (b) according as C is empty or not, its second gives (c). In its third, $\mathbf{B}$ has a nontrivial projective subuniverse P; by [14, Lem. 3.4], either P is a binary absorbing subuniverse of $\mathbf{B}$, which is (c) since P is nontrivial, or there is an essentially unary $\mathbf{U} \in \mathrm{HS}(\mathbf{B})$ with $|U| \geq 2$, which Lemma 7.17 forbids. □

Formalization note 7.19 (Both changes are necessary)**.** [13] states this with two alternatives, (b) and (c), citing [14, Thm. 6.15] — which has three. The elimination of the third is legitimate but is nowhere stated; it is the proof above, and Zhuk himself runs exactly this argument on exactly this trichotomy in the proof of [14, Thm. 4.15], step (4) ⇒ (5), without ever connecting it to Theorem 6.15.

Alternative (a) is also not optional. The two-alternative statement is true in [14] only because $\varnothing$ counts as central there; under Convention 2.4 it is false, as $\mathbf{A} = \mathbf{B} = \mathbf{Z}_p$ with $R = \{(x, x+1)\}$ shows — R is subdirect, $C = \varnothing$, and $\mathbf{Z}_p$ has no nontrivial binary absorbing subuniverse at all. Every one of the nine call sites discharges (a) by exhibiting a point of C first.

Lemma 7.20. *Let $R \leq_{\mathrm{sd}} \mathbf{A} \times \mathbf{B}$ with $\mathbf{A}$ BA and centre free,* $\mathrm{LeftLinked}(R) = A^2$*, and $C = \{c \in B : A \times \{c\} \subseteq R\}$. Then $C \neq \varnothing$ and $C \leq_{\text{BA,C}} \mathbf{B}$.*

Proof. For $k \geq 2$ put $W_k = \{(a_1, \dots, a_k) : \exists b \forall i\, R(a_i, b)\}$. If $W_{|A|} = A^{|A|}$ then $C \neq \varnothing$, since a witness b for the tuple listing all of A lies in C. Otherwise choose k minimal with $W_k \neq A^k$. Since $\mathrm{LeftLinked}(R) = A^2$ we get $\mathrm{LeftLinked}(W_k) = A^2$, and viewing W_k as a binary relation in $\mathbf{A} \times \mathbf{A}^{k-1}$ — subdirect by minimality of k — Theorem 7.1 and Lemma 7.7 produce a proper nonempty binary absorbing or central subuniverse of $\mathbf{A}$, contrary to hypothesis. So $C \neq \varnothing$.

By Lemma 7.18 applied to R^{-1} — alternative (a) being excluded — either C is central in $\mathbf{B}$ or $\mathbf{A}$ has a proper nonempty binary absorbing subuniverse; the latter is excluded, so C is central. For binary absorption, suppose not; by Lemma 7.4 there is $S \leq B \times B$ with $S \cap (C \times C) = \varnothing$, $S \cap (B \times C) \neq \varnothing$ and $S \cap (C \times B) \neq \varnothing$. Put

$$W = \{(a_1, \dots, a_{|A|}) : \exists (b, c) \in S\ \big(c \in C \wedge \forall i\, R(a_i, b)\big)\}.$$

By Lemma 7.3 $W <_{\text{C}} A^{|A|}$, so by Lemma 7.7 $\mathbf{A}$ has a proper nonempty central subuniverse — again a contradiction. □

Lemma 7.21. *If $C \lhd\!\lhd\!\lhd^{A} B \lhd\!\lhd\!\lhd A$ and $D <_{\text{BA,C}} B$ then $C \cap D \neq \varnothing$.*

Proof. Suppose not, and choose C'' minimal with $C \lhd\!\lhd\!\lhd^{A} C' <^{A}_{T(\sigma)} C'' \lhd\!\lhd\!\lhd^{A} B$ and $C'' \cap D \neq \varnothing$. By Lemma 7.13 $C'' \cap D <_{\text{BA,C}} C''$, so by Lemma 7.22 $T = \text{D}$; and then Lemma 7.5 gives $(C'' \cap D)/\sigma <_{\text{BA}} C''/\sigma$, contradicting the requirement in Definition 4.21 that C''/σ be BA and centre free. □

Lemma 7.22 ([14, Lem. 6.25])**.** *If $B <_{T_1} \mathbf{A}$, $C <_{T_2} \mathbf{A}$, $B \cap C = \varnothing$ and $T_1, T_2 \in \{\text{BA}, \text{C}, \text{S}\}$, then $T_1 = T_2 \in \{\text{BA}, \text{C}\}$.*

7.3 The intersection property

The goal is Lemma 7.30: if $B \lhd\lhd\lhd A$, $C \lhd\lhd\lhd A$ meet and δ is a dividing congruence for B, then $(B \cap C)/\delta$ is a single point or all of B/δ. Everything in §7 that produces a bridge goes through it.

Lemma 7.23 (The shift manoeuvre)**.** *Let $R \leq A^k$ be totally symmetric and closed under $(a_1, \ldots, a_k) \mapsto (a_1, a_1, a_2, \ldots, a_{k-1})$. If $\bar{a} \in R$ has entry multiset M, then R contains a tuple with entry multiset $M + \{u\} - \{v\}$ for any $u, v \in M$ occupying distinct positions. In particular, if $|\{\text{distinct entries of } \bar{a}\}| = r$ then R contains a tuple whose entries are any prescribed k-element multiset drawn from those r values, provided that multiset uses at least one of them.*

Proof. Total symmetry lets us place u first and v last; the shift then duplicates u and deletes v. Iterating and permuting gives every reachable multiset. □

Formalization note 7.24 (The shift deletes the last entry)**. [R]** [13] uses this twice with the words "by the conditions of the lemma we have…". A single application of the shift to $(a, b_2, \ldots, b_{k-1}, c)$ destroys c, so the claimed conclusion $(a, \ldots, a, c) \in R$ does not follow from one step; it needs $k - 2$ steps interleaved with permutations, which is what Lemma 7.23 packages.

Lemma 7.25. *Let $\mathbf{A}$ be BA and centre free and let $R \leq A^k$ be totally symmetric, closed under the shift, with $\mathrm{pr}_{1,2}(R) = A^2$. Then $R = A^k$.*

Proof. Induct on k; for $k = 2$ the hypothesis is the conclusion. Let $k > 2$. By total symmetry $\mathrm{pr}_{1,k}(R) = A^2$, so for any a, c there is a tuple of R with first entry a and last entry c; by Lemma 7.23 we may replace all other entries by a, giving $(a, \ldots, a, c) \in R$, and deleting c gives $(a, \ldots, a) \in R$. Hence, viewing $R \leq A^{k-1} \times A$, every left vertex $(a, \ldots, a)$ is adjacent to every c, so R is subdirect and $\mathrm{RightLinked}(R) = A^2$.

The relation $R' = \mathrm{pr}_{1,\ldots,k-1}(R)$ inherits total symmetry, has $\mathrm{pr}_{1,2}(R') = A^2$, and is shift-closed: from $\bar{a} \in R'$ extend to R, shift there, and project. By induction $R' = A^{k-1}$, so $R \leq_{\mathrm{sd}} A^{k-1} \times A$. If $R \neq A^k$ then Theorem 7.1 gives a proper nonempty binary absorbing or central subuniverse on $\mathbf{A}^{k-1}$ or on $\mathbf{A}$, and Lemma 7.7 moves the first case to $\mathbf{A}$ — contradicting the hypothesis either way. □

Lemma 7.26. *Let σ be a dividing congruence for $B \leq A$ and let $R \leq (\mathbf{A}/\sigma)^k$ be reflexive, totally symmetric and shift-closed. Then $(B/\sigma)^k \subseteq R$, or R is the set of constant tuples.*

Proof. If $\mathrm{pr}_{1,2}(R)$ is the equality relation then, by total symmetry, all coordinates of every tuple of R agree, and reflexivity supplies all constant tuples.

Otherwise $\mathrm{pr}_{1,2}(R)$ is a subuniverse of $(\mathbf{A}/\sigma)^2$ containing $0_{\mathbf{A}/\sigma}$ properly and stable under it, so by minimality of the cover (Proposition 3.23, transported by Lemma 3.25) it contains σ^*/σ, hence $(B/\sigma)^2$ — the last step by Definition 4.21(i). Now apply Lemma 7.25 to $R \cap (B/\sigma)^k$ over the algebra B/σ, which is BA and centre free by Definition 4.21(iii). Its hypothesis $\mathrm{pr}_{1,2}(R \cap (B/\sigma)^k) = (B/\sigma)^2$ is not immediate from $\mathrm{pr}_{1,2}(R) \supseteq (B/\sigma)^2$, since a witnessing tuple may leave B/σ; Lemma 7.23 repairs it, pulling $(x_1, x_2, y_3, \ldots, y_k)$ back to $(x_1, x_2, x_1, \ldots, x_1)$. □

Lemma 7.27 (Full fibre)**.** *Let F be a finite set with $|F| \leq N$ and let $W \subseteq F^N \times E$ satisfy $\mathrm{pr}_{1..N}(W) = F^N$ and be closed under coordinate substitution: if $(x_1, \ldots, x_N, e) \in W$ and $g : [N] \to [N]$ then $(x_{g(1)}, \ldots, x_{g(N)}, e) \in W$. Then $F^N \times \{d\} \subseteq W$ for some $d \in E$.*

Proof. Write $F = \{F_1, \ldots, F_t\}$ with $t \leq N$ and put $\bar{x} = (F_1, \ldots, F_t, F_t, \ldots, F_t) \in F^N$. By surjectivity pick d with $(\bar{x}, d) \in W$. Every $\bar{y} \in F^N$ is $\bar{x} \circ g$ for some $g : [N] \to [N]$, since every entry of $\bar{y}$ occurs among the entries of $\bar{x}$; closure gives $(\bar{y}, d) \in W$. □

Formalization note 7.28 (Why the arity is $|A|$)**.** **[R]** Three proofs in [13] contain, verbatim modulo names, "since $\mathrm{pr}_{1,\dots,|A|}(R') = (B/\delta)^{|A|}$, there exists d with $(B/\delta)^{|A|} \times \{d\} \subseteq R'$". Surjectivity of a projection does not produce a full fibre, and as stated this is a non sequitur. It is true because each such R' is cut out by a conjunction of unary and binary conditions on its first $|A|$ entries, hence closed under substituting entries for one another, and because $|B/\delta| \le |A|$. **The exponent $|A|$ is load-bearing**: it must be at least the number of blocks, so that one tuple can enumerate them. A rendering that normalises the arity away breaks all three proofs.

Lemma 7.29. Let $B_k <^A_{T_k(\sigma_k)} \cdots <^A_{T_1(\sigma_1)} B_0 = A$ with $T_i \in \{\textsc{ba}, \textsc{c}, \textsc{s}, \textsc{d}\}$, let δ be a congruence on $\mathbf{A}$, and let $m \in [k]$ with $T_m = \textsc{d}$. Then $\big((B_k \circ \delta) \cap B_{m-1}\big)/\sigma_m$ is either B_{m-1}/σ_m or B_m/σ_m; and in the second case $\sigma_m \supseteq \delta \cap \sigma_1 \cap \cdots \cap \sigma_{m-1}$.

Proof sketch. Induction on k, splitting on T_k and on whether $k > m$. The absorption cases run through Lemmas 7.3 and 7.5 to produce a strong subuniverse of B_{k-1}/σ_k and contradict Definition 4.21(iii); the $\textsc{d}$ cases build the auxiliary relation R' on $(B_{k-1}/\sigma_k)^{|A|} \times B_{n-1}/\sigma_n$, obtain a full fibre from Lemma 7.27, and finish with Lemma 7.18. Two steps must be supplied at each such finish: the centre is *proper*, the witness being the displayed non-containment two lines earlier, and the binary absorbing subuniverse produced lives on a *power* and is moved down by Lemma 7.7, which the source cites at one of the three sites and omits at the other two. □

Lemma 7.30 (The intersection property)**.** Let $B \lhd\!\lhd\!\lhd A$, $C \lhd\!\lhd\!\lhd A$ with $B \cap C \neq \varnothing$. Then

(d) if δ is a dividing congruence for $B \le A$ then $|(B\cap C)/\delta| = 1$ or $(B \cap C)/\delta = B/\delta$; and if $|(B\cap C)/\delta| = 1$ then $\delta \supseteq \delta_1 \cap \cdots \cap \delta_s$, where $\delta_1, \dots, \delta_s$ are the congruences of the *chosen* chains witnessing $B \lhd\!\lhd\!\lhd A$ and $C \lhd\!\lhd\!\lhd A$;

(s) if $G <_{\textsc{ba},\textsc{c}} B$ then $G \cap C \neq \varnothing$.

Proof outline, with the repaired steps in full. Fix chains $B = B_k < \cdots < B_0 = A$ and $C = C_\ell < \cdots < C_0 = A$, put $\sigma = \bigcap_i \sigma_i$ and $\omega = \bigcap_j \omega_j$, and induct on $k + \ell$. Every appeal to the inductive hypothesis lands at strictly smaller $k + \ell$.

(s). Split on the type $\mathcal{T}_\ell$ of the last step of C's chain. In the case $\mathcal{T}_\ell = \textsc{d}$ the source asserts $(G \cap C_{\ell-1})/\omega_\ell <_{\textsc{ba},\textsc{c}} C_{\ell-1}/\omega_\ell$ with a *proper* containment, where the cited lemmas supply only $\le$. **[R]** Split the case instead: if the containment is an equality then every ω_ℓ-block meeting $C_{\ell-1}$ meets $G \cap C_{\ell-1}$, in particular the block E with $C_\ell = C_{\ell-1} \cap E$, so $G \cap C_\ell \neq \varnothing$, which is what was to be proved.

(d). Put $S = \{(a_1/\delta, \dots, a_{|A|}/\delta) : \forall i,j \ (a_i, a_j) \in \sigma \cap \omega\}$, a reflexive, totally symmetric, shift-closed subuniverse of $(\mathbf{A}/\delta)^{|A|}$, and apply Lemma 7.26.

Case 1: S is the diagonal. Then $\sigma \cap \omega \subseteq \delta$, which is the "moreover" clause; and any $a, b \in B \cap C$ satisfy $(a,b) \in \sigma \cap \omega$, so $|(B \cap C)/\delta| = 1$.

Case 2: $(B/\delta)^{|A|} \subseteq S$. Refine S to $S_{m,n}$ by requiring $a_i \in B_m \cap C_n$, and take the least m (subcase 2A) or least n (subcase 2B) at which containment fails, splitting on the type of the corresponding step. **[R]** Three steps are supplied here that the source does not state. *Properness*: in 2A1 and 2B1 the containments $S_{m,0} \cap (B/\delta)^{|A|} <_{T_m} (B/\delta)^{|A|}$ are proper by the minimal choice of m, and nonempty because the constant tuples $(b/\delta, \dots, b/\delta)$ for $b \in B$ lie in $S_{m,0}$, since $B \subseteq B_m$ and $(b,b) \in \sigma \cap \omega$. *Which block*: in 2B3 the inductive hypothesis offers "size 1 or all of $C_{\ell-1}/\omega_n$", and the stronger reading "$= C_n/\omega_n$" used there is correct because $B \cap C \neq \varnothing$ and $C \subseteq C_n$ force $B_k \cap C_n \neq \varnothing$, so the single block is the one defining C_n. *Full fibres*: the appeals in 2A3 and 2B3 to a d with $(B/\delta)^{|A|} \times \{d\} \subseteq R'$ are Lemma 7.27. Subcase 2C survives and gives $(B \cap C)/\delta = B/\delta$. □

Formalization note 7.31 (What remains to be expanded)**.** The proof above is written at the level of its case structure, with every repaired step in full. Expanding the twelve subcases of (d) to the level of the rest of this section is the largest single piece of writing left; the source occupies 292 lines here and nothing in it is wrong. One loose end: the proof fixes *minimal* chains for B and C and I could find no step that consumes minimality — a shorter chain only makes the induction measure smaller. Either identify the use or drop the hypothesis.

Corollary 7.32. *Let $R \leq_{\mathrm{sd}} \mathbf{A}_1 \times \mathbf{A}_2$, $B_1 \lhd\!\lhd\!\lhd A_1$, $B_2 \lhd\!\lhd\!\lhd A_2$, and let σ be a dividing congruence for $B_1 \lhd\!\lhd\!\lhd A_1$. Then $\mathrm{pr}_1(R \cap (B_1 \times B_2))/\sigma$ is empty, of size 1, or equal to B_1/σ.*

Proof. Let $f_i : \mathbf{R} \to \mathbf{A}_i$ be the projections. By Lemma 6.4(b), $f_i^{-1}(B_i) \lhd\!\lhd\!\lhd \mathbf{R}$; apply Lemma 7.30(d) inside $\mathbf{R}$ to these two, with the congruence $f_1^{-1}(\sigma)$, and read the conclusion back through f_1. □

Formalization note 7.33. [13] writes $\delta = f^{-1}(\sigma)$ with f undefined; read f_1. The alternative "empty" is present here and absent from Lemma 7.30, which hypothesises $B \cap C \neq \varnothing$; the empty case needs its own line at every call site.

Lemma 7.34 (Parallelogram property for D)**.** *Let $B_1, B_2 \lhd\!\lhd\!\lhd A$, $C_1, C_1' <^{A}_{\mathrm{D}(\sigma_1)} B_1$ and $C_2, C_2' <^{A}_{\mathrm{D}(\sigma_2)} B_2$ with $C_1' \cap C_2$, $C_1 \cap C_2'$ and $C_1' \cap C_2'$ all nonempty. Then $C_1 \cap C_2 \neq \varnothing$.*

Proof. By Lemma 7.30(d), $(B_1 \cap B_2)/\sigma_1$ is of size 1 or all of B_1/σ_1. In the first case the single block is the one defining C_1, because $C_1 \cap C_2' \neq \varnothing$ puts a point of C_1 in $B_1 \cap B_2$; hence $B_1 \cap B_2 \subseteq C_1$ and $C_1 \cap C_2 = B_1 \cap C_2 \supseteq C_1' \cap C_2 \neq \varnothing$. The symmetric argument applies if $(B_1 \cap B_2)/\sigma_2$ has size 1.

So assume $(B_1 \cap B_2)/\sigma_1 = B_1/\sigma_1$ and $(B_1 \cap B_2)/\sigma_2 = B_2/\sigma_2$, and put $S = \{(a/\sigma_1, a/\sigma_2) : a \in B_1 \cap B_2\}$, which is then subdirect in $(B_1/\sigma_1) \times (B_2/\sigma_2)$. Apply Lemma 7.30(d) to each σ_1-class C of B_1 against B_2 with the congruence σ_2; the hypothesis $C \cap B_2 \neq \varnothing$ holds because S is subdirect. Each fibre of S is therefore a single point or the whole of B_2/σ_2.

If some fibre is full, then either S is full — and then the fibre of C_1/σ_1 contains C_2/σ_2, giving $C_1 \cap C_2 \neq \varnothing$ — or the centre of S is proper and nonempty, and Lemma 7.18 produces a proper nonempty binary absorbing or central subuniverse of B_1/σ_1 or B_2/σ_2, contradicting Definition 4.21(iii). If every fibre is a single point, then the fibre of C_1'/σ_1 is both C_2/σ_2 (witnessed by $C_1' \cap C_2$) and C_2'/σ_2 (witnessed by $C_1' \cap C_2'$), so $C_2 = C_2'$ and $C_1 \cap C_2 = C_1 \cap C_2' \neq \varnothing$. □

7.4 Bridges

This subsection is where the source is least reliable: of its five proofs, one rests on a formula that defines the wrong relation, one proves a statement that is false, and one asserts a hypothesis it does not have. All three are cured by a single missing property, and Lemma 3.33 supplies it free.

Formalization note 7.35 (Three defects, one cause)**.** **[R]** In [13]:

(a) the symmetrisation opening the proof of Lemma 7.38 is written $\sigma(x_1, x_2, x_3, x_4) = \exists x_5 \exists x_6\, \delta(x_1, x_2, x_5, x_5) \wedge \delta(x_3, x_4, x_5, x_6)$. Clause (d) of Definition 3.27 turns the repeated x_5 into $x_1 = x_2$, so $\mathrm{pr}_{1,2}(\sigma) \subseteq 0_A$ and the relation defined is *never* a bridge, for any σ_1. It is a one-character slip: the intended relation is δ° of Lemma 3.33, which the source writes correctly 210 lines later, in the proof of Lemma 4.12.

(b) Lemma 7.36 is stated there without pair-reflexivity, and is then *false*: on $\mathbf{Z}_3 \in \mathcal{V}_4$ the relation $\delta = \{(x_1, x_2, x_3, x_4) : x_1 - x_2 + x_3 - x_4 = 0\}$ satisfies every hypothesis, and none of the six bijections $\mathbb{Z}_3 \to \mathbb{Z}_3$ carries it to $\{x_1 - x_2 = x_3 - x_4\}$ (verified by exhaustion). The step that breaks is "composing δ with itself we get $\delta_0 \supseteq \delta$".

(c) Lemma 7.39 asserts that $\mathrm{pr}_{1,2}(\delta \cap B^4 \cap (\sigma^* \times \sigma^*))$ is linked "since $\widetilde{\delta} \supseteq \sigma^*$". That inclusion bears on a different hypothesis; linkedness does not follow, and without pair-reflexivity the projection can collapse to 0_B.

Pair-reflexivity (Definition 3.32) repairs all three, and costs nothing: Lemma 7.39 is invoked exactly once in [13], on a δ°, and Lemma 7.36 exactly once from inside that proof.

Lemma 7.36. *Let σ be a congruence on $\mathbf{A}$ and δ a symmetric,* pair-reflexive *bridge from σ to σ with $\mathrm{pr}_{1,2}(\delta) = \widetilde{\delta} = A^2$. Then there is an abelian group $(G; +, -)$ with $(A/\sigma; \delta/\sigma) \cong (G; x_1 - x_2 = x_3 - x_4)$.*

Proof. By Lemma 7.12 and Hobby–McKenzie, $\mathbf{A}/\sigma$ is affine, hence polynomially equivalent to a module $\mathbf{G}$. Composing δ with itself gives $\delta_0 = \delta * \delta$, an equivalence relation on $\mathbf{A}^2$, and $\delta \subseteq \delta_0$ by Lemma 3.33 — this is the step that needs pair-reflexivity. Then δ_0/σ is a congruence of $\mathbf{G}^2$ having the diagonal as a block, so it is $\{((x_1, y_1), (x_2, y_2)) : x_1 - x_2 = y_1 - y_2\}$. Finally δ/σ is preserved by the Maltsev operation $x - y + z$ and has $\widetilde{\delta} = \mathrm{pr}_{1,2}(\delta) = A^2$, so $\mathrm{pr}_{1,2,3}(\delta) = A^3$; since the last entry of δ_0/σ is determined by the first three, counting gives $\delta = \delta_0$. ☐

Remark 7.37 (What the untwisted form really is)**.** Without pair-reflexivity the correct conclusion is $\delta = \{(x_1, x_2, x_3, x_4) : \varphi(x_1 - x_2) = x_3 - x_4\}$ for an automorphism φ of G: factoring out the diagonal exhibits δ as the preimage of a subgroup $K \leq G \times G$ that meets $G \times 0$ and $0 \times G$ trivially and is subdirect, hence a graph. Symmetry says $\varphi^2 = \mathrm{id}$; pair-reflexivity says $\varphi = \mathrm{id}$. The counterexample of Hazard 7.35(b) is $\varphi = -\mathrm{id}$, which is why it needs p odd.

Lemma 7.38. *Let δ be a bridge from $0_{\mathbf{A}}$ to $0_{\mathbf{A}}$ with $\mathrm{pr}_{1,2}(\delta) \subseteq \widetilde{\delta}$ and $\mathrm{pr}_{1,2}(\delta)$ linked. Then $\mathbf{A}$ is BA and centre free.*

Proof outline, with the supplied steps. Replace δ by $\sigma := \delta^\circ$ of Lemma 3.33 and put $\omega = \mathrm{pr}_{1,2}(\sigma) = \mathrm{pr}_{1,2}(\delta)$. Note first that $\widetilde{\delta}$ is *reflexive*, because $0_A \subseteq \mathrm{pr}_{1,2}(\delta) \subseteq \widetilde{\delta}$ by clause (c); hence $\omega \subseteq \widetilde{\delta} \subseteq \widetilde{\sigma}$, and $\widetilde{\sigma}$ is symmetric, which is what legitimises the "without loss of generality" below. Induct on $|A|$.

Suppose $B <_T \mathbf{A}$ with $T \in \{\textsc{ba}, \textsc{c}\}$. Since ω is linked, ω meets $(A \setminus B) \times B$ or $B \times (A \setminus B)$; say the first, and pick (a, b) there.

Case 1: some subalgebra $D \lneq A$ contains a and b. Let E be the block containing a, b of the transitive closure of $(\omega \cup \omega^{-1}) \cap D^2$; that closure is a congruence on $\mathbf{D}$, because the transitive closure of a subalgebra of D^2 is a subalgebra, so E is a subuniverse, and $\omega \cap E^2$ is linked by construction. *This is the existence of E, which the source asserts.* Moreover $\mathrm{pr}_{1,2}(\sigma \cap E^4) = \omega \cap E^2$: the inclusion $\subseteq$ is trivial and $\supseteq$ holds because pair-reflexivity puts (x_1, x_2, x_1, x_2) in σ for every $(x_1, x_2) \in \omega$. So the inductive hypothesis applies to $\sigma \cap E^4$ and makes E BA and centre free, while Lemma 7.13 makes $B \cap E$ a proper nonempty subuniverse of E of type T — proper because $a \in E \setminus B$, nonempty because $b \in B \cap E$. Contradiction.

Case 2: no such D. Then $\{a\} \circ \omega = A$, because $\{a\}$ is a subuniverse by idempotence, hence so is $\{a\} \circ \omega$, and it contains a and b. Put $\xi(x_1, x_2) = \sigma(a, x_1, x_2, b)$; from $(a, b, a, b), (a, a, b, b) \in \sigma$ both projections of ξ contain a and b, hence equal A. Put $C = B \circ \xi$, so $C \leq_T \mathbf{A}$ by Lemma 7.3, with $a \in C$ and $b \notin C$ — the latter because $b \in C$ would give $(a, c, b, b) \in \sigma$ with $c \in B$, and clause (d) would force $a = c \in B$.

If $T = \textsc{ba}$, applying the absorbing term to (a, b, a, b) and (a, a, b, b) gives $(a, b_1, b_2, b) \in \sigma$ with $b_1, b_2 \in B$, so $C \cap B \neq \varnothing$. If $T = \textsc{c}$, the three tuples (a, b, a, b), (a, a, a, a) and (a, a, b, b) witness

$$\sigma \cap (\{a\} \times A \times C \times B), \quad \sigma \cap (\{a\} \times C \times C \times A), \quad \sigma \cap (\{a\} \times C \times A \times B) \neq \varnothing,$$

which is hypothesis (4) of Corollary 6.14 for the ternary relation $\{(x_2, x_3, x_4) : \sigma(a, x_2, x_3, x_4)\}$ with $n = 3$, all types C and all ambients A — legitimate because $\lhd\lhd\lhd$ is reflexive. *The source displays the three tight boxes $\{a\}\times B\times C\times B$, $\{a\}\times C\times C\times C$, $\{a\}\times C\times B\times B$ instead, which are subsets of these and so do not match the corollary's shape; widening them is the missing step.* Since $n = 3$ and all types are C, every alternative of the corollary fails, so hypothesis (3) must fail: $\sigma \cap (\{a\} \times C \times C \times B) \neq \varnothing$. Put $C' = \mathrm{pr}_2$ of that set. Then $C' \leq_{\mathrm{C}} A$ by Lemma 7.3, using $\{a\} \circ \omega = A$, and hence $C' \leq_{\mathrm{C}} C$ by Lemma 7.13; and by clause (d) either $a \notin C'$ or $C \cap B \neq \varnothing$.

So in every case we have $\varnothing \neq C' <_T C <_T A$ — taking $C' = B\cap C$ when $C\cap B \neq \varnothing$. Then $|C| > 1$, and applying the inductive hypothesis to $\sigma \cap C^4$ contradicts $C' <_T C$: its hypotheses hold because $\{a\} \circ \omega = A$ makes $\omega \cap C^2$ a star centred at a, hence linked. □

Lemma 7.39. Let σ be an irreducible congruence on $\mathbf{A}$ and δ a *pair-reflexive* bridge from σ to σ with $\widetilde{\delta} \supsetneq \sigma$. Then

(a) σ^* is a congruence;
(b) B/σ is BA and centre free for every block B of σ^*;
(c) if δ is also symmetric then there is a prime p such that every block B of σ^* satisfies $(B/\sigma; (\delta \cap B^4)/\sigma) \cong (\mathbb{Z}_p^{n_B}; x_1 - x_2 = x_3 - x_4)$ for some $n_B \geq 0$.

Proof. By Lemma 3.25 we may assume $\sigma = 0_{\mathbf{A}}$. Since $\widetilde{\delta}$ is a subuniverse of $\mathbf{A}^2$ properly containing σ and stable under it, minimality gives $\sigma^* \subseteq \widetilde{\delta}$; the same argument applied to $\mathrm{pr}_{1,2}(\delta)$, which properly contains σ by clause (c), gives $\sigma^* \subseteq \mathrm{pr}_{1,2}(\delta)$.

Let B be a block of LeftLinked(σ^*) with $|B| > 1$ and put $\delta' = \delta \cap B^4 \cap (\sigma^* \times \sigma^*)$. Then $\mathrm{pr}_{1,2}(\delta') = \sigma^* \cap B^2$: for $(x_1, x_2) \in \sigma^* \cap B^2$ pair-reflexivity puts (x_1, x_2, x_1, x_2) in δ — legitimate because $\sigma^* \subseteq \mathrm{pr}_{1,2}(\delta)$ — and that tuple lies in B^4 and in $\sigma^* \times \sigma^*$. *This is the step the source asserts.* The resulting relation is linked precisely because B is a block, and is contained in $\widetilde{\delta'}$ because $\sigma^* \subseteq \widetilde{\delta}$; so Lemma 7.38 makes B BA and centre free. Then $\sigma^* \cap B^2$, being linked and subdirect on B, must be all of B^2 by Theorem 7.1. Blocks of size 1 satisfy $B^2 \subseteq \sigma^*$ trivially, so $\sigma^* =$ LeftLinked(σ^*) is an equivalence relation and hence a congruence, giving (a); and (b) follows, one-element algebras being vacuously BA and centre free.

For (c), fix a block B of σ^* with $|B| \geq 2$ and apply Lemma 7.36 to $\delta \cap B^4$, whose hypotheses now hold: $\mathrm{pr}_{1,2}(\delta \cap B^4) = B^2$ by the same pair-reflexivity computation, and $\widetilde{\delta \cap B^4} = B^2$ because $B^2 \subseteq \sigma^* \subseteq \widetilde{\delta}$. This gives an abelian group G_B with $(B; \delta \cap B^4) \cong (G_B; x_1 - x_2 = x_3 - x_4)$.

It remains to make every G_B elementary abelian for one common prime. Put $\omega = \delta \cap (\sigma^* \times \sigma^*)$, which has the same properties. From $x_1 - x_2 = x_3 - x_4$ one pp-defines $(k+1)x_1 = k\,x_2 + x_3$ for every k, by

$$((k+1)x_1 = k\,x_2 + x_3) = \exists x_4\ (k\,x_1 = (k-1)x_2 + x_4) \wedge (x_1 - x_2 = x_3 - x_4),$$

and hence, identifying x_3 with x_1, the relation $q\,x_1 = q\,x_2$ for every q. Let S be what that pp-definition defines when $x_1 - x_2 = x_3 - x_4$ is replaced by ω. If some G_B had an element of order p_1^2, or elements of coprime orders, or if two blocks had elements of different prime orders, then for a suitable q the relation S would be a subuniverse of $\mathbf{A}^2$ with $S \supsetneq \sigma$ and $\sigma^* \not\subseteq S$, contradicting the minimality of σ^*. Hence every element of every G_B has one and the same prime order p, and $G_B \cong \mathbb{Z}_p^{n_B}$. □

Formalization note 7.40 (It is minimality, not irreducibility)**.** The contradiction at the end is with the *minimality of* σ^*, which is a consequence of irreducibility; [13] says "contradicts the irreducibility of σ". The distinction matters in a formalization, where σ^* is data attached to a proof of irreducibility (Hazard 3.22).

Proof of Lemma 4.12. $(a) \Rightarrow (b)$: take $\delta = S$ of Definition 4.10, which by Lemma 4.11 is a bridge with $\widetilde{S} = \sigma^* \supsetneq \sigma$.

$(b) \Rightarrow (a)$: given a bridge δ with $\widetilde{\delta} \supsetneq \sigma$, which we may take reflexive, form δ° of Lemma 3.33; it is symmetric and pair-reflexive, and $\widetilde{\delta^\circ} \supseteq \widetilde{\delta} \supsetneq \sigma$. Now Lemma 7.39 gives items (b) and (c) of Definition 4.10, with the witness $S = \delta^\circ \cap (\sigma^* \times \sigma^*)$. □

Lemma 7.41. Let σ be a congruence on $\mathbf{A}$ and δ a reflexive bridge from σ to σ with

(i) $\delta(x_1, x_2, x_3, x_4) = \delta(x_3, x_4, x_1, x_2)$;
(ii) $(a, b, a, b), (b, a, b, a) \in \delta$ for every $(a, b) \in \mathrm{pr}_{1,2}(\delta)$;
(iii) $\mathrm{RightLinked}(\mathrm{pr}_{1,2,3}(\delta)) = A^2$.

Then $(a, a, b, b) \in \delta$ for some $a \neq b$.

Proof. Factor by σ and assume $\sigma = 0_{\mathbf{A}}$; clause (c) of Definition 3.27 then forces $|A| \geq 2$. Write $P = \mathrm{pr}_{1,2}(\delta)$, which is symmetric by (ii), and $W = \mathrm{pr}_{1,2,3}(\delta) \leq_{\mathrm{sd}} \mathbf{P} \times \mathbf{A}$. Induct on $|A|$.

Case 1: some $B <_T \mathbf{A}$ with $|B| > 1$, $T \in \{\text{BA}, \text{C}\}$. **[R]** Put $W' = \mathrm{pr}_{1,2,3}(\delta \cap B^4)$ and write $W(x_1, x_2, x_3) = \exists x_4\, \delta(x_1, x_2, x_3, x_4) \wedge A(x_1) \wedge A(x_2) \wedge A(x_3) \wedge A(x_4)$ with A the full unary relation. Replacing A by B — a *single* application of Lemma 7.3, which replaces every occurrence, so no transitivity of $\leq_{\text{C}}$ is needed — gives $W' \leq_T W$. It is nonempty, containing (b, b, b) for every $b \in B$, and its projections are $\mathrm{pr}_{1,2}(W') = P \cap B^2$, by (ii), and $\mathrm{pr}_3(W') = B$. So Lemma 7.15 makes W' linked, which is (iii) for $\delta \cap B^4$. Conditions (i) and (ii) are inherited and $\delta \cap B^4$ is reflexive. If it is a bridge, the inductive hypothesis applies and produces the required pair inside B. If it is not, then $\mathrm{pr}_{1,2}(\delta \cap B^4) = 0_B$, so by clause (d) every tuple of $\delta \cap B^4$ has the form (x, x, y, y); linkedness of W' on $|B| \geq 2$ points then forbids the perfect matching, so some $(a, a, b, b) \in \delta$ with $a \neq b$.

Case 2: some $\{a\} <_T \mathbf{A}$, $T \in \{\text{BA}, \text{C}\}$, and no such B of size > 1. By (i) the relation $\mathrm{pr}_{1,3}(\delta)$ is symmetric, and by (iii) it is linked, so $\{a\} \circ \mathrm{pr}_{1,3}(\delta) \neq \{a\}$; pick $(a, b, c, d) \in \delta$ with $c \neq a$. If $a = b$ then $c = d$ by clause (d) and we are done. If $\{a\} \circ \mathrm{pr}_{1,3}(\delta) \neq A$ then that set is a strong subuniverse containing a and c, so of size > 1, which is Case 1. Otherwise choose $(a, e, b, f) \in \delta$. By (ii), $(b, a, b, a) \in \delta$. Let g be a ternary absorbing operation for $\{a\}$ — available for $T = \text{C}$ by Lemma 4.4, which the source does not cite. Applying g to (a, e, b, f), (b, a, b, a) and (a, a, a, a) gives $(a, a, g(b, b, a), a) \in \delta$, whence $g(b, b, a) = a$ by clause (d); applying g to (b, a, b, a), (b, a, b, a) and (a, e, b, f) then gives $(a, a, b, a) \in \delta$, whence $b = a$, a contradiction.

Case 3: $\mathbf{A}$ is BA and centre free. Put $C = \{(a, b) \in P : \{(a, b)\} \times A \subseteq W\}$. By Lemma 7.20, applied to the transpose of W, $C \leq_{\text{BA}} \mathbf{P}$. If $C = P$ then in particular $(a, a) \in C$ for any a; otherwise $C <_{\text{BA}} P$ and Lemma 7.42 gives $C \cap 0_{\mathbf{A}} \neq \varnothing$. Either way $(a, a) \in C$ for some a, so for every c there is d with $\delta(a, a, c, d)$, and clause (d) forces $d = c$. Taking $c \neq a$ finishes the proof. □

Lemma 7.42 ([12, Lem. 7.2])**.** *If $0_{\mathbf{A}} \subseteq \sigma \leq \mathbf{A}^2$ and $\omega <_{\text{BA}} \sigma$, then $\omega \cap 0_{\mathbf{A}} \neq \varnothing$.*

Proof of Lemma 4.15. Put $\xi = \delta * \delta^{-1}$, which is symmetric and pair-reflexive. Each of $\mathrm{RightLinked}(\mathrm{pr}_{1,2,3}(\xi))$ and $\mathrm{RightLinked}(\mathrm{pr}_{1,2,4}(\xi))$ is a congruence containing σ, so by minimality of σ^* each is σ or contains σ^*; this gives three cases.

If both equal σ, then for $(a, b, c, d) \in \xi$ the classes c/σ and d/σ are determined by $a/\sigma, b/\sigma$; since $(a, b, a, b) \in \xi$ we get $(a, c), (b, d) \in \sigma$, so ξ is trivial. *The source then asserts that the same determinacy holds for δ, "since $\mathrm{pr}_{1,2}(\delta) = \mathrm{pr}_{3,4}(\delta)$". The bridge between the two is a counting argument, and this is where that hypothesis is spent:* on σ^*/σ, which is finite, δ induces a bipartite relation whose left neighbourhoods are nonempty, pairwise disjoint — that is what the case gives — and cover the right side; equal finite cardinalities force every neighbourhood to be a singleton, so the induced relation is a bijection and determinacy holds in both directions. With that, the two auxiliary relations ζ_1, ζ_2 of the source are bridges when they need to be, and $\widetilde{\zeta_1} = \widetilde{\zeta_2} = \sigma$ — forced, since σ is PC — yields $\mathrm{pr}_{1,3}(\delta) = \sigma$ or $\mathrm{pr}_{1,4}(\delta) = \sigma$, and by the same argument with x_1, x_2 interchanged, $\mathrm{pr}_{2,4}(\delta) = \sigma$ or $\mathrm{pr}_{2,3}(\delta) = \sigma$. *Two of the four combinations must now be excluded, which the source does not do:* $\mathrm{pr}_{1,3} = \mathrm{pr}_{2,3} = \sigma$ would give $\mathrm{pr}_{1,2}(\delta) \subseteq \sigma$, against $\mathrm{pr}_{1,2}(\delta) = \sigma^* \supsetneq \sigma$, and likewise for $\mathrm{pr}_{1,4} = \mathrm{pr}_{2,4}$. The two surviving combinations are the conclusion, the reverse inclusion following from stability under σ.

If $\mathrm{RightLinked}(\mathrm{pr}_{1,2,3}(\xi)) \supsetneq \sigma^*$, choose a block B of it that is not a σ-block. Then $a, b, d \in B$ whenever $c \in B$ and $(a, b, c, d) \in \xi$, so $\xi \cap (A^2 \times B \times A) \subseteq B^4$ and therefore $\mathrm{pr}_{1,2,3}(\xi \cap B^4) = \mathrm{pr}_{1,2,3}(\xi) \cap (A^2 \times B)$, which is linked. *If $\sigma^* \cap B^2 \supsetneq \sigma \cap B^2$ then $\xi \cap B^4$ is a bridge and Lemma 7.41 applies, contradicting PC-ness via Lemma 4.12. If not, the lemma cannot be cited — it is not a bridge — and the case must be inlined: clause (d) then puts both pairs of every tuple in σ, and linkedness on a B that is not a σ-block produces $(x_1, x_1, x_3, x_3) \in \xi$ with $(x_1, x_3) \notin \sigma$ directly.* The third case is the mirror image, using $\mathrm{pr}_{1,2,4}$. □

Formalization note 7.43 (Two more hedges)**.** The third case above is "this case can be considered in the same way as Case 2" in [13]. It is genuinely the mirror: $(a, b, a, b) \in \xi$ places b in the block, and $\sigma^* \subseteq \mathrm{RightLinked}$ places a and c. Writing it out is mechanical.

7.5 Types interaction

Lemma 7.44 ([12, Lem. 8.19])**.** *Let $\omega, \sigma_1, \sigma_2$ be congruences on $\mathbf{A}$ with $\omega \cap \sigma_1 = \omega \cap \sigma_2$ and $\omega \setminus \sigma_1 \neq \varnothing$. Then there is a bridge δ from σ_1 to σ_2 with $\widetilde{\delta} = \sigma_1 \circ \sigma_2$.*

Proof. Put $\delta(x_1, x_2, y_1, y_2) = \exists z_1 \exists z_2\, \sigma_1(x_1, z_1) \wedge \sigma_1(x_2, z_2) \wedge \omega(z_1, z_2) \wedge \sigma_2(z_1, y_1) \wedge \sigma_2(z_2, y_2)$. Clauses (a) and (b) are visible. For (d): if $(x_1, x_2) \in \sigma_1$ then $(z_1, z_2) \in \sigma_1 \cap \omega = \sigma_2 \cap \omega$, so $(y_1, y_2) \in \sigma_2$; *and conversely*, if $(y_1, y_2) \in \sigma_2$ then $(z_1, z_2) \in \omega \cap \sigma_2 = \omega \cap \sigma_1$, so $(x_1, x_2) \in \sigma_1$ — the direction the source omits, and the one that spends the other half of the hypothesis. For (c): $\sigma_1 \subseteq \mathrm{pr}_{1,2}(\delta)$ by taking $z_i = y_i = x_1$, and choosing $(a, b) \in \omega \setminus \sigma_1$ gives $(a, b, a, b) \in \delta$ with $(a, b) \notin \sigma_1$ and, since $\omega \setminus \sigma_1 = \omega \setminus \sigma_2$, also $(a, b) \notin \sigma_2$. Finally $\widetilde{\delta} = \sigma_1 \circ \sigma_2$: the inclusion $\supseteq$ by taking $z_1 = z_2$, using reflexivity of ω, and $\subseteq$ by reading off $(x, z_1) \in \sigma_1$, $(z_1, y) \in \sigma_2$. □

Lemma 7.45. Let $C_1 <^{A}_{T_1(\sigma_1)} B_1 \lhd\lhd\lhd A$ and $C_2 <^{A}_{T_2(\sigma_2)} B_2 \lhd\lhd\lhd A$ with $T_1, T_2 \in \{\textsc{ba}, \textsc{c}, \textsc{s}, \textsc{d}\}$, $C_1 \cap B_2 \neq \varnothing$, $B_1 \cap C_2 \neq \varnothing$ and $C_1 \cap C_2 = \varnothing$. Then $T_1 = T_2 \in \{\textsc{ba}, \textsc{c}, \textsc{d}\}$; and if $T_1 = T_2 = \textsc{d}$ there is a bridge δ from σ_1 to σ_2 with $\widetilde{\delta} = \sigma_1 \circ \sigma_2$.

Proof. No S. If $T_1 = \textsc{s}$, pick $S_1 \subseteq C_1$ with $S_1 <_{\textsc{ba},\textsc{c}} B_1$. Since $C_2 \lhd\lhd\lhd A$ — the chain for B_2 extended by one D step — and $B_1 \cap C_2 \neq \varnothing$, Lemma 7.30(s) gives $S_1 \cap C_2 \neq \varnothing$, hence $C_1 \cap C_2 \neq \varnothing$. Symmetrically for T_2.

Both absorbing. If $T_1, T_2 \in \{\textsc{ba}, \textsc{c}\}$ then by Lemma 7.13 $C_1 \cap B_2 \leq_{T_1} B_1 \cap B_2$ and $B_1 \cap C_2 \leq_{T_2} B_1 \cap B_2$; both are proper, since $C_1 \cap B_2 = B_1 \cap B_2$ would put $B_1 \cap C_2$ inside $C_1 \cap C_2 = \varnothing$. Their intersection is empty, so Lemma 7.22 gives $T_1 = T_2 \in \{\textsc{ba}, \textsc{c}\}$.

Mixed. Let $T_1 \in \{\textsc{ba}, \textsc{c}\}$ and $T_2 = \textsc{d}$ (the mirror case, which [13] does not mention, is identical). By Lemma 7.30(d), $(B_1 \cap B_2)/\sigma_2$ has size 1 or is all of B_2/σ_2. In the first case the single block is the one defining C_2, because $B_1 \cap C_2 \neq \varnothing$; then $B_1 \cap B_2 = B_1 \cap C_2$, so $C_1 \cap B_2 \subseteq C_1 \cap C_2 = \varnothing$, a contradiction. In the second, Lemmas 7.13 and 7.5 give $(C_1 \cap B_2)/\sigma_2 \leq_{T_1} B_2/\sigma_2$, and this is *proper*: equality would make the σ_2-block defining C_2 meet C_1, i.e. $C_1 \cap C_2 \neq \varnothing$. So B_2/σ_2 has a proper nonempty strong subuniverse, contradicting Definition 4.21(iii).

Both D. Let ω be the intersection of the dividing congruences of the chosen chains for $B_1 \lhd\lhd\lhd A$ and $B_2 \lhd\lhd\lhd A$. Apply Lemma 7.30(d) to B_1 and C_2 with the congruence σ_1 — *to C_2, not to B_2; against B_2 the step does not follow.* Its second alternative, $(B_1 \cap C_2)/\sigma_1 =$

B_1/σ_1, would put a point of C_2 in the σ_1-block defining C_1; so $|(B_1 \cap C_2)/\sigma_1| = 1$ and the "moreover" clause gives $\sigma_1 \supseteq \omega \cap \sigma_2$, because the chain witnessing $C_2 \lhd\lhd\lhd A$ is the chain for B_2 extended by the step $C_2 <_{\mathrm{D}(\sigma_2)} B_2$ (Hazard 4.25). Symmetrically $\sigma_2 \supseteq \omega \cap \sigma_1$, so $\omega \cap \sigma_1 = \omega \cap \sigma_2$. Finally, for $c_1 \in C_1 \cap B_2$ and $c_2 \in B_1 \cap C_2$ we have $(c_1, c_2) \in \omega$, since both lie in $B_1 \cap B_2$ and each B_i sits inside a single block of every dividing congruence of its own chain; and $(c_1, c_2) \notin \sigma_1$, since $c_2 \in B_1$ and $c_2\, \sigma_1\, c_1 \in C_1$ would give $c_2 \in C_1 \cap C_2$. Now Lemma 7.44 applies. □

7.6 Factorisation

Lemma 7.46. *Let $R \leq_{\mathrm{sd}} \mathbf{A}_1 \times \mathbf{A}_2$ with $R \cap (B_1 \times B_2) \neq \varnothing$, $B_i \lhd\lhd\lhd A_i$, σ a congruence on $\mathbf{A}_1$ with B_1/σ BA and centre free. If some $c \in A_2$ has $(E \times \{c\}) \cap R \neq \varnothing$ for every $E \in B_1/\sigma$, then such a c may be chosen in B_2.*

Proof outline. If c may already be chosen in B_2 there is nothing to do; otherwise walk down a chain for $B_2 \lhd\lhd\lhd A_2$ and let $B_2'' <_{T(\delta)} B_2'$ be the first step at which the choice fails, so $B_2 \subseteq B_2''$. Put S' and S'' for the sets of σ-class tuples of length $|A_1|$ witnessed by some c in B_2', resp. B_2''. A single good $c \in B_2'$ makes $S' = (B_1/\sigma)^{|A_1|}$, and $S'' \neq \varnothing$ because $R \cap (B_1 \times B_2) \neq \varnothing$ and $B_2 \subseteq B_2''$ — *this is where that hypothesis is spent.* For $T \in \{\mathrm{BA}, \mathrm{C}\}$, Lemmas 7.3 and 7.7 then put a strong subuniverse on B_1/σ, a contradiction; for $T = \mathrm{S}$ the same follows once $R \cap (B_1 \times D_2) \neq \varnothing$, which Lemma 7.30(s) supplies inside $\mathbf{R}$.

For $T = \mathrm{D}$ one builds $S \leq (B_1/\sigma)^{|A_1|} \times B_2'/\delta$ and finds a full column over $d = c/\delta$, while $(B_1/\sigma)^{|A_1|} \times B_2''/\delta \not\subseteq S$ by minimality. Corollary 7.32 then computes $\mathrm{pr}_{|A_1|+1}(S)$. **[R]** It offers three alternatives and the source takes the third without comment. "Empty" is excluded by $R \cap (B_1 \times B_2) \neq \varnothing$. "Size one" is excluded thus: the single δ-class would have to be $[c]_\delta$, since c has an R-partner in B_1; and $B_2'' = B_2' \cap E$ for a δ-block E, so either $E = [c]_\delta$, putting c in B_2'' against the minimal choice of B_2', or $E \cap [c]_\delta = \varnothing$, making $R \cap (B_1 \times B_2'') = \varnothing$ against $B_2 \subseteq B_2''$. So S is a central relation, and Lemmas 7.18 and 7.7 contradict the hypotheses. □

Lemma 7.47. *Under the hypotheses of Lemma 7.46, if* $(\mathrm{LeftLinked}(R) \cap B_1^2)/\sigma = (B_1/\sigma)^2$ *then* $\mathrm{LeftLinked}(R \cap (B_1 \times B_2))/\sigma = (B_1/\sigma)^2$.

Proof outline. Write $R_n = (R \circ R^{-1})^n$, which is reflexive and symmetric because R is subdirect, so $R_{2n} = R_n \circ R_n^{-1}$ — *this is why the source restricts to $n = 2^k$.* If $(R_1 \cap B_1^2)/\sigma$ is already full, Lemma 7.20 applied to $S = \{(a/\sigma, b) : a \in B_1, (a,b) \in R\}$, subdirect over $(B_1/\sigma) \times \mathrm{pr}_2(S)$ rather than over $(B_1/\sigma) \times A_2$, produces the required c, which Lemma 7.46 moves into B_2; two classes are then linked through c. Otherwise take the largest $n = 2^k$ at which fullness fails, apply Lemma 7.20 to R_n, push c into B_1, and conclude by Lemma 7.18 that $(R_n \cap B_1^2)/\sigma$ is full after all. □

Lemma 7.48. Let σ be a dividing congruence for $B \lhd\lhd\lhd A$, let δ be a congruence on $\mathbf{A}$ with $(\delta \cap B^2)/\sigma \neq B^2/\sigma$, and let ω be the intersection of the dividing congruences of the chosen chain for $B \lhd\lhd\lhd A$. Then (1) $\delta \cap B^2 \subseteq \sigma \cap B^2$; (2) $\sigma \supseteq \delta \cap \omega$; (3) $(\delta \vee (\sigma \cap \omega)) \cap B^2 = \sigma \cap B^2$; (4) $(\delta \vee (\sigma \cap \omega)) \cap \omega = \sigma \cap \omega$.

Proof. Throughout we use $B^2 \subseteq \omega$, which holds because each B_i in the chain lies inside a single block of its own dividing congruence, and B is contained in every B_i. *The source uses this at least four times and never states it.*

(2). The hypothesis gives σ-classes C_1, C_2 meeting B with $((C_1 \cap B) \times (C_2 \cap B)) \cap \delta = \varnothing$. Then $C_1 \cap B <_{\mathrm{D}(\sigma)} B$, and Lemma 7.29 with $m = k$ and $B_k = C_1 \cap B$ gives $((C_1 \cap B) \circ \delta \cap B)/\sigma = \{C_1\}$, since the value B/σ is excluded by the choice of C_2; its second clause is $\sigma \supseteq \delta \cap \omega$.

(1) follows, since $\delta \cap B^2 \subseteq \delta \cap \omega \subseteq \sigma$.

(4). Put $\delta' = \delta \vee (\sigma \cap \omega)$, which is LeftLinked$(\delta \circ (\sigma \cap \omega))$. If $(\delta' \cap B^2)/\sigma$ were full then Lemma 7.47 would make LeftLinked of the restricted relation full, so some single linking step would cross σ-classes: there would be $a_1, a_2, c \in B$ and $b_1, b_2 \in A$ with $(a_i, b_i) \in \delta$, $(b_i, c) \in \sigma \cap \omega$ and $(a_1, a_2) \notin \sigma$. But $a_1, a_2, c \in B$ gives $(a_i, c) \in \omega$, hence $(a_i, b_i) \in \omega$, hence $(a_i, b_i) \in \sigma$ by (2), hence $(a_1, a_2) \in \sigma$ — a contradiction. So $(\delta' \cap B^2)/\sigma \neq (B/\sigma)^2$, and applying (2) to δ' gives $\sigma \supseteq \delta' \cap \omega$; with $\delta' \supseteq \sigma \cap \omega$ this is (4).

(3) follows from (4) by intersecting with $B^2 \subseteq \omega$. □

Lemma 7.49. ***[R]*** *Let $B \lhd\lhd\lhd A$ and let σ be a congruence on* $\mathbf{A}$ *with $|B/\sigma| > 1$ and B/σ BA and centre free. Then there is a dividing congruence $\delta \supseteq \sigma$ for $B \leq A$.*

Proof. Choose $\delta \supseteq \sigma$ maximal with $|B/\delta| > 1$; it exists because σ qualifies. Then B/δ is BA and centre free, since the preimage under $B/\sigma \twoheadrightarrow B/\delta$ of a proper nonempty strong subuniverse would be one of B/σ.

It remains to see that δ is irreducible. Suppose $\delta = S_1 \cap \cdots \cap S_k$ with each $S_i \supsetneq \delta$ a subuniverse of $\mathbf{A}^2$ *stable under* δ. Some S_i has $B^2 \not\subseteq S_i$, else $B^2 \subseteq \delta$. Now LeftLinked(S_i) is a congruence containing $S_i \supsetneq \delta \supseteq \sigma$, so maximality of δ forces $B^2 \subseteq$ LeftLinked(S_i), i.e. (LeftLinked$(S_i) \cap B^2)/\delta = (B/\delta)^2$. By Lemma 7.47, applied *with the congruence* δ, LeftLinked$(S_i \cap B^2)/\delta = (B/\delta)^2$, so $(S_i \cap B^2)/\delta$ is linked. It is also *proper*: stability under δ makes S_i a union of products of δ-blocks, so $(S_i \cap B^2)/\delta = (B/\delta)^2$ would give $B^2 \subseteq S_i$. Theorem 7.1 then produces a proper nonempty strong subuniverse of B/δ, a contradiction. □

Formalization note 7.50 (Two slips, one cause)**.** [13] factors by σ in the last four lines and states the conclusion for B/δ; those do not match, and with σ the properness step has no support. Reading δ throughout is plainly what was meant — B/δ was shown BA and centre free two lines earlier for exactly this use. Properness then needs the stability clause, which the source drops when it introduces the S_i; stability is the only part of Definition 3.21 doing work here.

Lemma 7.51. *Let δ be a congruence on* $\mathbf{A}$ *and $C <^{A}_{T(\sigma)} B \lhd\lhd\lhd A$ with $T \in \{\textsc{pc}, \textsc{l}\}$. Then $C/\delta = B/\delta$, or $C/\delta <_{\textsc{s}} B/\delta$, or $C/\delta <^{A/\delta}_{T} B/\delta$; and in the last case $\delta \cap B^2 \subseteq \sigma \cap B^2$.*

Proof outline. Put $R = \{(a/\sigma, a/\delta) : a \in B\}$ and note $R \circ R^{-1} = (\delta \cap B^2)/\sigma$, so LeftLinked$(R) = (B/\sigma)^2$ exactly when $(\delta \cap B^2)/\sigma = B^2/\sigma$. In that case Lemma 7.20 yields $U \leq_{\textsc{ba,c}} B/\delta$ with $(B/\sigma) \times U \subseteq R$, and $U \subseteq C/\delta$, giving the first two alternatives.

Otherwise Lemma 7.48(4) gives $\sigma \cap \omega = \delta' \cap \omega$ for $\delta' = \delta \vee (\sigma \cap \omega)$, and $B/\delta' \cong B/\sigma$ by (3), so B/δ' is BA and centre free and Lemma 7.49 produces a dividing congruence $\delta'' \supseteq \delta'$ for B. Applying Lemma 7.48(2) to δ'' — legitimate because $\sigma \cap B^2 \subseteq \delta''$ and $|B/\delta''| > 1$ give $(\delta'' \cap B^2)/\sigma \neq B^2/\sigma$ — yields $\delta'' \cap \omega = \sigma \cap \omega$, hence $\delta'' \cap B^2 = \sigma \cap B^2$, so the δ''-classes and σ-classes agree on B and $C/\delta <^{A/\delta}_{\mathcal{T}(\delta''/\delta)} B/\delta$. Finally $\mathcal{T} = T$: $\omega \setminus \delta'' \neq \varnothing$, since $\omega \supseteq B^2$ and $|B/\sigma| > 1$, so Lemma 7.44 gives a bridge from δ'' to σ, and Lemma 4.14 in both directions makes the two congruences of the same type. The last clause holds because $\delta \cap B^2 \subseteq \delta' \cap B^2 = \sigma \cap B^2$. □

7.7 The headline properties

Proof of Lemma 6.2. If $\mathbf{B}$ has a proper nonempty binary absorbing or central subuniverse, take it. Otherwise $\mathbf{B}$ is BA and centre free and $|B| > 1$, so Lemma 7.49 with $\sigma = 0_{\mathbf{A}}$ gives a dividing congruence δ for B; take $C = B \cap E$ for any block E *of* δ with $B \cap E \neq B$, which exists because $|B/\delta| > 1$. Its type is L or PC according to δ. □

Proof of Lemma 6.12. (t). For $T \in \{\textsc{ba}, \textsc{c}\}$ this is Lemma 7.13. For $T = \textsc{s}$, take $E \subseteq C$ with $E <_{\textsc{ba,c}} B$; if $C \cap D = \varnothing$ there is nothing to prove, and otherwise $B \cap D \neq \varnothing$, so

Lemma 7.30(s) gives $E \cap D \neq \varnothing$ and Lemma 7.13 gives $E \cap D <_{\text{BA,C}} B \cap D$. For $T \in \{\text{PC}, \text{L}\}$, Lemma 7.30(d) makes $(B \cap D)/\sigma$ empty, a point, or all of B/σ; the three cases give $C \cap D = \varnothing$, $C \cap D \in \{\varnothing, B \cap D\}$, and $C \cap D <_{T(\sigma)} B \cap D$ respectively.

(i). Apply (t) to each step of a chain for $B \lhd\lhd\lhd^A A$, obtaining $B \cap D \;\dot\lhd\dot\lhd\dot\lhd^A\; D$; with $D \lhd\lhd\lhd A$ this gives $B \cap D \;\dot\lhd\dot\lhd\dot\lhd\; A$. □

Formalization note 7.52 (The statement is weaker than the proof)**.** **[R]** The proof of (i) establishes $B \cap D \;\dot\lhd\dot\lhd\dot\lhd^A\; D$ and then weakens it. Three later arguments use the stronger form while citing (i): Corollary 6.9(b) and (m), and Lemma 6.17. State (i) as $B \cap D \;\dot\lhd\dot\lhd\dot\lhd^A\; D$ and derive $B \cap D \;\dot\lhd\dot\lhd\dot\lhd\; A$ as a corollary.

Proof of Lemma 6.4. (bt) is the reverse-homomorphism lemma and (b) follows by iteration. (fs) is Lemma 7.5. (ft) is Lemma 7.51 for $T \in \{\text{PC}, \text{L}, \text{D}\}$ and follows from (fs) for $T \in \{\text{BA}, \text{C}, \text{S}\}$ — *the source cites only the former, which does not cover the absorption types*; (f) then follows by iteration, all three outcomes of (ft) staying inside the type list of Definition 4.24.

(fm). Let $C = C_1 \cap \cdots \cap C_t$ with $C_i <^A_{T(\sigma_i)} B$ and induct on t. For $t = 1$ this is (ft), and *the hypothesis that $f(B)$ is* S*-free is exactly what kills its middle alternative*, which $\mathcal{M}T$ cannot absorb. If $C_i/\delta = B/\delta$ for some i, say $i = t$, put $D_j = C_j \cap C_t$; by Lemma 6.12(t) $D_j \leq^A_{T(\sigma_j)} C_t$, and the inductive hypothesis applied to $C \leq_{\mathcal{M}T} C_t$ gives the claim. Otherwise every $C_i/\delta <^{A/\delta}_T B/\delta$, and $\delta \cap B^2 \subseteq \sigma_i \cap B^2$ for every i by the last clause of Lemma 7.51. That is what makes the images meet correctly: $(C_i \circ \delta) \cap B = C_i$, so a δ-class E lying in every C_i/δ has $E \cap B \subseteq C_1 \cap \cdots \cap C_t$, whence

$$(C_1 \cap \cdots \cap C_t)/\delta = C_1/\delta \cap \cdots \cap C_t/\delta,$$

the inclusion $\subseteq$ being trivial. This is the one place where the failure of images to commute with intersections (§2.1, item viii) is repaired, and it should be extracted as a lemma in its own right. (bm) follows from (bt) by intersecting. □

Theorem 7.53 (Stable intersection)**.** Let $C_i <^A_{T_i(\sigma_i)} B_i \lhd\lhd\lhd A$ with $T_i \in \{\text{BA}, \text{C}, \text{S}, \text{L}, \text{PC}\}$ for $i \in [n]$, $n \geq 2$, and suppose $\bigcap_i C_i = \varnothing$ while $B_j \cap \bigcap_{i \neq j} C_i \neq \varnothing$ for every j. Then all T_i are BA; or all are L and for all k, ℓ there is a bridge δ from σ_k to σ_ℓ with $\widetilde{\delta} = \sigma_k \circ \sigma_\ell$; or $n = 2$ and $T_1 = T_2 = \text{C}$; or $n = 2$, $T_1 = T_2 = \text{PC}$ and $\sigma_1 = \sigma_2$.

Proof. For $n = 2$ combine Lemmas 7.45 and 4.14. For general n put $B_2' = B_2 \cap C_3 \cap \cdots \cap C_n$ and $C_2' = C_2 \cap B_2'$; then $C_2' <_{T_2(\sigma_2)} B_2' \lhd\lhd\lhd A$ by Lemma 6.12, properly because $C_2' = B_2'$ would make $C_1 \cap B_2'$ empty. The three hypotheses of the pair case hold, so $T_1 = T_2$ and, for the dividing types, the bridge exists.

If $T_1 = \text{PC}$ then $\sigma_1 \circ \sigma_2 = \sigma_1 = \sigma_2$, since otherwise composing the bridge with its transpose gives a bridge from σ_1 to itself with $\widetilde{\ } \supsetneq \sigma_1$, making σ_1 linear. With $\sigma_1 = \sigma_2 =: \sigma$, the blocks E_1, E_2 defining C_1, C_2 are *distinct* — equal blocks would put the point of $B_1 \cap C_2'$ in $C_1 \cap C_2'$ — so $C_1 \cap C_2 = \varnothing$; applying this to every pair and then the hypothesis at $j = 3$ forces $n = 2$.

[R] For $T_1 = \text{C}$ the source offers nothing, and the argument above has no analogue: there is no congruence and no pair of blocks. Here is the missing proof. Suppose $n \geq 3$, put $D = \bigcap_i B_i$ and $E_i = C_i \cap D$. Then D is a nonempty subuniverse, containing $B_1 \cap C_2 \cap \cdots \cap C_n$; each $E_i \leq_{\text{C}} D$ by Lemma 7.13, and properly, since $E_j = D$ would give $\bigcap_i E_i = \bigcap_{i \neq j} E_i \neq \varnothing$. Moreover $\bigcap_i E_i = \varnothing$ and, because $C_i \subseteq B_i$,

$$\textstyle\bigcap_{i \neq j} E_i = D \cap \bigcap_{i \neq j} C_i = B_j \cap \bigcap_{i \neq j} C_i \neq \varnothing.$$

Now the diagonal $R = \{(a, \dots, a) : a \in D\} \leq D^n$ is a subuniverse by idempotence, and the two displays say exactly that R is $(E_1, \dots, E_n)$-essential. By [14, Lem. 6.11] no such relation exists for $n \geq 3$ when the E_i are central. Hence $n = 2$. □

Formalization note 7.54 (What the repair costs)**.** [13] argues $n = 2$ only for PC, then asserts case (c). The repair needs [14, Lem. 6.11], whose printed proof cites *itself* three times where it means the preceding lemma; a self-contained rendering must supply both. Every ingredient is already formalised in the predecessor Lean development: essentiality, the Barto–Kazda criterion, and "central implies ternary absorbing".

Proof of Corollary 6.14. Pull back along the projections $f_i : \mathbf{R} \to \mathbf{A}_i$ using Lemma 6.4(b),(bt), apply Theorem 7.53 inside $\mathbf{R}$, and translate $\widetilde{\delta} = \sigma'_k \circ \sigma'_\ell$ back as $\sigma_k \circ \mathrm{pr}_{k,\ell}(R) \circ \sigma_\ell$. ☐

Proof of Lemma 6.17. With $C = \bigcap_{i \le n} C_i$ and $D_j = \bigcap_{i \le j} C_i$, Lemma 6.12(t) gives $D_{j+1} \le^A_T D_j$, using $D_j \lhd\lhd\lhd A$ from part (i); collapsing the equalities gives the chain. ☐

Proof of Lemma 4.17. **[R]** By Lemma 4.12 there is a bridge δ from σ to σ with $\widetilde{\delta} \supsetneq \sigma$, which we may take reflexive; replace it by δ°, symmetric and pair-reflexive. Lemma 7.39(c) applies, and $\sigma^* = A^2$ is a congruence with the single block A, so $A/\sigma \cong \mathbb{Z}_p^m$ as a group. Now $m = 0$ gives $\sigma = A^2$, impossible for an irreducible congruence (§2.1, item iv). And $m \ge 2$ contradicts irreducibility: A/σ is affine, so the basic operation acts as $\sum_i c_i x_i$ with $\sum_i c_i = 1$, and the coordinate congruences $\theta_i = \{(x, y) : x_i = y_i\}$ are linear subspaces of $(\mathbb{Z}_p^m)^2$, hence subuniverses of $(\mathbf{A}/\sigma)^2$, each properly containing $0_{\mathbf{A}/\sigma}$, with $\bigcap_i \theta_i = 0_{\mathbf{A}/\sigma}$ — so $0_{\mathbf{A}/\sigma}$ is reducible, and by Lemma 3.25 so is σ. Hence $m = 1$, and the group isomorphism is an isomorphism of algebras by Proposition 2.7. ☐

Lemma 7.55. *Let $C_1 <^A_{\mathcal{M}T} B_1 \lhd\lhd\lhd A$ with $T \in \{\mathrm{PC}, \mathrm{L}, \mathrm{D}\}$, let $B_2 \lhd\lhd\lhd A$, $C_1 \cap B_2 = \varnothing$ and $B_1 \cap B_2 \ne \varnothing$. Then $(C_1 \circ (\omega_1 \cap \cdots \cap \omega_s)) \cap B_2 = \varnothing$, where $\omega_1, \dots, \omega_s$ are all congruences of type T on $\mathbf{A}$ with $\omega_i^* \supseteq B_1^2$.*

Proof. Downward induction on $|B_1|$, starting at $B_1 = A$. Write $C_1 = \bigcap_i C_1^i$ with $C_1^i <^A_{\mathrm{D}(\sigma_i)} B_1$, so each σ_i is among the ω_j and $C_1^i = (C_1^i \circ \sigma_i) \cap B_1$; by Corollary 6.8(e),(f), $C_1^i \circ \sigma_i <^A_{\mathrm{D}(\sigma_i)} B_1 \circ \sigma_i \lhd\lhd\lhd A$. Since $\bigcap_i (C_1^i \circ \sigma_i) \cap B_1 \cap B_2 = C_1 \cap B_2 = \varnothing$, walk down a chain from A to B_1: either the intersection with B_2 is already empty at A, and then $C_1 \circ \bigcap_j \omega_j \subseteq \bigcap_i (C_1^i \circ \sigma_i)$ finishes; or there is a first step $B'_1 <_{\mathrm{D}} B''_1$ at which it becomes empty, and the inductive hypothesis at B''_1, together with $C_1 \subseteq B'_1$ and $\bigcap_j \omega_j \subseteq \sigma_i$, finishes. ☐

Formalization note 7.56 (A citation that does no work)**.** [13] invokes Theorem 7.53 for the dichotomy above. No theorem is consumed: the intersection is empty at B_1 by hypothesis, so either it is empty at A or there is a first step where it becomes so.

Proof of Lemma 6.19. **[R]** Write $\bar{X}$ for X/σ and $C_1 = \bigcap_{j \le t} C_1^j$ as above. By Lemma 6.4(ft) each $\bar{C}_1^j$ is $\bar{B}_1$, or $<_{\mathrm{S}} \bar{B}_1$, or $<_T \bar{B}_1$. Choose $F \subseteq \{\bar{C}_1^1, \dots, \bar{C}_1^t\}$ *minimal* with $\bigcap F \cap \bar{B}_2 = \varnothing$; it exists because the whole family works, it is nonempty because $\bigcap \varnothing = \bar{B}_1$ meets $\bar{B}_2$, and no member of it equals $\bar{B}_1$.

Walk down a chain for $\bar{B}_2 \lhd\lhd\lhd \bar{A}$ to the first step $\bar{B}'_2 <_{T'} \bar{B}''_2$ with $\bigcap F \cap \bar{B}'_2 = \varnothing$ and $\bigcap F \cap \bar{B}''_2 \ne \varnothing$. Apply Theorem 7.53 to the family $F \cup \{\bar{B}'_2\}$, with ambient $\bar{B}_1$ for the members of F and $\bar{B}''_2$ for $\bar{B}'_2$. Its leave-one-out hypothesis holds in both shapes: omitting a member of F is minimality of F together with $\bar{B}_2 \subseteq \bar{B}'_2$, and omitting $\bar{B}'_2$ is the choice of the step. So all the types coincide, and the theorem's four conclusions admit only BA, C, L and PC: *no member of F has type* S.

Hence every member of F is $<_T \bar{B}_1$, so $\bigcap F <_{\mathcal{M}T} \bar{B}_1$ — properly, since $\bigcap F \cap \bar{B}_2 = \varnothing$ while $\bar{B}_1 \cap \bar{B}_2 \ne \varnothing$ — with the ambient still $\bar{B}_1$. Lemma 7.55 inside $\mathbf{A}/\sigma$ now returns congruences δ_i of type T with $\delta_i^* \supseteq \bar{B}_1^2$ and $(\bigcap F \circ \bigcap_i \delta_i) \cap \bar{B}_2 = \varnothing$. Their preimages $\omega_i \supseteq \sigma$ satisfy $\omega_i^* \supseteq B_1^2$ and, since C_1 is contained in the preimage of $\bigcap F$, also $(C_1 \circ \bigcap_i \omega_i) \cap B_2 = \varnothing$. Maximality of σ gives $\sigma = \bigcap_i \omega_i$. ☐

Formalization note 7.57 (What the source does instead). [13] splits on whether $\bar{B}_1$ is S-free and, in the negative case, argues that a BA-and-central $E <_{\mathrm{S}} \bar{B}_1$ meets both $\bar{C}_1$ and $\bar{B}_2$, "hence" those two meet. They need not: two sets meeting a third need not meet each other, and the configuration reproduces itself on E, so no iteration of those facts can close it. Passing to a minimal subfamily avoids the split altogether — minimality is precisely the leave-one-out hypothesis the stable-intersection theorem wants, and its conclusion has no room for S.

8 The main statements

This section is Zhuk's §§3.3–3.4. Everything converges here. Theorem 8.1 is the single induction that carries the whole proof; the three theorems the algorithm consumes are corollaries of it.

8.1 The three claims the algorithm needs

The algorithm of §9 is justified by exactly three facts. Informally:

(IC1) every domain of size ≥ 2 has a strong subset, or an equivalence σ with $D_x/\sigma \cong \mathbb{Z}_p$;
(IC2) reducing a variable's domain to a strong subset of it does not destroy solvability, for a consistent enough instance;
(IC3) for a linked, consistent enough, strong-subset-free instance, the set of "linear cosets" containing a solution is empty, full, or a codimension-one affine subspace.

(IC1) is Lemma 6.2 together with Lemma 4.17. (IC2) is Theorem 8.6. (IC3) is Theorem 8.8.

8.2 The main induction

Theorem 8.1 (Main inductive claim)**.** Let $\mathcal{I}$ be an irreducible cycle-consistent instance and let $D^{(1)}$ be a 1-consistent reduction for $\mathcal{I}$ with $D^{(1)} \lhd\lhd\lhd D$.

(1) If $\mathcal{I}$ is crucial in $D^{(1)}$ then (1a) holds, and (1b) or (1c) holds:
 (1a) every constraint of $\mathcal{I}$ has the parallelogram property;
 (1b) $\mathcal{I}$ is a connected linear-type instance whose solution set is subdirect;
 (1c) there is an expanded covering $\mathcal{J}$ of $\mathcal{I}$, crucial in $D^{(1)}$, with a linked connected subinstance Υ whose solution set is not subdirect.

(2) If $D^{(2)} \leq_T D^{(1)}$ is a 1-consistent reduction for $\mathcal{I}$ with $T \in \{\text{BA}, \text{C}\}$, and $\mathcal{I}^{(1)}$ has a solution, then $\mathcal{I}^{(2)}$ has a solution.

Formalization note 8.2 (The induction, stated precisely)**.** The proof reads "we prove the claim by induction on the size of $D^{(1)}$", and this one line hides the entire well-foundedness argument. Making it precise is the first thing a formalization must do, because *the two halves of the conclusion are proved by mutual induction and each invokes the other at a smaller reduction.*

The measure is

$$\mu(D^{(1)}) = \sum_{x \in \mathrm{Var}(\mathcal{I})} |D_x^{(1)}| \in \mathbb{N},$$

and the statement being proved by strong induction on μ is the conjunction of (1) and (2), *universally quantified over* $\mathcal{I}$:

$$P(k): \quad \forall \mathcal{I},\ \forall D^{(1)} \text{ with } \mu(D^{(1)}) \leq k,\ (1) \text{ and } (2) \text{ hold.}$$

The quantifier over $\mathcal{I}$ must be *inside*, because every use of the inductive hypothesis changes the instance: (2) applies it to a weakening $\mathcal{I}'$ of $\mathcal{I}^{(2)}$; Case 1 of (1) applies it to $\mathcal{I}$ with $D^{(2)}$; Case 2 applies it to weakenings and to expanded coverings. An induction that fixed $\mathcal{I}$ and inducted on $D^{(1)}$ alone does not go through.

Two further points that the one-line phrasing conceals.

(a) *The instance grows.* Expanded coverings have *more* variables than $\mathcal{I}$. So μ is not monotone in the instance, and the induction is genuinely on $\mu(D^{(1)})$ alone — with the reduction of a covering being the extension of $D^{(1)}$ along the parent map (Proposition 5.15(h)), whose μ is *larger*. Every appeal to the inductive hypothesis at a covering must therefore be checked to occur at a strictly smaller reduction, not merely at a different instance. In Case 2 of (1) the reductions used are the $D^{(B,\perp)}$, which are contained in $D^{(1)}$ restricted at z to a proper subset B; that is where the decrease comes from.

(b) *Which appeals decrease.* (2) invokes the hypothesis at $D^{(2)} \subsetneq D^{(1)}$: strict because $D^{(2)} <_T D^{(1)}$ has $C \lneq B$ at some variable. Case 1 of (1) likewise. Case 2 invokes it at $D^{(B,\perp)} \subseteq D^{(B,\top)}$, which is $D^{(1)}$ with the z-domain cut to $B \lneq D_z^{(1)}$. In every case the decrease is at a single variable and by at least one element. A formalization should extract this as a lemma: *if $D' \leq_T D$ with T any type and $D' \neq D$, then* $\mu(D') < \mu(D)$, and use it uniformly.

Proof of (2). Suppose $\mathcal{I}^{(2)}$ has no solution. Weaken $\mathcal{I}^{(2)}$ to an instance $\mathcal{I}'$ crucial in $D^{(2)}$ (Remark 5.13). By the inductive hypothesis applied to $\mathcal{I}'$ and $D^{(2)}$, $\mathcal{I}'$ satisfies (1a) and (1b) or (1c).

Case (1c). There is an expanded covering $\mathcal{J}$ of $\mathcal{I}'$, crucial in $D^{(2)}$, with a linked connected subinstance Υ. Let x be a variable of a constraint $C \in \Upsilon$. By Lemma 8.14(p), $\mathrm{Con}_1(C, x)$ is a perfect linear congruence; choose $\zeta \subseteq \mathbf{D}_x \times \mathbf{D}_x \times \mathbf{Z}_p$ with $(y_1, y_2, 0) \in \zeta \iff (y_1, y_2) \in \mathrm{Con}_1(C, x)$ and $\mathrm{pr}_{1,2}(\zeta) = \mathrm{Con}_1(C, x)^*$.

Replace the variable x of C in $\mathcal{J}$ by a fresh x' and add the constraint $\zeta(x, x', z)$ with z fresh of domain $\mathbf{Z}_p$; call the result Θ, and extend $D^{(1)}, D^{(2)}$ by $D_{x'}^{(1)} = D_{x'}^{(2)} = D_{x'}$. Let E_i be the set of $b \in \mathbf{Z}_p$ such that Θ has a solution in $D^{(i)}$ with $z = b$. By Lemma 6.10, $E_2 \dot{\leq}_T E_1$.

Since $\mathcal{J}$ is crucial in $D^{(2)}$, $\Theta^{(2)}$ has a solution but none with $z = 0$; since $\mathcal{I}^{(1)}$ has a solution, so does $\mathcal{J}^{(1)}$, hence $\Theta^{(1)}$ has one with $z = 0$. Thus $0 \in E_1$, $0 \notin E_2$, and E_1 has at least two elements. As $\mathbf{Z}_p$ has no proper subalgebra of size > 1, $E_1 = \mathbf{Z}_p$; then $E_2 <_T \mathbf{Z}_p$ with $T \in \{\textsc{ba}, \textsc{c}\}$, contradicting Lemma 6.24.

Case (1b). Corollary 8.21 gives a contradiction directly. □

Formalization note 8.3 (Three gaps in the case (1c) argument)**.** (a) "E_1 contains at least two different elements, $0 \in E_1$ and $0 \notin E_2$, hence $E_1 = \mathbf{Z}_p$" uses that E_1 is a subuniverse of $\mathbf{Z}_p$ — true because $E_1 = \Theta^{(1)}(z)$ is a relation defined by an instance (Definition 5.3) — *and* that the subuniverses of $\mathbf{Z}_p$ are the singletons and the whole set. The latter needs $w^{\mathbf{Z}_p}$ to generate: from $a \neq b$ one reaches every element. Prove it; it is one line but it is the crux.

(b) $E_2 \dot{\leq}_T E_1$ is applied with the *dotted* conclusion of Lemma 6.10, and the argument then treats E_2 as a proper subuniverse. The case $E_2 = \varnothing$ must be handled: it is excluded because $\Theta^{(2)}$ has a solution. Say so.

(c) The choice of ζ is by Definition 3.35, which supplies ζ with $\mathrm{pr}_{1,2}\,\zeta = \sigma^*$ and $(a_1, a_2, b) \in \zeta \Rightarrow ((a_1, a_2) \in \sigma \iff b = 0)$. The proof uses the *biconditional* $(y_1, y_2, 0) \in \zeta \iff (y_1, y_2) \in \mathrm{Con}_1(C, x)$, which is stronger: it needs, for each $(y_1, y_2) \in \sigma$, that $(y_1, y_2, 0) \in \zeta$. This follows from $\mathrm{pr}_{1,2}\,\zeta = \sigma^* \supseteq \sigma$ plus the implication, but the step is suppressed.

Proof of (1), *sketch of the structure.* First, $|D_x^{(1)}| > 1$ for some x: otherwise $\mathcal{I}^{(1)}$ has all domains singletons and 1-consistency makes it solvable, contradicting cruciality.

Case 1: some $D_x^{(1)}$ has a nontrivial BA *or central subuniverse.* Lemma 8.17 yields a 1-consistent reduction $D^{(2)} \leq_T D^{(1)}$, $T \in \{\textsc{ba}, \textsc{c}\}$. One checks $\mathcal{I}$ is crucial in $D^{(2)}$: for a weakening $\mathcal{J}$ of a constraint, $\mathcal{J}^{(1)}$ has a solution, so by the inductive hypothesis (part (2) at $D^{(2)}$) $\mathcal{J}^{(2)}$ has one. Applying the inductive hypothesis (part (1)) at $D^{(2)}$ gives the conclusion.

Case 2: otherwise. Lemma 6.2 gives $E <_{T(\sigma)}^{D_z} D_z^{(1)}$ for some z and $T \in \{\textsc{pc}, \textsc{l}\}$. Property (1a) is obtained by choosing, for each x, a minimal $D_x^{(2)} \leq_{\mathcal{M}T} D_x^{(1)}$ containing the value at x of a solution produced by weakening a fixed constraint C; Lemma 6.17 gives $D^{(2)} \lhd\lhd\lhd^D D^{(1)}$

and Lemma 8.11 splits into two subcases, in one of which the inductive hypothesis applies and in the other of which Lemma 8.18 applies directly.

For (1b)/(1c) one forms, for each B in the family $\mathcal{B} = \{B : B <^{D_z}_{T(\sigma)} D^{(1)}_z\}$, the reduction $D^{(B,\top)}$ that equals $D^{(1)}$ off z and B at z, and the maximal 1-consistent reduction $D^{(B,\perp)} \subseteq D^{(B,\top)}$ (possibly empty). Lemma 8.16 supplies tree-coverings $\Upsilon_{B,x}$ computing $D^{(B,\perp)}_x$, and their conjunction Υ_x works uniformly in B. Writing $\mathcal{B}_0$ for the B with $D^{(B,\perp)}$ nonempty, the argument splits:

- $\mathcal{B}_0 = \varnothing$: the type must be L, and Lemma 8.19 applied to a suitably weakened tree-covering produces, for any two constraints sharing a variable, a reflexive bridge between their Con_1's. Hence $\mathcal{I}$ is connected, and (1b) or (1c) holds according as its solution set is subdirect.
- $\mathcal{B}_0 \neq \varnothing$: one constructs a family Ω of weakenings of $\mathcal{I}$ that is *critically* unable to solve all of $\mathcal{B}_0$ (conditions 1–4 of Construction 8.5) and argues by whether some member fails (1b).

□

Formalization note 8.4 (Construction 8.5 is the hardest object)**.** The set Ω is specified by four conditions, one of which (condition 3) is a minimality condition "if we replace any instance in Ω by all weaker instances then 2 is not satisfied". The paper obtains Ω by starting from $\{\mathcal{I}\}$ and repeatedly weakening, justified by "we cannot weaken forever". Condition 4 is then *derived* from condition 3.

For a formalization this is the single most demanding step in §8, for three reasons. (i) The iteration weakens a *set* of instances, so the termination measure is a measure on finite sets of instances, not on instances — see Hazard 5.11. (ii) The derivation of condition 4 from condition 3 is a two-line argument that instantiates condition 3 at $\Omega \cup \{\mathcal{J}_C : C \in \mathcal{J}\} \setminus \{\mathcal{J}\}$ and reads off an existential; the existential witness B depends on $\mathcal{J}$, so condition 4 is a $\Pi\Sigma$ statement and must be stated as such. (iii) The subsequent Subcase 1 builds a *second* descending sequence $\mathcal{B}_1 \supseteq \mathcal{B}_2 \supseteq \dots$ inside the first, again terminating by finiteness. Nested well-founded constructions of this shape are where informal proofs most often hide a circularity, and this one should be isolated and stated as a standalone lemma before Theorem 8.1 is attempted.

Remark 8.5 (Construction of Ω)**.** Ω is a finite set of weakenings of $\mathcal{I}$ with:

1. every member is a weakening of $\mathcal{I}$;
2. $\bigcap_{\mathcal{J}\in\Omega} \mathrm{Sol}(\mathcal{J}) = \varnothing$, where $\mathrm{Sol}(\mathcal{J}) = \{B \in \mathcal{B}_0 : \mathcal{J}^{(B,\perp)} \text{ has a solution}\}$;
3. replacing any member by all weaker instances destroys 2;
4. for every $\mathcal{J} \in \Omega$ there is $B \in \mathcal{B}_0$ with (a) $\mathcal{J}$ crucial in $D^{(B,\perp)}$ and (b) $B \in \mathrm{Sol}(\mathcal{J}')$ for every $\mathcal{J}' \in \Omega \setminus \{\mathcal{J}\}$.

8.3 The three consumed theorems

Theorem 8.6 (Reductions are safe)**.** *Let Θ be a cycle-consistent irreducible instance and $B <^{D_x}_{T} D_x$ with $T \in \{\text{BA}, \text{C}, \text{PC}\}$. Then Θ has a solution if and only if Θ has a solution with $x \in B$.*

Proof. For $T = \text{PC}$ this is Theorem 8.7. For $T \in \{\text{BA}, \text{C}\}$, Lemma 8.17 gives a 1-consistent reduction $D^{(1)} \leq_T D$ with $D^{(1)}_x \subseteq B$; by Theorem 8.1(2), $\Theta^{(1)}$ has a solution, hence Θ has one with $x \in B$. □

Theorem 8.7 (PC reductions do not kill all solutions)**.** *If $\mathcal{I}$ is cycle-consistent and irreducible, $B <^{D_y}_{\text{PC}(\sigma)} D_y$ for some $y \in \mathrm{Var}(\mathcal{I})$, and $\mathcal{I}$ has a solution, then $\mathcal{I}$ has a solution with $y \in B$.*

Theorem 8.8 (Codimension one)**.** Suppose

(i) $\mathcal{I}$ is a linked cycle-consistent irreducible instance with $\mathrm{Var}(\mathcal{I}) = \{x_1, \ldots, x_m\}$;
(ii) $\mathbf{D}_{x_i}$ is S-free for every i;
(iii) weakening all constraints of $\mathcal{I}$ gives an instance with subdirect solution set;
(iv) σ_{x_i} is the intersection of all linear congruences σ on $\mathbf{D}_{x_i}$ with $\sigma^* = D_{x_i}^2$;
(v) $L_{x_i} = \mathbf{D}_{x_i}/\sigma_{x_i}$;
(vi) $\phi\colon \mathbf{Z}_{q_1} \times \cdots \times \mathbf{Z}_{q_k} \to L_{x_1} \times \cdots \times L_{x_m}$ is a homomorphism, q_i prime;
(vii) weakening any one constraint of $\mathcal{I}$ gives an instance with a solution in $\phi(\bar{a})$ for every $\bar{a} \in \mathbf{Z}_{q_1} \times \cdots \times \mathbf{Z}_{q_k}$.

Then $\Delta = \{\bar{a} : \mathcal{I}$ has a solution in $\phi(\bar{a})\}$ is empty, full, or an affine subspace of codimension 1 — the solution set of a single linear equation.

Proof. If Δ is full we are done. Otherwise pick $\bar{b} \notin \Delta$ and regard $\phi(\bar{b})$ as a reduction $D^{(1)}$ for $\mathcal{I}$; condition (vii) says $\mathcal{I}$ is crucial in $D^{(1)}$.

Step 1: some $\mathrm{Con}_1(C, x)$ *is a perfect linear congruence.* By Lemma 8.11, either $C_0^{(1)} = \varnothing$ for some constraint C_0, or $D^{(1)}$ is 1-consistent.

- If $C_0^{(1)} = \varnothing$: cruciality forces $\mathcal{I} = \{C_0\}$, say $C_0 = R(y_1, \ldots, y_t)$. By Lemma 8.18, R has the parallelogram property and $\mathrm{Con}_1(R, 1)$ is linear with $\mathrm{Con}_1(R, 1)^* = D_{y_1}^2$. By Lemma 4.17, $\mathbf{D}_{y_1}/\delta \cong \mathbf{Z}_p$; with ψ the quotient map, $\zeta = \{(a_1, a_2, b) : \psi(a_1) - \psi(a_2) = b\}$ witnesses perfection.
- If $D^{(1)}$ is 1-consistent: by Theorem 8.1(1) every constraint has the parallelogram property and (1b) or (1c) holds. In either case Lemma 8.14(p) applies — under (1c) to the linked connected subinstance, whose containing a constraint of $\mathcal{I}$ is forced by condition (iii); under (1b) to $\mathcal{I}$ itself.

Step 2: add a $\mathbf{Z}_p$ *coordinate.* Let ζ witness perfection of $\mathrm{Con}_1(C, x)$. Add a variable z with domain $\mathbf{Z}_p$, replace x in C by a fresh x', and add $\zeta(x, x', z)$, giving $\mathcal{I}'$. Put

$$L = \{(\bar{a}, b) : \mathcal{I}' \text{ has a solution with } z = b \text{ in } \phi(\bar{a})\} \leq \mathbf{Z}_{q_1} \times \cdots \times \mathbf{Z}_{q_k} \times \mathbf{Z}_p.$$

By condition (vii), $\mathrm{pr}_{1..k}(L)$ is full; and $(\bar{b}, 0) \notin L$. So L has dimension k and is the solution set of one linear equation. If that equation is $z = c$ with $c \neq 0$ then $\Delta = \varnothing$; otherwise setting $z = 0$ gives an equation defining Δ, of dimension $k - 1$. ☐

Formalization note 8.9 (The linear algebra of Step 2)**.** Step 2 uses four facts about subgroups of $\mathbf{Z}_{q_1} \times \cdots \times \mathbf{Z}_{q_k} \times \mathbf{Z}_p$ that are stated as if obvious:

(a) L is a subuniverse, hence (the q_i, p being prime and w being a sum) a subgroup;
(b) "dimension" is well defined for a subgroup of a product of $\mathbf{Z}_q$ with *different* primes q. It is not a vector space; the correct statement is that such a group splits as a product over primes of $\mathbb{F}_q$-vector spaces, and dimension is the tuple of dimensions. The footnote in [13] about "J may contain only variables on the same field" is the same issue surfacing in the algorithm;
(c) a subgroup of codimension one is the kernel of a single homomorphism to $\mathbb{F}_q$, i.e. the solution set of one linear equation;
(d) "$\mathrm{pr}_{1..k}(L)$ full and $(\bar{b}, 0) \notin L$ imply $\dim L = k$".

Item (b) is the one to be careful about: it is the reason Theorem 8.8 is stated with mixed primes $q_1, \ldots, q_k$ and the reason (c) must be proved prime by prime. In Mathlib the substrate exists (`ZMod`, `Module` over `ZMod p`) but the mixed-prime dimension notion and the codimension-one characterisation are not there; see §13.

8.4 The supporting lemmas

The lemmas invoked above, in dependency order.

Lemma 8.10 ([12, Lem. 6.1]). *An expanded covering of a cycle-consistent irreducible instance is cycle-consistent and irreducible.*

Lemma 8.11. *Suppose $D^{(1)}$ is a 1-consistent reduction for $\mathcal{I}$, every $D^{(1)}_x$ is $\textsc{s}$-free, $T \in \{\textsc{pc}, \textsc{l}, \textsc{d}\}$, $D^{(1)} \lhd\lhd\lhd D$, and $D^{(2)}_x \leq_{\mathcal{M}T} D^{(1)}_x$ is a minimal $\mathcal{M}T$ subuniverse for every x. Then either $C^{(2)} = \varnothing$ for some constraint C, or $\mathcal{I}^{(2)}$ is 1-consistent.*

Lemma 8.12. *A constraint crucial in a reduction of a cycle-consistent irreducible instance is rectangular, and its* Con_1*'s are irreducible congruences.*

Lemma 8.13. *A relation whose restriction along strong subsets becomes empty yields a bridge between the witnessing congruences.*

Lemma 8.14 (Connected instances). *Let $\mathcal{I}$ be a cycle-consistent connected instance. Then*

(a) *any two constraints with a common variable are adjacent;*
(b) *for constraints C_1, C_2, variables $x_i \in \mathrm{Var}(C_i)$, and any path from x_1 to x_2, there is a bridge δ from $\mathrm{Con}_1(C_1, x_1)$ to $\mathrm{Con}_1(C_2, x_2)$ with $\widetilde{\delta}$ containing all pairs connected by that path;*
(p) *if $\mathcal{I}$ is linked then $\mathrm{Con}_1(C, x)$ is a perfect linear congruence for every $C \in \mathcal{I}$, $x \in \mathrm{Var}(C)$.*

Formalization note 8.15. Clause (p) is the bridge between the bridge machinery and the linear algebra; it is what makes Theorem 8.8 possible. Its proof is (b) followed by Lemma 6.22: linkedness of the instance makes $\widetilde{\delta}$ linked as a relation, which upgrades the bridge. The two senses of "linked" (Hazard 3.12) meet exactly here, and the step where one becomes the other should be isolated.

Lemma 8.16 ([14, Lem. 5.6]). *For a 1-consistent instance and a reduction, there is a tree-covering whose projection onto a given variable computes the maximal 1-consistent subreduction at that variable.*

Lemma 8.17. *Let $D^{(1)}$ be a 1-consistent reduction of a cycle-consistent instance $\mathcal{I}$ with $D^{(1)} \lhd\lhd\lhd D$, and $B <^{D_x}_{T} D^{(1)}_x$ with $T \in \{\textsc{ba}, \textsc{c}, \textsc{pc}\}$. Then there is a nonempty 1-consistent reduction $D^{(2)} \lhd\lhd\lhd D^{(1)}$ with $D^{(2)}_x \subseteq B$. Moreover, if $T \in \{\textsc{ba}, \textsc{c}\}$ then $D^{(2)} \leq_T D^{(1)}$; and if $T = \textsc{pc}$ and every $D^{(1)}_y$ is $\textsc{s}$-free then $D^{(2)} \leq_{\mathcal{M}\textsc{pc}} D^{(1)}$.*

Lemma 8.18. *A constraint that becomes empty under a minimal multi-type reduction, in an instance crucial there, has the parallelogram property; and in the linear case its first induced congruence σ satisfies $\sigma^* = D^2$.*

Lemma 8.19 (Bridges from a subdirect PC/linear instance)**.** Suppose

(i) $\mathcal{I}$ has subdirect solution set;
(ii) $D^{(1)}$ is a reduction with $D^{(1)}_x \lhd\lhd\lhd D_x$ for every x;
(iii) $C \in \mathcal{I}$ is a constraint of type $T \in \{\textsc{pc}, \textsc{l}\}$;
(iv) $B <^{D_z}_{\mathcal{T}(\xi)} D^{(1)}_z$ for some variable z, with $\mathcal{T} \in \{\textsc{ba}, \textsc{c}, \textsc{s}, \textsc{pc}, \textsc{l}\}$;
(v) if $T = \textsc{pc}$ then $\mathcal{T} \in \{\textsc{pc}, \textsc{l}\}$;
(vi) $\mathcal{I}^{(1)}$ has a solution;
(vii) $\mathcal{I}^{(1)}$ has no solution with $z \in B$;
(viii) weakening C in $\mathcal{I}$ gives an instance with a solution in $D^{(1)}$ having $z \in B$.

Then $\mathcal{T} = T$, and for every variable x of C there is a bridge δ from ξ to $\mathrm{Con}_1(C, x)$ with $\widetilde{\delta} \supseteq \mathcal{I}(z, x)$.

Formalization note 8.20 (The longest proof in the section)**.** Lemma 8.19 carries eight numbered hypotheses and its proof runs to roughly ten pages. It is the only source of bridges in §8, and every use of connectedness routes through it. Its conclusion has two independent parts — "$\mathcal{T} = T$" and "there is a bridge with $\widetilde{\delta} \supseteq \mathcal{I}(z, x)$" — and different call sites use different parts; splitting it into two lemmas is tempting but the proof produces them together.

A formalization should attack this lemma *last* among the §3.2 material and *first* among the things budgeted generously. It, together with Theorem 7.53 (which it consumes) and Theorem 8.1 (which consumes it), is where the majority of the effort will go.

Corollary 8.21. *In the situation of Theorem 8.1(2) with $\mathcal{I}'$ satisfying (1b), the type $\mathcal{T}$ of the reduction and the type of the constraints must agree, which contradicts $\mathcal{T} \in \{\textsc{ba}, \textsc{c}\}$.*

9 The algorithm

[13] proves the correctness statements but does not contain the algorithm; that is in [12]. This section states it, because target (T1) of §1 is a proposition *about* it and cannot be written down otherwise.

9.1 Solve

```
function Solve(I):
    repeat
        I := ForceConsistency(I)
        if I = false: return "No"
        if some D_x has a strong subset B:
            I := ReduceDomain(I, x, B)
    until nothing changed
    return SolveLinear(I)
```

ForceConsistency enforces cycle-consistency and irreducibility, returning `false` if some domain empties. The reduction step is licensed by Theorem 8.6: restricting to a strong subset cannot destroy solvability. When no strong subset remains, Lemma 6.2 together with Lemma 4.17 gives, for each domain of size at least 2, a congruence σ_x with $D_x/\sigma_x \cong \mathbb{Z}_{q_1} \times \cdots \times \mathbb{Z}_{q_{n_x}}$; choose σ_x minimal. That case is SolveLinear.

9.2 SolveLinear

Write $\phi\colon \mathbb{Z}_{p_1} \times \cdots \times \mathbb{Z}_{p_k} \to D_{x_1}/\sigma_{x_1} \times \cdots \times D_{x_m}/\sigma_{x_m}$ for a linear map and

$$\phi^{-1}(\mathcal{I}) = \{\bar{a} : \mathcal{I} \text{ has a solution in } \phi(\bar{a})\}.$$

Computing $\phi^{-1}(\mathcal{I})$ would solve $\mathcal{I}$; the algorithm cannot do that, but it can do four things:

(p0) decide $\bar{a} \in \phi^{-1}(\mathcal{I})$, by reducing each domain to the corresponding block and recursing on a smaller domain;
(p1) decide $\phi^{-1}(\mathcal{I}) = \mathbb{Z}_{p_1} \times \cdots \times \mathbb{Z}_{p_k}$, by testing $\bar{0}$ and the k unit vectors with (p0) — correct because $\phi^{-1}(\mathcal{I})$ is a subgroup;
(p2) compute $\phi^{-1}(\mathcal{I})$ when $\mathcal{I}$ is not linked, by splitting into linked pieces, recursing, and taking the union;
(p3) compute $\phi^{-1}(\mathcal{I})$ when it is empty or of codimension one, by finding $\bar{a} \notin \phi^{-1}(\mathcal{I})$, then for each i a b_i with $\bar{a}[i \mapsto b_i] \in \phi^{-1}(\mathcal{I})$ if one exists, and reading off the equation $\sum_{i \in J}(y_i - a_i)/(b_i - a_i) = 1$ over the i for which b_i exists.

```
function SolveLinear(I):
    m := dim(D_{x_1}/s_1 x ... x D_{x_n}/s_n)
    phi := a bijective linear map Z_{p_1}x...xZ_{p_m} -> that product
    while phi^-1(I) != Z_{p_1}x...xZ_{p_m}:              // (p1)
        I' := I
        for C in I':                                  // drop what can be dropped
            if phi^-1(I' \ {C}) != Z_{p_1}x...xZ_{p_m}:   // (p1)
                I' := I' \ {C}
        F := phi^-1(I')                                  // (p2) or (p3)
        if F = {}: return "No"
        m := dim F ;  psi := a bijective linear map onto F ;  phi := phi o psi
    return "Yes"
```

The inner loop makes $\mathcal{I}'$ minimal with $\phi^{-1}(\mathcal{I}')$ not full. At that point Theorem 8.8 applies — its hypothesis (vii) is exactly minimality — so $\phi^{-1}(\mathcal{I}')$ is empty or of codimension one, and (p3) computes it; or $\mathcal{I}'$ is not linked and (p2) does. Since $\phi^{-1}(\mathcal{I}) \subseteq \phi^{-1}(\mathcal{I}')$, the image of ϕ shrinks while still containing all solutions. The dimension strictly decreases, so the loop terminates.

9.3 What is honestly formalizable

Formalization note 9.1 (Three obstacles between the pseudocode and a Lean function)**.** (a) *Termination is not structural.* Solve recurses through (p0) on strictly smaller domains, and SolveLinear loops on a strictly decreasing dimension. Neither is a structural recursion; both need an explicit well-founded measure, and the two interleave — SolveLinear calls Solve on a smaller domain, which calls SolveLinear again. The measure is lexicographic: $(\sum_x |D_x|,\ \dim \phi)$.

(b) *Several steps quantify over objects that are only finitely many by accident.* "If some D_x has a strong subset B" quantifies over subsets of D_x, over congruences on D_x, and — through absorption — over *term operations*, of which there are infinitely many. Decidability comes from the fact that on a finite set there are finitely many operations of each arity and finitely many subsets, so "B absorbs A with some term" is decidable by exhaustive search over Clo_k for $k \leq |A|^{|A|}$; but that bound is a theorem, not a definition, and it is the only reason the algorithm is an algorithm.

(c) *Polynomial time is a different subject.* The bound in [12] counts calls and bounds the recursion depth by $|A|$; making that formal needs a cost model. See §12.

Theorem 9.2 (Target (T1))**.** *There is a function* Solve *from instances of* CSP(Γ) *to Booleans such that for every instance* $\mathcal{I}$*,*

$$\textsc{Solve}(\mathcal{I}) = \texttt{true} \iff \mathcal{I} \text{ has a solution.}$$

Theorem 9.2 mentions no machine, is provable from §8, and is the honest formal content of "the algorithm works". It is strictly stronger than decidability of CSP(Γ), which is trivial for a finite template: it says that *this* algorithm — the one whose existence is the dichotomy — is correct.

10 The hard half

If no weak near-unanimity operation preserves Γ, then $\mathrm{CSP}(\Gamma)$ is NP-complete. This half is older, shorter, and better understood than the tractable half, and it factors cleanly into algebra and complexity.

10.1 The chain

(1) *Reduction to a core, and to an idempotent algebra.* One may assume Γ contains all singleton unary relations, so that $\mathrm{Pol}(\Gamma)$ is idempotent, at the cost of a polynomial-time reduction.
(2) *Maróti–McKenzie.* A finite idempotent algebra has a Taylor term if and only if it has a weak near-unanimity term [9].
(3) *No Taylor term $\Rightarrow$ pp-interpretation.* If a finite idempotent algebra has no Taylor term, then the variety it generates contains a two-element algebra whose only operations are projections; equivalently, Γ primitively positively interprets every finite constraint language, in particular a template for 3-Sat or Nae-3-Sat.
(4) *Bulatov–Jeavons–Krokhin.* A primitive positive interpretation of Δ in Γ yields a polynomial-time many-one reduction $\mathrm{CSP}(\Delta) \leq_p \mathrm{CSP}(\Gamma)$ [1].
(5) *Cook–Levin.* Nae-3-Sat is NP-hard.

Steps (1)–(3) are algebra; step (4) is a syntactic construction plus a complexity-theoretic wrapper; step (5) is pure complexity theory.

10.2 The layered targets

Theorem 10.1 (Target (H0)). *Let* $\mathbf{A}$ *be a finite idempotent algebra with no weak near-unanimity term operation. Then* $\mathrm{Inv}(\mathbf{A})$ *primitively positively interprets the constraint language* $\{\textsc{Nae}\}$ *on* $\{0,1\}$*, where* $\textsc{Nae} = \{0,1\}^3 \setminus \{(0,0,0),(1,1,1)\}$.

Theorem 10.2 (Target (H1)). *If* Γ *primitively positively interprets* Δ *then* $\mathrm{CSP}(\Delta) \leq_p \mathrm{CSP}(\Gamma)$*. Consequently, under* (H0), $\mathrm{CSP}(\Gamma)$ *is* NP*-hard.*

Theorem 10.1 mentions no machine. Theorem 10.2 is where the complexity layer enters, and it is small — the reduction is a syntactic substitution of formulas — but it cannot be stated at all without a formalized notion of polynomial-time many-one reduction.

10.3 Definitions

Definition 10.3 (pp-formula, pp-definability). A *primitive positive formula* over Γ is $\exists y_1 \ldots \exists y_s \bigwedge_j R_j(\bar{z}_j)$ with $R_j \in \Gamma \cup \{=\}$. A relation is *pp-definable* from Γ if some pp-formula defines it.

Definition 10.4 (pp-interpretation). Γ on A *pp-interprets* Δ on B if there are $d \geq 1$, a pp-definable $F \subseteq A^d$, and a surjection $\pi\colon F \to B$ such that the π-preimage of every relation of Δ, and of the equality relation on B, is pp-definable from Γ.

Theorem 10.5 (Geiger; Bodnarchuk–Kalužnin–Kotov–Romov). *For a finite* A*, a relation is pp-definable from* Γ *if and only if it is preserved by every polymorphism of* Γ.

Formalization note 10.6 (The Galois connection is the expensive import). Theorem 10.5 is what converts the algebraic hypothesis ("no WNU polymorphism") into a syntactic conclusion ("pp-defines a hard relation"), and it is the only genuinely substantial import of §10. The easy direction is a one-line calculation. The hard direction — every relation preserved by all polymorphisms is pp-definable — goes through the free algebra on $|A|^k$ generators and an indicator-instance construction; it is roughly four printed pages and is the single largest item in the (H0) budget. It is also of independent value and would be a reasonable Mathlib contribution on its own.

Remark 10.7 (Why the hard half is not the hard half)**.** Despite its name, (H0) is far smaller than (T0): perhaps twenty printed pages against a hundred. The asymmetry is intrinsic. Showing a problem is hard requires exhibiting one construction; showing it is easy requires an algorithm that works on every instance, and the correctness proof must survive every configuration the instance can present.

11 Defects in the sources

A blueprint earns its keep by finding the places where the source cannot be transcribed. This section is now a register rather than an argument: every item has been resolved, and the reasoning lives beside the statement it repairs. They are not criticisms of [13], which is unusually careful prose — across nine thousand lines of TEX it says "obvious" three times and "clearly" never. They are the residue that any informal exposition leaves.

Ten places where the source is consistent but leaves the reader to supply a reading are legislated in §2.1; two of the ten turned out to be theorems rather than choices. Fourteen places where it is wrong, incomplete, or states less than its own proof gives are listed below. All fourteen are repaired, and none needed new mathematics: three needed a citation the later paper had dropped, one was a typographical slip, one was a statement that is false as written, and the rest were omitted steps.

Where	**What, and where the repair is**
Propagation modulo a congruence	Stated, used five times, never proved; it is Lemma 6.4 at the canonical surjection. A mechanical audit of all 81 labelled statements shows it is the *only* statement in the source that is cited while being neither proved nor attributed.
Stable intersection, type C	$n = 2$ is argued only for PC; the argument has no analogue for C. Theorem 7.53 and Hazard 7.54.
Bridges from relations	Applied without its third hypothesis, which can fail; cruciality supplies it. §8.
Main induction, (1c)	The endgame proves *connected* where (1c) demands *linked*, and Case 1 derives (1c) for the wrong reduction. Hazard 8.3.
Minimal $\mathcal{MT}$ subuniverses	The bridging lemma was deleted from the source along with its citation. Hazard 6.16.
Symmetrisation of a bridge	The displayed formula defines a relation that is never a bridge. Hazard 7.35(a).
Bridges and abelian groups	*False as stated*; a three-element counterexample. Hazard 7.35(b) and Remark 7.37.
Nontrivial reflexive bridges	Linkedness asserted from an inclusion bearing on another hypothesis. Hazard 7.35(c).
The PC inductive step	Cites the box form of absorption-preserves-linkedness for a restriction that is not a box. Lemma 7.15, Hazard 7.16.
Existence of an irreducible congruence	Factors by the wrong congruence, hiding an unsupported properness step. Hazard 7.50.
Linear congruences on top	The conclusion is asserted; proved here by a route that also removes an external import. Hazard 4.19.
Maximal multi-type extension	Concludes that two sets meet because each meets a third. Hazard 7.57.
The central-relation citation	Drops a third alternative present in the source it cites, and needs an "empty" alternative under Convention 2.4. Hazard 7.19.
Full fibres from surjectivity	A non sequitur at three sites; true for a reason that makes the arity $\|A\|$ load-bearing. Hazard 7.28.

Two further patterns are worth naming, because they recur rather than occurring once. The first is a *proper* containment asserted where the cited lemma gives only $\subseteq$, usually after factoring by a congruence, which can close the gap; it appears at least five times and is repaired the same way each time, by producing the witness that the source has already displayed two lines earlier. The second is an upgrade from "the image has size one" to "the image is *this* block"; it appears four times and should become a named lemma.

Finally, two statements are weaker than their own proofs and should be restated: Lemma 6.12(i), whose proof gives $B \cap D \;\dot{\lessdot}\!\lessdot\!\lessdot^{A}\; D$ (Hazard 7.52), and Lemma 7.1, which needs the words *proper nonempty* (Hazard 7.2).

Remark 11.1 (A retracted item, and what replaced it)**.** An earlier draft of this section carried an eleventh blocking item: that the recursion-depth bound $|A| + |\Gamma|$ of [12, Lem. 5.2] was "not established for the CheckTuple $\to$ SolveNonlinked path", and that "without the repair the algorithm is not polynomial". That was wrong, and it is withdrawn.

Lemma 5.2 bounds the depth by arguing that every recursion site except CheckWeakerInstance "reduce[s] all domains of size greater than 1". At the SolveNonlinked site this needs the instance to be neither linked nor fragmented — otherwise the linked-component congruence can be trivial on some domain, that domain does not shrink, and the bound degrades from the constant $|A|$ to $O(|A| \cdot n)$, which is fatal, since the total work is $\mathrm{poly}(N)^{\mathrm{depth}}$. The source does state the hypothesis and does discharge it. It is simply not in one place:

- the precondition "an instance that is not linked and not fragmented" is in the prose introducing the function, and *not* in the `Input:` line of its pseudocode;
- it is discharged at the call site by "Θ_0 is not fragmented because it is crucial" — itself a one-line argument the source does not give: a fragmented unsatisfiable instance has an unsatisfiable part, and no constraint of the *other* part is then crucial;
- the step from "not fragmented" to "*every* domain splits" — propagate the partition along a path, which exists by non-fragmentedness and is value-preserving by 1-consistency — appears once, inside the proof of a *different* lemma, and is never cited here.

So the mathematics is present and the theorem is not in doubt. What is absent is any single place where the six recursion sites of Lemma 5.2 are discharged, and that is a blueprint's job rather than a defect of the source. The same pattern holds for the Γ-order argument, whose case (3) needs the projections to be unchanged under weakening, which holds because the instance is 1-consistent when CheckWeakerInstance runs — true, and unstated.

The one step that really is missing is supplied by Lemma 11.2 below. It is two lines, it is not in the source, and without it the $|A|$ bound at this site is unjustified.

Lemma 11.2 (The SolveNonlinked call site)**.** *Let Θ_0 be cycle-consistent, non-fragmented and not linked, and let Θ_0' be Θ_0 with each domain restricted to a subset $D_x' \subseteq D_x$. Then every domain of Θ_0 of size greater than one is strictly shrunk by the time SolveNonlinked(Θ_0') recurses into Solve.*

Proof. Write $\tau_x = \mathrm{LinkedCon}(\Theta_0, x)$, a congruence by [12, Lem. 6.2]. First, τ_x is proper at *every* variable. Since Θ_0 is not linked there are at least two connected components of the graph on $\bigsqcup_z \{z\} \times D_z$; and every component meets every variable, because from any (x, a) non-fragmentedness supplies a path to y and 1-consistency extends a along it. So each D_y is partitioned into at least two nonempty τ_y-blocks, each therefore proper.

Now split on whether Θ_0' is linked.

Not linked. Then SolveNonlinked recurses on a component, whose trace on each D_y is a τ_y-block intersected with D_y' — a proper subset of D_y by the previous paragraph.

Linked. Then for every z and all $a, b \in D_z'$ some path of Θ_0' connects them; every such path is a path of Θ_0, so $D_z' \times D_z' \subseteq \tau_z$, i.e. D_z' lies inside a single τ_z-block. That block is proper, so $D_z' \subsetneq D_z$ — every domain has *already* been shrunk, at the CheckTuple step, before SolveNonlinked was called at all. □

Remark 11.3 (Why this is the right argument)**.** Two things are worth extracting.

First, the question this replaces was the wrong one. The natural worry is whether the restriction preserves SolveNonlinked's stated precondition "not linked", and one then starts asking how the minimal linear congruence ConLin relates to LinkedCon — a relation the source never states, and which is not needed. The dichotomy above uses no relation between them: *either* the precondition survives and the component split does the shrinking, *or* it fails and the restriction has already done it.

Second, it follows that SolveNonlinked's precondition is not required for the depth bound at all. The function is correct on a linked instance too — there is then one component and it recurses on the whole of it — and Lemma 11.2 covers that case. The precondition should be read as documentation of the intended use, not as an obligation the caller must discharge.

Remark 11.4 (What this list is evidence of)**.** Nine blocking items in a fifty-page paper is not a high rate; it is roughly what one should expect, and it is the reason blueprints exist. Note the distribution: one is a genuine misstatement (D1), one is an omission (D2), two are gaps in the mathematics that need new arguments (D3, D4), three are hypotheses that are too weak or missing (D5, D6, D8), and two are artefacts of the interface with the older paper (D7, D9). None of them threatens the theorem. All of them cost time.

Remark 11.1 is the counterweight, and the more instructive entry. A tenth item was listed, survived a review round, and was published — and was wrong, because it was taken from a reader who worked from the pseudocode while the hypothesis it needed was in the surrounding prose. A list like this one is exactly as reliable as the reading behind each entry, and the failure mode is not carelessness but reading the formal-looking part of a paper and not the sentence before it.

12 The complexity layer

Targets (T2) and (H1) are stated here and not proved. The point of stating them is to be precise about what is *not* being claimed.

Definition 12.1 (What a cost model would have to supply)**.** To state (T2) one needs: an encoding of instances of CSP(Γ) as strings; a machine model with a step count; a predicate "f is computed in time bounded by a polynomial in the input length"; and closure of that predicate under composition. To state (H1) one needs, in addition, a polynomial-time many-one reduction relation $\leq_p$ and the class NP.

Mathlib supplies the machine (`Turing.TM2`), an encoding (`Computability.Encoding`), and a time-bounded computation predicate (`Turing.TM2ComputableInPolyTime`). It does not supply a complexity *class*, P, NP, NP-completeness, $\leq_p$, or SAT; and the composition lemma `TM2ComputableInPolyTime.comp` is an open `proof_wanted` in Mathlib itself. Moreover `TM2ComputableInPolyTime` applies to total functions $\alpha \to \beta$, never to languages, so even the existing predicate would need reshaping.

Remark 12.2 (This layer now exists, and it was cheap)**.** An earlier draft of this remark said that building the complexity layer was "a substantial project — comparable in size to the algebraic core" and that attempting it as part of this project "would be a scoping error". That was wrong, and it has been tested rather than argued: the vocabulary — words, a cost model, P, NP by verifiers, $\leq_p$, NP-hardness, NP-completeness, and a generic NP-hard problem — is about 850 lines of Lean, `sorry`-free, and took an afternoon.

That figure is what the literature should have led one to expect. The class layer is 702 lines in Balbach's Isabelle Cook–Levin development and 992 in Gäher–Kunze's Coq one, against totals of 56 514 and ≈ 16 500 respectively. Defining the classes has never been the expensive part; the expensive part is connecting them to a concrete machine, which is a different activity.

So the reason to keep (T2) out of scope is *not* that complexity theory is unavailable. It is Hazard 1.3: (T2) requires exhibiting SOLVE as a program with a proved step bound, and that is the largest program-construction task in the project. (H1), by contrast, needs only a syntactic substitution and should be built.

Remark 12.3 (What the exercise cost, and what it caught)**.** Two defects surfaced in those 850 lines, both in *statements* rather than proofs, and neither of a kind a type-checker reports. The generic problem was first defined with a single budget bounding both the certificate length and the running time, which makes the theorem false — the two directions pull the bound opposite ways. And the language was first given only the tests "register is empty" and "register's first bit is 1", with which no loop that consumes its input can be written at all; that was found by trying to write the first program.

The ratio is worth noting for the rest of this document. Two statement-level errors in 850 lines, against ten found in Zhuk's fifty pages by readers who did not formalize (§11). The defect list should be expected to grow under formalization, not to be complete.

13 Formalization

13.1 What is reused

The predecessor development `ZhukLean` is a complete, `sorry`-free Lean 4 formalization of Zhuk's centre theorem, built on `Mathlib.ModelTheory`. It is consumed here as a `lake` dependency, not copied. It supplies:

- products of structures (`piStructure`, `prodStructure`, coordinatewise realization, reindexing and evaluation homomorphisms) — the one part of the universal-algebra vocabulary Mathlib lacks;
- absorption in the tuple form (`Witnesses`, `Absorbs`, `BinAbsorbs`), Taylor terms (`IsTaylorOn`), and central absorption (`CentrallyAbsorbs`);
- the relational description of absorption by essential relations (Barto–Kazda), which is what converts absorption at some arity into absorption at arity 3;
- Zhuk's centre theorem itself (`zhuk_center`) and the ternary collapse (`exists_ternary_witnesses`) — which is precisely Lemma 4.4, cited without proof in [13].

Three of [13]'s sixteen black-box imports are therefore already discharged.

13.2 What is built here

Module	Contents	Status
`CSP/Product`	products of structures	complete
`CSP/Congruence`	`Congruence L M`, order, `sInf`, quotient	complete
`CSP/Irreducible`	irreducibility, existence and uniqueness of σ^*	complete
`CSP/Bridge`	bridges, collapse, composition, linear/PC	2 imports open
`CSP/Types`	the six types, multi-types, $\lhd\lhd\lhd$	complete
`CSP/Instance`	instances, reductions, consistency, cruciality	complete
`CSP/Main`	the main statements	targets

Everything below `CSP/Main` is `sorry`-free. The two open items in `CSP/Bridge` are Lemma 3.30 and its collapse identity, both imported from [12].

13.3 Design decisions

Build on `Mathlib.ModelTheory`, keep the language generic. A bespoke "finite algebra with one WNU operation" type would bundle the standing hypotheses as fields and avoid threading them, which is a real cost in the predecessor development. But it would discard `Substructures.lean` — 985 lines of exactly the Sg API needed, including `mem_closure_iff_exists_term` with the variable type equal to the generating set, an interface no hand-rolled clone matches — and it would discard the predecessor's 1600 lines wholesale. Congruences and quotients are also *cheaper* in `ModelTheory`, not dearer. Instantiating to $\mathcal{V}_n$ is a thin layer on top.

$\lhd\lhd\lhd$ is `Relation.ReflTransGen`. Its induction principle is the one the proofs use.

Dotted relations are defined as disjunctions. `DotLll C B := C = ∅ ∨ Lll C B`, so the case split is forced at every use site rather than resolved by the reader.

The bridge criterion is the definition of linearity. Zhuk defines a linear congruence by a per-block isomorphism to $\mathbb{Z}_p^m$ and derives the bridge criterion. We reverse this: the criterion is first-order over finite data and is what every consumer uses; the block description becomes a structure theorem. See Hazard 4.13.

σ^* is data attached to irreducibility. `Irreducible.exists_isCover` produces it; there is no standalone function, because there is no cover without irreducibility. A pleasant by-product: with the definition taken literally, irreducibility *implies* $\sigma \neq A^2$, since the empty family intersects to the universe. The source never says this.

Instances are lists; scopes are tuples; the variable type is a parameter. See Hazards 5.2 and 5.16. The last of these is the decision that has not yet been cashed out, and it should be made before anything about coverings is written.

Nothing is stubbed with `sorry` at the level of a definition. `IsConnectedInstance` and `IsExpandedCovering` are *absent* rather than defined as `sorry`, because a `def : Prop := sorry` is a junk constant that downstream proofs can silently lean on. Absence forces the design decision; a stub hides it.

13.4 Order of attack

1. Lemma 6.10. Small, self-contained, used everywhere, and adjacent to what is already formalized.
2. The mixed-prime linear algebra of Hazard 8.9. Independent of everything else; can be done in parallel.
3. Lemma 3.30. Twelve source lines of relational algebra; closes the two open items in `CSP/Bridge`.
4. The coproduct design for coverings, then Proposition 5.15.
5. Theorem 7.53. The one hard statement in the interface, and the sole source of bridges.
6. Lemma 8.19. Eight hypotheses, ten pages.
7. Theorem 8.1. Last.

13.5 Scope

The calibration point is that the predecessor formalized about 3.5 printed pages in about 1670 lines of Lean — roughly 480 lines per page. Excluding §12 and [13]'s §4, which is an independent second result and not needed for the dichotomy, the transitive closure of what must be proved is on the order of 100–130 printed pages, so 35 000–60 000 lines of Lean.

Line count, however, is not the binding constraint, and an earlier draft of this section was wrong to convert it into person-years at conventional rates. The pipeline this project belongs to has since produced roughly 33 000 verified, `sorry`-free Lean lines in about ten active hours on the Jordan–Schönflies formalization, by fanning independent modules into separate git worktrees and landing them in waves; the rate held on the hard content, not merely on the foundation.

What binds is instead the *critical path*: the depth of the serial chain, and the number of places where the source is wrong and new mathematics is needed. Here the serial chain is Lemma 8.19 → Theorem 7.53 → Theorem 8.1 — each a single large proof that cannot be split across workers — and the new mathematics is the four blocking items D3, D4, D6 and D7 of §11. Those are not throughput-limited. Schedule around them, not around the line count.

What is finished here is the design: the route, the statements, the vocabulary at formalization granularity, the hazard list, the defect list, the missing-layer inventory, and a building development whose foundations are proved and whose summit is stated.

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